Department of Mechanical Engineering, National Taiwan University
Academic Year 114

Mechanical Engineering Practice Final Analysis Report

Group 5

Members: CHENG-HSUAN LEE, Lin Yueti, Yang Zhenguo, Li Wangmao, Liao Zhixiang, Lu Zhaoyu

Directory

AI Assisted Use Statement:

During the execution of this topic, Google Gemini is used as a collaboration aid to assist in the following tasks:

1. Report Structuring
   Produce a complete report outline based on topic requirements to ensure all necessary sections are covered and properly organized.
2. Conceptual Refinement
   Assist in discussing and comparing different mechanism design options (such as bridge structure and robotic arm structure), and improve the reasons and basis for the solution selection based on the topic constraints.
3. Analytical Verification
   Cross-check engineering calculation results such as torque, leverage ratio, and safety factor to verify the rationality and correctness of design decisions.
4. Technical Writing
   Assist in writing clear and professional technical descriptions, including design concepts (Design Intent) and Design for Manufacturing and Assembly considerations (DFMA, Design for Manufacturing and Assembly) and other related content.

All design decisions, final engineering calculations, and CAD model creation were independently made and verified by members of this team. AI tools are provided for reference purposes only and do not replace any critical engineering judgment or design work.

Summary

This report takes the final task of "Mechanical Engineering Practice" as the core and records the complete engineering development process of this group from task analysis, program evaluation, mechanism design, material selection to actual testing and correction. After initially comparing various options such as shooting balls, climbing, building bridges, hot air balloons and drones, the team considered mission success rate, operational error tolerance, production feasibility, cost constraints and subsequent scalability, and finally chose to use drones as the main mobile platform. During the development process, the team first tried to use the gripper to directly pick up the ball. However, after actual testing, they found that the hovering accuracy was insufficient, the propeller downwash would interfere with the ball's position, and the gripper mechanism was affected by weight and space constraints, making it difficult to complete the task stably. Therefore, the system architecture was further converted into a "child-mother system", where the drone is responsible for crossing the height difference and transporting, and the child vehicle is responsible for completing the ball retrieval on the platform to improve the overall fault tolerance rate and mission stability.

In terms of design and production, this team adopts modular design thinking and designs the machine arm, sub-cart, ball-retrieving mechanism, control panel holder and connectors separately, so that parts can be quickly replaced and iterated repeatedly. In terms of material selection, MDF was selected for the machine arm after analysis of strength and vibration characteristics to take into account rigidity, processing speed and maintenance convenience; 3D printed PLA was used for the fuselage and part of the structure to enhance design flexibility. The flight control system was also converted from Betaflight to INAV, significantly improving altitude control and landing stability. Finally, the team gradually improved the system reliability through thrust tests, propeller comparisons, sub-car ball-taking tests and multiple flight corrections. Overall, this topic not only completed a set of UAV subsystems that can perform pick-and-place tasks, but also demonstrated the complete implementation process of "analyzing problems, establishing hypotheses, production verification, correction and iteration" in engineering design.

Part One, Engineering Design Thinking and Design Evolution

1.1 Task analysis and establishment of design criteria

After getting the competition rules in the first week, we first broke down the task into specific engineering requirements, and then started discussing the plan.

The physical constraints of the task: it needs to cross the height difference (there is a significant drop from area B to area A), complete multiple pick-and-place cycles within the time limit, and each cycle includes four sub-tasks: flying to, waiting for or operating the ball, flying back, and releasing the ball. As can be seen from this structure, the task score is determined by two things: the success rate of a single loop, and how many loops can be completed within the time limit.

This means that maximizing your score is not about having the fastest speed or the highest success rate on a single pass, but about maximizing the expected value of both. A system that takes only 10 seconds each time but has a 50% success rate may not be as good in expected score as a system that takes 25 seconds each time but has a 95% success rate. This understanding became key when it was later decided to abandon the three-ball mechanism - the three-ball mechanism increased the speed, but also reduced the success rate.

Operational uncertainty of the task:

The exact position of the ball on the platform in Zone A is not fixed during the game (it may have been moved due to previous operations), and the operator's actual operation accuracy under the pressure of the game will be worse than during practice. This means that the system design must include sufficient room for error, rather than assuming that it will operate under optimal conditions every time.

Task timing constraints:

There is a fixed reset time between each attempt, and the retest mechanism is triggered after two consecutive zero scores. This makes "constantly getting partial points" more strategically valuable than "occasionally getting full points", especially in early attempts.

Based on these analyses, we established a five-dimensional framework for design evaluation and applied it consistently throughout the development process:

(1) Mission success rate:

The probability of the system completing a single complete pick-and-place cycle under reasonable operating conditions (not optimal, not worst). This assessment needs to consider the combined effects of operational difficulty, environmental disturbances (downwash, ball displacement uncertainty, landing position errors) and hardware reliability. Problems with any of the three will cause the success rate to plummet.

(2) Operational error tolerance:

When any sub-step fails (the landing position is deviated, the ball is not aligned, the flight path deviates), does the system have a remedial path or must it return to zero and try again. In a competition situation, "there are still opportunities for recovery after a mistake" and "the task ends immediately after a mistake" have a great impact on the operation results.

(3) Production feasibility:

Can it be completed with the equipment we actually have (FDM printer in the laboratory, laser cutting machine, hand welding equipment), technical ability (no one has complete drone DIY experience), material pipeline (normal procurement process, one to two weeks of waiting for arrival), and available time (about 2 to 3 class times per week plus independent time). Many designs that are completely feasible under the conditions of "complete equipment and skilled personnel" are impossible under our conditions.

(4) Cost limit:

The budget directly determines the upper limit of selection of core parts. In particular, the four components that determine the upper limit of flight system performance, including flight control, motors, batteries, and remote controls, each have a choice span from a few hundred to several thousand yuan, and high-end options are not necessarily more suitable for our mission needs than mid-level options.

(5) Subsequent expansion:

Whether the structure of the plan supports the improvement of details later without overturning the overall design. This dimension was not easy to quantify in the early days, but it had a great impact later - when we discovered in the sixth week that directly clamping the ball was not possible, whether we could quickly switch to the parent-child system directly depended on whether the original architecture had room for expansion.

1.2 Evaluation and selection of five options

We spent nearly two complete discussions evaluating the five options, and we did not converge on a certain direction in the first discussion.

1. Shooting mechanism: an extreme trade-off of speed for fault tolerance
   The initial appeal of the ball-shooting scheme was the simplicity of its concept - use a catapult to launch the ball from area B to area A, or more likely, take the ball from area A and shoot it directly back to area B, or devise some device to allow the ball to fly over the height difference.
   In terms of technical feasibility, mechanisms such as air pressure ejection, spring ejection, and magnetic acceleration are all mature technologies, and there is no need to invent new principles. If debugged in place, a single execution time can be completed in seconds, much faster than any scenario that requires movement.
   However, after evaluation, it was found that there are too many sources of uncertainty in shooting accuracy, and most of them are uncontrollable. The initial velocity depends on the repeatability of the launching mechanism (spring fatigue, air pressure stability), the angle depends on the initial alignment accuracy, and the ball's landing point is also affected by air resistance and rotation. Even if each individual source of error is small, their cumulative effect can reach ten centimeters or more over a range of several meters.
   The more fundamental problem is the dynamic conditions of the game: the position of the ball on the A-zone platform may change after each operation, which means that it needs to be realigned before each ball is taken; and "realignment" itself is a time-consuming and uncertain step.
   In the end, zero error tolerance is a killer. The ball missed without any remedy, and the mission failed immediately. In a game situation with multiple attempts, a system that succeeds 80% of the time but is completely irremediable when it fails is less likely to be expected to score than a system that succeeds 60% of the time but can retry after a failure.

Conclusion: From a cost-effectiveness perspective, the technical investment (accurate launching mechanism, precision alignment system) of the ball-shooting solution may be higher than that of a drone, but the fault tolerance rate is much lower than that of a drone. exclude.

2. Climbing mechanism: Production bottlenecks are more fatal than technical difficulties

   The theoretical advantage of a climbing mechanism is that once climbed, it is stable and does not require hovering or resisting gravity.

   We conducted a detailed feasibility analysis on the climbing mechanism. The design of the climbing mechanism requires: a reliable gripping mechanism (suction cup, hook, friction wheel), sufficient driving torque, stability verification under different load conditions, and an additional mechanism that allows the machine to complete the ball retrieval after it is deployed at the top.

   The problem is that the complexity of each subproblem is underestimated. The suction cup needs to maintain sufficient negative pressure and is very sensitive to surface material and cleanliness; the hook needs the platform to have appropriate geometric features to engage and cannot slip when climbing; the friction wheel relies on contact friction and is very sensitive to the surface condition of the platform. Under race conditions we have no control over the surface state of the platform.

   Even if the climbing problem is solved, the climbing machine's activity space on the top of the platform is strictly limited - the orientation of the machine after climbing up is uncertain, and the ball-retrieving mechanism needs to be able to complete its work in any orientation, which further increases the complexity of the mechanism design.

   The most decisive factor was time estimation: designing a climbing system that could perform all of these maneuvers reliably under our production conditions would conservatively require more than twice the time we had available throughout the semester. Production feasibility is the hard limit. exclude.

3. Bridging plan: site adaptability issues

   The concept of bridging is to erect a structure connecting areas B and A before or

during the game, and have wheeled or crawling robots drive along the bridge.
We carefully evaluated how the bridge could be erected: it could be a folding structure, deployable trusses, slide rail extensions, etc. Among them, slide rail extension is technically the simplest, requiring only a retractable structure that is long enough.
The problem lies in the adaptability of the playing surface. The erection of the bridge requires that the distance and relative position from area B to area A be known and fixed, so that the target end of the bridge can accurately land on the platform in area A. But if there is any change in the field configuration on the day of the competition (a slightly different distance, a deviation in the platform position, an unstable landing point at the bridge end), the entire solution will be invalid - and this failure cannot be repaired immediately during the competition.
In addition, the action of "building a bridge" itself requires time and precision. If the bridge fails to be built (the bridge end is not stable), a lot of time will have been spent in the first cycle without scoring.
The bridging plan has fundamental problems at two sub-levels: "site adaptability" and "fault tolerance during the erection process."

4. Hot air balloon project: the risks of technological unfamiliarity
   The idea of a hot air balloon is to use buoyancy to overcome gravity and does not require a power system to maintain altitude. In theory, energy consumption is the lowest, and size restrictions are relatively loose (large size does not necessarily mean overweight).
   However, the biggest problem we faced during the evaluation was not "What are the technical difficulties of hot air balloons", but "We simply don't know what are the technical difficulties of hot air balloons." No member of the team has practical experience in balloon control, heating system design, or buoyancy calculations, and the teaching resources on the Internet are mainly focused on ornamental hot air balloons rather than precise and controllable mission platforms.
   When doing tasks that require precision in a completely unfamiliar technical field, risk assessment itself is extremely unreliable. We may underestimate difficult points, miss critical design constraints, or discover completely unexpected problems during testing. Once we encounter such a problem, we don't have the

background knowledge to quickly find a solution.

From the perspective of "subsequent scalability", the hot air balloon solution has almost no room for iteration - if a certain subsystem does not perform as expected, we don't know how to improve it, because we don't even know the direction of improvement.

For us, choosing a completely unfamiliar technical field is high-risk - not only is it difficult, but we don't even know what the difficulty is. Rather than choosing a direction that seems easy but has no idea where the risks are, it is better to choose a direction that knows where the difficulties lie and can solve them accordingly.

5. Drones: reasons for selection and known challenges

We selected drones because the risks are ones we can understand and manage. Before choosing, we made an honest difficulty checklist:

(1) The flight control system needs to learn and adjust parameters from scratch, and there is no ready-made "PID parameter that is best for this task"
(2) The peak current of the four motors may exceed the safe discharge limit of the battery, requiring power management.
(3) Propeller downwash will interfere with ball retrieval, and the intensity of the interference is difficult to accurately predict in advance.
(4) The center of gravity and inertia distribution of flying with a load are completely different from those of an empty aircraft, and the parameters need to be readjusted.
(5) Landing accuracy is affected by ground effect and is particularly difficult to control when close to the platform
(6) The electromechanical integration complexity of the entire system is several times that of other solutions

After knowing these difficulties, we still choose drones for the following reasons:

First, every failure of the drone has a corresponding remedial action (you can re-align it if it flies incorrectly, you can pull it up and try again if it fails to land, you can try again if it fails to retrieve the ball). Most of the failures of other plans are one-time endings.

Second, the difficulties of drones are "decomposable" - flight control parameter adjustment, power management, ball retrieval mechanism, communication system. Each sub-problem has a large amount of data resources and community support. We can learn and draw on it without inventing a solution from scratch.

Third, the architecture of the drone is "scalable" - if it is later found that it is necessary to install a ground subsystem, replace the propeller, or switch the firmware, these can be completed without overturning the overall design. This later proved to be the most critical prediction.

After selecting the drone, we also established a principle that was repeatedly reminded in the future: the best performance of a single subsystem does not mean the best performance of the entire mission system.

1.3 Flight end architecture and parts selection

1. rack form

   After selecting a drone, the first specific question is the form of the frame. We evaluated the following options

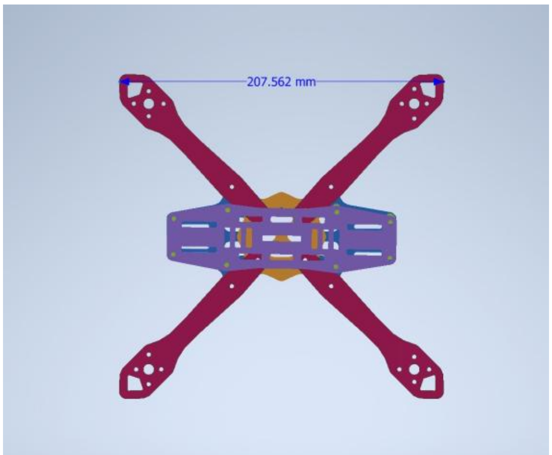


Figure 1. Rack top view CAD

X-type four-axis (final choice): X-type four-axis is the most mainstream configuration of FPV and flying machines in the world. Accordingly, the mixing matrix presets of the flight control firmware (Betaflight, INAV, ArduPilot), PID parameter adjustment guide, and arm strength design of the motor position all have the most complete information for this configuration. The flight control software knows the physical characteristics of the four-axis X-type, and the starting point parameters for parameter adjustment are all credible.

Six-axis (Hexacopter): More motors provide a higher overall thrust upper limit and better single-motor failure tolerance, but also bring higher weight, more complex mixing matrix calculations, more welding points (each joint is a potential failure point), and higher costs. For our mission, the advantages of six-axis (extra thrust redundancy) are not necessary, while the disadvantages (weight, complexity, cost) are real.

Coaxial propellers (Coaxial): The smallest size, but the aerodynamic interference of the upper and lower propellers significantly reduces efficiency, and the flight control parameters are more difficult to adjust and there is less data.

Asymmetric configuration: Some asymmetric frames (such as front and rear arms with unequal lengths) specially designed for specific tasks can improve the dynamic characteristics in specific directions, but you need to modify the mixing matrix yourself, and it is difficult to find reference adjustment information.

The reason for choosing the X-type four-axis is very straightforward: it has the most adjustment data and the most complete firmware support, and we don't need to verify the control logic ourselves. Time is saved on ball retrieval system and control integration.

2. Flight control board selection: Multiple considerations for SpeedyBee F405 AIO

   The selection of flight control board considers the following main options:

   SpeedyBee F405 AIO (final choice): The F405 core (STM32F405) is an MCU fully supported by both Betaflight and INAV, and firmware compatibility is guaranteed. AIO (All-In-One) integrates a 40A four-in-one ESC, allowing the flight controller and ESC to be on one board, eliminating the complexity of wiring from the flight controller to four independent ESCs, and also reducing wire weight and potential wiring errors. Four sets of full-featured UART ports allow us to have enough serial port resources to connect SBUS receivers, GPS (if needed later), and other peripherals. The Bluetooth module allows us to use the mobile app to quickly view and adjust parameters on site, eliminating the need to bring a laptop every time to adjust parameters. The Blackbox function allows flight data to be automatically recorded, and the specific data during the flight can be analyzed afterwards, instead of just relying on memory and subjective feelings to make adjustments.

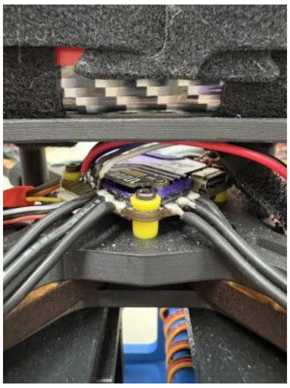


Figure 2. Close-up of SpeedyBee F405 AIO flight control board installation

Matek F405 series: It is also the core of F405 and has similar functions, but the ESC needs to be purchased and wired separately, making the integration more complex.

SpeedyBee F7 V3: The F7 core is more powerful, but it doesn't provide the necessary performance boost for our tasks, it's more expensive, and some F7 boards don't have INAV support as complete as the F405.

Betaflight F4 board: Lower-end option, missing some features we later used (like black box, full UART support).

There is an important detail that was discovered later and is worth recording: SpeedyBee F405 AIO has a hard-wired physical signal inverter on the UART2 RX pin, specifically for the SBUS protocol (SBUS uses an inverted serial signal, and most MCUs require software or hardware inversion to read it). In the early days, we connected the SBUS receiver to UART6 and spent nearly half a day investigating why there was no signal at all on the Betaflight receiver page - the receiver light was normal and the remote controller was paired successfully, but the flight controller could not read it. Finally, I found out that UART6 does not have a hardware inverter, and the MCU cannot interpret the SBUS inverted signal at all. This detail illustrates: You can't just look at "SBUS supported" on the spec sheet, you also need to confirm which specific UART pin has hardware inverter support.

3. Motor selection: selection logic of RS2205 KV2300

Motor selection requires simultaneous consideration of four interrelated parameters: KV value (speed coefficient), stator size (determines torque output capability), battery voltage coordination, and selection of supporting propellers. There are strict physical constraints between these four, and they cannot be optimized independently.

Our design starting point is the thrust-to-weight ratio requirement: the total system weight is estimated to be about 950 to 1050 g, and the target thrust-to-weight ratio is set to 3.5:1 to 4.0:1 (enough control margin, while not exceeding the current capacity of the 4S battery), which means the total thrust target of the four motors is 3325 to 4200 gf.

Given this thrust target, we evaluated several motor options:

RS2205 KV2300 (final choice): 2205 is a commonly used stator size for 5-inch propellers, and the torque output is a good match for 5-inch propellers. The operating speed of the KV2300 under the 4S battery is about 34,000 RPM. With the 5045 four-blade propeller, the thrust of a single propeller can reach 900~1000 gf, and the total thrust of the four propellers is 3600~4000 gf, which fully covers our target. More importantly, the RS2205 community has very rich information, with complete records from thrust curves to common faults, giving us a credible reference when debugging and troubleshooting.

RS2306 KV2400: The larger stator (2306) provides higher torque output and is suitable for heavier propellers (such as 5-inch tri-blade propellers or larger propeller diameters). But for our configuration using 5-inch propellers, the advantages of 2306 are not reflected, and instead bring heavier motor weight (approximately 8 to 10 g more for each motor, and 30 to 40 g more for four motors).

2204 KV2300: Smaller stator, lighter weight, but the torque output is not enough to drive the 5-inch propeller to the thrust level we need, and the motor heats up more seriously under high load.

Imported high-end brand motors: have better performance, but the price is usually 3 to 5 times that of RS2205, and there is no necessary performance improvement for our task requirements.

Motor damage incident: During the initial assembly process, we encountered a damage incident caused by information out of synchronization. The screws that fixed the motor were of a slightly longer type. After tightening, the ends of the screws touched the motor stator coil, causing slight damage to the coils. After disassembly, the damage was confirmed and a new motor was replaced. This incident allowed us to establish a standard operation of "confirming screw specifications before assembly" in subsequent assembly: the length of the motor fixing screws must be strictly limited to a range that does not touch the coil. Usually the manufacturer will give the maximum allowable screw length in the instruction manual, which must be followed.

4. Battery selection: Complete calculation basis for 4S 2700mAh 35C
   Battery selection is the part that involves the most calculations in the entire power system, and we have done a detailed quantitative analysis here.
   Voltage level selection: 4S (14.8V)
   Comparison of 3S (11.1V) vs 4S (14.8V): Under the same thrust requirement, the 4S battery allows the motor to operate at a lower current ($P = V \times I$, a high voltage results in a low current at the same power), which reduces the heat loss of the line and ESC, and also reduces the risk of the instantaneous peak current exceeding the battery discharge limit. In addition, 4S allows us to choose a motor with a higher KV value (the speed of the KV2300 is within the appropriate range only at 4S), providing the propeller with a higher speed and better efficiency.
   6S (22.2V) was not considered: 6S batteries and accompanying motors (which typically require lower KV values) cost more and were clearly overkill for the scale of our mission.
   Capacity selection: 2700mAh
   Flight time requirements: The competition requires multiple round trips within the time limit. We estimate that a single complete cycle takes about 30 seconds. Together with the preparation time and buffer, the required continuous flight capability is about 5 minutes.
   Calculation of power consumption: Based on the average hovering current

consumption of a single motor of 3~4A (total 12~16A), adding the power consumption of the sub-vehicle servo (each continuous rotating servo is about 700~800 mA when working, a total of three of them is about 2.4A), the power consumption of the control circuit (about 0.5A), and the peak current demand in flight operations (up to 2~3 times the average during take-off and rapid climb), it is conservatively estimated that the average total current is about 20A.

Figure 3. Weighing of the complete drone

Power consumption in 5 minutes: 20A × (5/60) hours = 1667 mAh, a 2700 mAh battery has a safety margin of approximately (2700-1667)/2700 = 38%.

Discharge rate selection: 35C

Maximum continuous discharge current: 2700 mAh × 35C = 94.5A.

The problem is: each of the four motors consumes about 30A at full throttle, totaling about 120A, which exceeds the safe discharge limit of the battery. This means we can't operate at full throttle during the race

Solution: Set the upper throttle output limit (Motor Output Limit) in the flight control software to limit the maximum throttle to about 80%. The corresponding motor current is much lower than the full throttle value. In our mission, full throttle takeoff is not required - the system is designed with a thrust-to-weight ratio of over 3.5:1, and it still has sufficient flight capability at 80% throttle.

5. Remote control system: AT9S Pro + R9DS choice

   The selection of the remote control system seems unimportant, but we later discovered that it actually has many details that need to be paid attention to.

   AT9S Pro remote control: This is a remote control originally designed for

fixed-wing aircraft. It has 9 channels (the basic control channels we need plus AUX mode switching are enough), and supports SBUS output. The main reasons for choosing it are that the price is within an acceptable range and there are detailed instructions for use.
Challenge: Fixed-wing to multi-rotor configuration conversion
The factory default of AT9S Pro is fixed-wing mode. The channel mapping, stroke volume, and mixing control settings are all for fixed-wing aircraft and require significant resetting before it can be used for multi-rotor aircraft. This setting process requires understanding the allocation method of SBUS channels and manually mapping the Roll, Pitch, Yaw, Throttle, and mode switching AUX channels to the correct SBUS channel positions. The job took longer than we expected.
R9DS receiver: supports SBUS output, matching the communication band of AT9S Pro. The pairing process requires confirmation that the receiver and transmitter are operating in the same frequency band and mode.
Channel neutral correction: The remote control's hardware rocker has mechanical errors, and the SBUS value of the neutral point may not be accurately aligned with 1500 us (standard neutral value). During the initial test flight, we encountered a problem where the joystick was centered but the aircraft continued to drift. After troubleshooting, it was confirmed that the channel neutral value shifted, and a sub-fine adjustment correction was required in the remote control settings. This step requires the use of Betaflight's receiver page to read the SBUS values in real time, confirm that the neutral values of all channels are accurate at 1500 us, and add appropriate Deadband settings (usually 10 to 15 us) to absorb residual mechanical jitter.

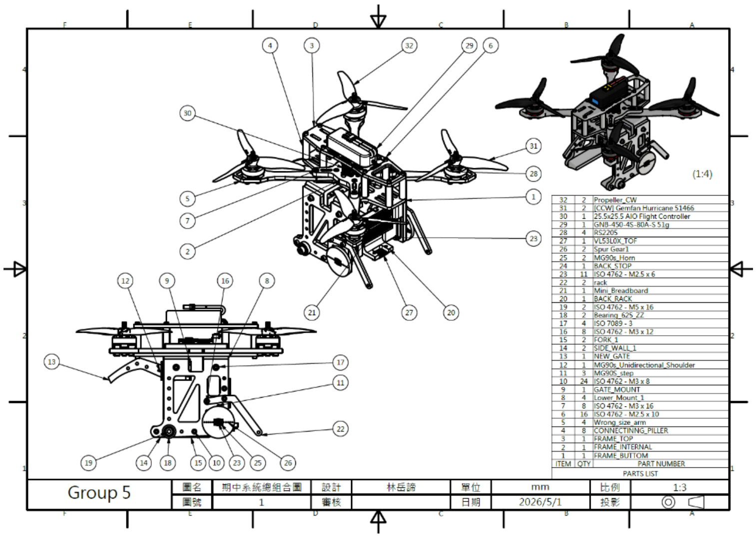


Figure 4. Mid-term system overall combination diagram

6. SBUS communication protocol selection

   There are several options for remote control signal transmission protocols: PWM (one wire per channel), PPM (single-wire serial transmission, but lower accuracy), SBUS (high-speed inverting serial, single-wire transmission of all channels, millisecond-level delay).

   Reasons for choosing SBUS: higher transmission frequency (usually 14ms update rate, faster than PPM's 20ms), more channels (up to 16 channels), digital signals immune to electromagnetic interference (the electromagnetic environment of drones is complex, and analog signals are susceptible to interference).

   Special features of SBUS: SBUS uses inverted serial signals, and most flight control MCUs require special processing. As mentioned earlier, the SpeedyBee F405 only has a hardware inverter on UART2, which was an important limitation that we later discovered during debugging.

7. ESC Signaling Protocol: DShot's Choice

   The communication protocol from the flight control to the ESC uses DShot (digital signal) instead of the traditional analog PWM.

   The advantage of DShot: pure digital signal, no signal quantization error and electromagnetic interference problems like PWM. DShot600 in the DShot series provides a fast enough update rate (updated every 1.67ms) in our system, allowing the flight controller to transmit control commands to the ESC in a timely manner.

DShot supports two-way communication: Bidirectional DShot is selected to allow the ESC to report rotational speed data to the flight control. This function was used in the thrust test bench experiment - we can read the actual rotational speed directly from Betaflight without the need for an additional rotational speed sensor.

1.4 Design and testing of the first version of the gripper

1. Design background and concepts

   When discussing ball retrieval mechanisms in Week 2, we suggested several possible ways to retrieve the ball:

   (1) Gripper: Actively controls two or more grippers to surround the ball and hold it with friction or shape constraints. The advantage is strong certainty, but the disadvantage is high alignment accuracy requirements.

   (2) Funnel/passive ball guide: Place a funnel-shaped entrance above the ball, allowing the ball to "roll" into the storage cavity as it moves with the drone. The advantage is that it allows for larger alignment errors, but the disadvantage is that the drone needs to make precise horizontal movements to get the ball into the funnel, and some kind of mechanism is needed to prevent the ball from rolling out after it enters.

   (3) Adsorption: Use vacuum adsorption or magnetism (if the ball is ferromagnetic, but tennis balls do not) to pull the ball. Not feasible in this task, excluded.

   (4) Net bag: unfold a net and let the ball fall into it, similar to the concept of catching fish. The alignment error rate is high, but the mechanism to control the expansion and retraction is complicated, and the fixation of the ball in the net is uncertain.

   After evaluation, the first version chose the gripper design because the concept is the most intuitive, the mechanism is the most certain (successful clamping means it is clamped, unlike a net bag where you need to confirm that the ball is in the correct position), and it can directly refer to existing mechanical claw design information.

2. Detailed design of the first version of the gripper

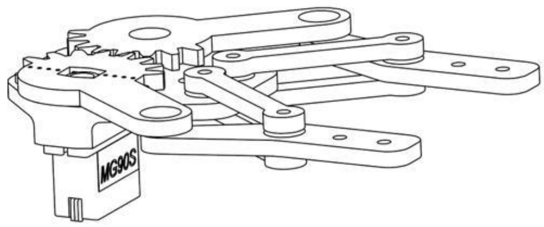


Figure 5. The first version of the gripper CAD drawing

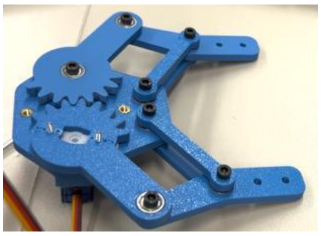


Figure 6. The first version of the gripper entity

Actuator: Use MG90S metal gear steering gear. The reasons for choosing the MG90S were its light weight (approximately 14 g), sufficient travel, simple control (standard PWM or digital control), and low price, making it suitable for use in the prototype stage. Metal gears have better durability than plastic gears.

Linkage design: A simple four-bar linkage mechanism is designed to convert the rotational motion of the servo into the opening and closing action of the gripper. The four-bar linkage mechanism is more flexible in shape design than gear transmission, allowing the clamping jaw to open to a large enough angle to surround the tennis ball.

Console: Use ESP32 development board. The reasons for choosing ESP32 are: it has enough PWM output pins to drive the servo, has Wi-Fi and Bluetooth communication capabilities (although the first version used wired testing), low power consumption, and the development tools of the Arduino ecosystem allow us to quickly verify the program logic.

Structural materials: All structural parts are made of 3D printed PLA, mainly because of the fast development speed (geometric design can be quickly iterated), low cost (cheap materials), and no need for special processing tools.

3. Problems discovered during testing

(1) Question 1: MG90S's nominal torque vs. actual demand

The nominal torque of MG90S is about 2.2 kg·cm at 4.8V. At first glance, this force seems quite strong, but it is not enough in the actual clamping mechanism. After the clamping claw clamps the ball, the reaction force of the ball is amplified through the connecting rod and then acts on the servo output shaft. The equivalent torque requirement far exceeds the servo specifications. In the repeated action test, we observed that the clamping force weakened significantly after several opening and closing times, and the connecting rod joint became loose due to wear of the holes in the PLA parts.

If you want to continue to use the servo-driven gripper solution, you need to switch to a servo with higher torque (such as DS3218, torque about 20 kg·cm), but this means that the weight increases to about 60 g/piece, and the size of the corresponding gripper mechanism also needs to be expanded.

(2) Issue 2: Position uncertainty in hover

During static tests on the ground, the jaws were able to align normally and grip a ball placed in a fixed position. However, under the condition of the drone hovering, the body itself has a natural drift of about ±5 cm, and the position of the ball is not fixed due to the downwash. These two uncertainties add up to a point where the actual operable window of gripper alignment with the ball is much smaller than static testing would indicate.

(3) Question 3: Geometric limitations of the mechanism in the belly

The gripper mechanism needs to be installed under the belly of the aircraft, but the belly space is very limited. The opening action of the linkage mechanism requires a certain amount of lateral space. Under the geometric limitations of the aircraft belly, the opening angle of the clamping jaws is limited, and the allowable error in alignment is further reduced. Moreover, the lower part of the belly is the intended installation location for the later sub-car system - if you continue to use direct ball clamping, there will be no installation space for the sub-car system.

(4) Problem 4: Downwash actively interferes with the ball’s position

This problem is not visible in static testing and is only noticeable when simulating flight conditions. After the rotor downwash hits the platform surface, it forms a wall jet that spreads outward and blows the ball away from the center. We did a preliminary test on the ground using an electric fan to simulate downwash, and confirmed that the airflow intensity at normal flight speeds was enough to move the ball several centimeters on the platform. This means that even if the jaws are aligned with the ball's original position, the ball may no longer be there during the fall.

4. The first version of the gripper provides the basis for subsequent designs
   Although the first version of the gripper did not ultimately enter official use, it laid several foundations for subsequent development:
   (1) Verification of ESP32 control framework: We verified the complete control link of ESP32 reading remote control input and controlling the servo through PWM on the first version of the gripper. This framework was directly used in subsequent sub-vehicle systems and did not need to be redesigned from scratch.
   (2) Assembly specifications for 3D printed parts: After discovering the sliding teeth problem of direct locking screws of PLA parts, we began to evaluate the use of hot-melt nuts and fully adopted them in subsequent sub-car designs.
   (3) Size perception: After completely fabricating a link-claw mechanism, we had a real perception of "how big and heavy this mechanism is." This helped avoid some dimensional misjudgments when designing the sub-car system later.
   (4) The establishment of design closed-loop thinking: The first version of the gripper gave us the first complete experience of "design → production → testing → problem discovery → analysis → decision-making", and we learned a lesson: the design looks good on paper, but it does not mean that it will be good under actual operating conditions.

1.5 Architecture conversion: complete design demonstration of child-mother system

After confirming that there are systemic problems with direct ball clamping, the choice we faced was: continue to try to improve the clamping jaw (change to a larger servo, improve hovering accuracy), or fundamentally change the structure.

1. Fundamental Limitations of the Direct Patch Scheme

Continuing to improve the direct ball clamping solution requires solving three problems at the same time: hovering accuracy, downwash interference, and steering gear torque. These three problems have a common characteristic: they are not problems of design details, but limitations of physical conditions.

The upper limit of hovering accuracy is determined by the sensor resolution of the flight control, the random disturbance of the airflow, and the delay of the control loop. It cannot be fundamentally solved by adjusting the PID better. Downwash interference will definitely exist as long as the rotor is rotating; reducing the speed can weaken the downwash, but it also reduces the thrust. This contradiction is more prominent in the ground effect area close to the ground. The servo torque problem can be solved by replacing the servo with a larger one, but the resulting increase in weight will make the thrust-to-weight ratio problem more serious. Solving these three problems at the same time requires paying costs in three different subsystems, and these costs compete with each other for the same resources (weight margin, control capabilities). It is unrealistic to solve three problems simultaneously under our conditions.

2. The cost of the child-mother system architecture

   A key question was raised during the discussion: "Under what conditions is it easiest to retrieve the ball?" The answer is that it is easiest on the ground, under conditions where there is no downwash interference, a stable reference frame, and the ability to retry alignment multiple times. These conditions are completely opposite to "get the ball while hovering in the air".

   If you can make retrieval happen on the ground, the problem changes from "retrieving the ball under the worst conditions" to "retrieving the ball under the best conditions." The trade-off is the need for a subsystem that can reach the platform and act on the ground.

   This thought process naturally led us to the architecture of the parent-child system: the drone is responsible for moving (what it does best), and the ground child vehicle is responsible for fetching the ball (the easiest thing on the ground).

3. Design responsibilities of each subsystem of the subsystem

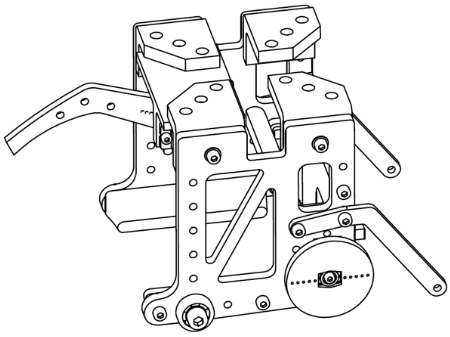


Figure 7. 3D draft of sub-car subsystem

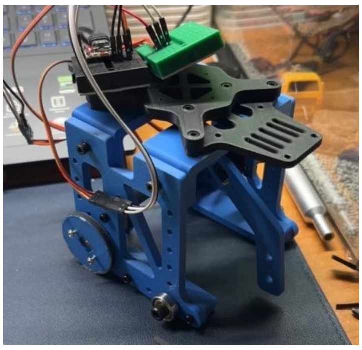


Figure 8: Sub-car subsystem physical finished product

Design responsibilities of UAV (parent system):

It is no longer necessary to hover accurately to retrieve the ball, only to roughly land in the safe landing zone of the platform (the allowable error is relaxed to ±15 cm)

Need to reliably bear the weight of the sub-car system (approximately 295 g)

Need to remain stable after putting down the sub-car (the center of gravity will change due to the release or movement of the sub-car)

It is necessary to avoid the rotor touching the edge of the platform during landing and take-off

Design responsibilities of ground subcar (subsystem):

After landing, it deploys from the bottom of the drone and searches for and approaches the ball on the platform.

Guide the ball into the storage cavity through the ball guide mechanism (front fork)

After fixing the ball, wait for the drone to take off, and be taken back to Area B together.

The ball is released in area B by manual operation or automatic mechanism

The core advantage of this division of responsibilities is that each subsystem only needs to solve its own problems and does not need to take into account the needs of another subsystem. The flight control parameters of the drone can be optimized for "stable flight and reliable landing" without having to consider "accurate hovering to ±2 cm"; the control logic of the sub-car can be designed for "ground movement and ball retrieval", without being affected by flight dynamics.

4. Accurate analysis of the weight cost of the mother-child system
   The final weight of the sub-car system: frame + steering gear + wheel set + control panel + wires is about 295 g, plus 57 g of tennis balls, the total take-off weight increases from about 685 g without a plane to about 1037 g with a load.
   Thrust-to-weight ratio change: Estimated based on the maximum thrust of the four motors of about 3800 gf:
   - Empty machine: thrust-to-weight ratio = 3800/685 ≈ 5.5:1
   - Strap cart: push-to-weight ratio = 3800/1037 ≈ 3.7:1 (reduced to about 3.5:1 after wearing tennis balls)

   The thrust-to-weight ratio of 3.5:1 is still within the safe range (it is generally considered that stable flight can be achieved if it is above 2.5:1, and comfortable control margin is available if it is above 3:1). But the hovering throttle point rises from about 20% (empty) to about 28% (with load), which means that at a setting of full throttle limit of 80%, the margin for controlling upward movement shrinks from 60% to 52%, which is still sufficient.

   Increased moment of inertia: The sub-car system is installed directly under the belly of the aircraft, about 9.5 cm in the negative direction of the Z-axis of the aircraft body. According to the parallel axis theorem, the addition of sub-cars significantly increases the pitch and roll moments of inertia of the entire machine. The increase in rotational inertia slows down the response of the airframe to the angular velocity command, which is the main reason for the low-frequency swing in the early stage of the test flight. The PID parameters need to be adjusted accordingly (increasing the I gain).

5. Installation interface design of sub-mother system

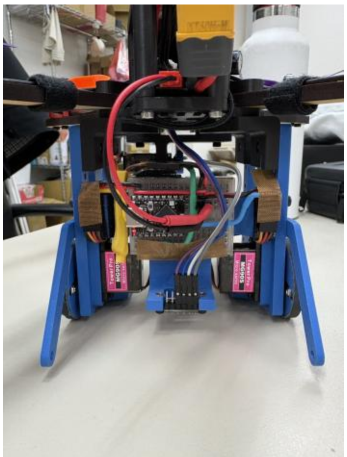


Figure 9. The combination of the back of the whole machine

The subcart is fixed to the mounting plate under the frame with 8 M3 screws. The selection of 8 screws was based on evaluation: 4 may not be enough - vibrations during flight may cause the screws to gradually loosen; 6 screws have geometric asymmetry in their arrangement; 8 screws, 2 on the front, rear, left and right, provide even fixation, and the disassembly and assembly time is within an acceptable range (about 3 minutes with tools).

Reasons for screw fixation instead of quick-release design: Although the quick-release mechanism is more convenient, it adds extra structural weight and potential mechanism reliability issues. In a competition situation, the most reliable connection is required for each flight fixation of the sub-car, and the 8 screws are superior to any quick release design in terms of reliability.

1.6 Comprehensive practice of modular design

After determining the architecture of the component-child system, we made a clear choice in the design methodology: fully adopt modular design, with rapid iteration as the highest priority.

1. Modular design strategy

   The core of modular design is to "design parts that may need modification into modules that can be replaced independently." This requires predicting "where problems are most likely to go wrong" at the beginning of the design, and then making modular designs for those places.

   Our prediction is:

      Arm: The arm is the first to receive force during a crash and is most easily damaged. It needs to be replaced quickly.

Ball-retrieving mechanism: the core innovative part with the most frequent iterations

Connectors: Affected by the impact of landing, the strength needs to be repeatedly verified and modified.

Control panel installation: Frequent wiring is required during the debugging phase, and the installation design must support repeated disassembly and assembly.

Battery slot: requires frequent disassembly and recharging

Parts that do not require modularization (can be integrally formed or fixedly connected):

Motherboard and ESC installation (no modifications required once soldering is complete)

Motor installation (no need to disassemble unless damaged)

Main power line (fixed connections are more reliable)

2. CAD Design Strategy: Principles of Partial Design

In the CAD design of the sub-car, we consciously modeled the parts of each functional area separately:

Body frame (floor): The basic structure that carries all other parts. This part is designed with multiple sets of mounting holes reserved (instead of only the holes that are actually needed), so that the frame does not need to be redesigned when new parts are added later. The arrangement of the reserved holes follows a rule: they are arranged at fixed intervals along the long and horizontal axes, so that parts of different sizes can find suitable installation positions.

Front fork ball retrieval mechanism: designed as an independent module, connected to the frame through a standardized screw interface. The specific geometry of the front fork (opening angle, guide ball shape) was modified several times during the test. Each time only the front fork itself needed to be reprinted, and the frame was not affected.

Baffle mount: The mount for fixing the tailgate, also designed independently.

The early version of the baffle seat was easy to break during landing impact. The second version thickened the key section, and the third version introduced a rounded corner design. Each version only needs to reprint the bezel seat, without affecting other parts.

Control board mounting base: The mounting frame of the ESP32-C3 and servo control board is designed to be installed in a slot that can be slid in/out. It is convenient to remove the wiring during debugging, and no tools are required for fixing.

Wheel set holder: The mounting frame for the continuously rotating steering gear and wheel set. Wheel sets of different diameters require different mounting seat geometries, and are designed as independent parts so that when replacing the wheel set, you only need to replace the mounting seat.

3. Comprehensive promotion of hot melt nut system

Figure 10: Actual hot-melt nut

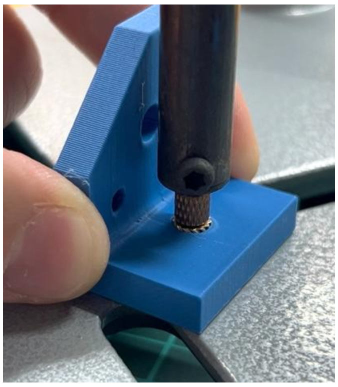


Figure 11. Hot melt nut insertion operation

The application of hot-melt nuts in this system goes far beyond the usual usage of 3D printed parts. We use hot-melt nuts in all screw holes that require repeated disassembly and assembly for the following reasons:

(1) The lifespan of direct tapping of PLA: Direct tapping of FDM-printed PLA parts (self-tapping screws into holes) usually has no problem the first time it is used, but after the 3rd to 5th disassembly and assembly, the threads of the screw holes begin to wear, and the locking ability is often completely lost after the 8th to 10th time. On average, each part required disassembly and assembly more than 10 times during our testing, making self-tapping unfeasible.

(2) FDM hole position accuracy problem: 3D printed hole positions are usually slightly smaller than the design value (shrinkage of the outer wall of the FDM part) or slightly elliptical (anisotropy in the printing direction). The hot-melt nut naturally adapts to the actual shape of the hole through thermoplastic deformation during insertion, eliminating locking uncertainty caused by dimensional deviations.

(3) Geometric restrictions on the installation location: Due to space constraints, the screw hole locations of some parts cannot be placed with nuts from the back. Hot-melt nuts are used to ensure reliable threads at these locations. There is no need to modify the geometry to allow the nuts to be placed.

Operating skills for electric soldering iron embedding: The hot-melt nut is heated with an electric soldering iron and then pressed into the hole. The heating time and pressing depth require certain operating skills. If the heating is insufficient, the plastic around the nut will not be completely melted, and the nut may loosen; if the heating is excessive, the plastic around the nut will melt excessively, affecting the surrounding structure. We confirmed suitable operating parameters through some test samples during the first batch of embeddings.

4. Specific implementation of machine arm quick replacement design

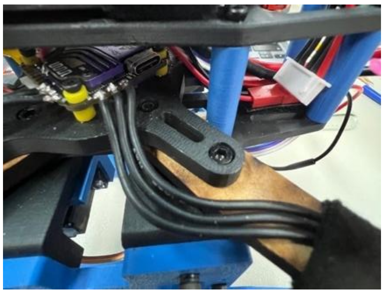


Figure 12. Arm locking screw 1 (two fixing screws, quick disassembly and replacement of the arm)

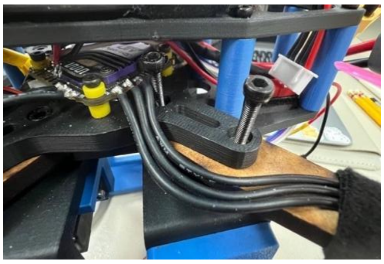


Figure 13. Machine arm locking screw 2 (sandwich structure side view)

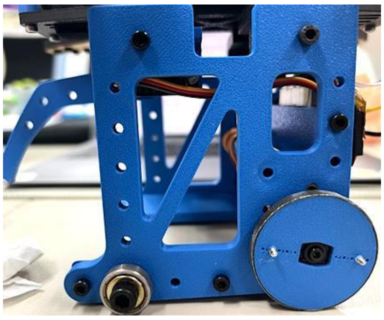


Figure 14. Reserved holes in the fuselage (supports can be quickly added)

The quick-change arm design involved several specific design decisions:

Number of fixing points: two screws. Fewer fixing points make replacement faster, but reduce reliability; four fixing points are more reliable, but replacement time doubles. We choose two and add dowel pins between the two fixed points to ensure that the position of the machine arm is accurately reproduced every time it is installed, without the need to re-measure the alignment.

Spare parts management: The laser-cut MDF arm can cut 8 to 10 pieces at the same time from one sheet, and the cost is very low (the cost of one sheet is less than tens of yuan). We confirm the number of spare parts before each major test to ensure that there are enough spare arms that can be replaced immediately during the test. We replaced more than 20 arms throughout development (mostly damage from collisions during test flights), and testing was never interrupted due to lack of spare parts.

1.7 Arm Material Selection: Experiment-Driven Design Decisions

1. Experimental design ideas

   The selection of arm materials involves three competing goals: light (directly affects thrust-to-weight ratio), hard (prevents vibration from being transmitted to the fuselage), and strong (prevents breakage in a crash). Different materials have different performance on these three goals, and these three goals cannot be optimized at the same time.

   The goal of our experiments is to compare the performance of candidate materials under the force modes most relevant to the task, rather than comparing generic material properties on a spec sheet. The aircraft arm is mainly subjected to cantilever beam bending load during flight (the bending moment generated by the motor thrust and gravity at the fixed end of the aircraft arm), so the core of the test is to simulate this force mode.

2. Reasons for selection of candidate materials

   5 mm PLA (3D printing): the most natural choice because we already have printers and operating experience, the highest degree of geometric design freedom (can print complex curved surfaces), and the lowest cost. Disadvantages are anisotropy (inter-laminar strength is much lower than intra-laminar strength), sensitivity to moisture, and possible softening at sustained high temperatures (e.g. near ESC heating).

   5.5 mm MDF (dense board, laser cutting): Wooden composite material, more isotropic than FDM parts, and laser cutting allows high shape accuracy (better than the dimensional accuracy of 3D printed parts). Laser cutting spare parts are produced quickly (a batch can be cut in a few minutes). The disadvantage is that it can only make flat shapes (complex three-dimensional geometry cannot be made), and it is sensitive to moisture (it may expand slightly in high humidity environments).

Carbon Fiber Tube/Plate: Lightest and strongest, standard material for commercial drone arms. However, carbon fiber cutting requires special tools (ordinary laser cutting machines are not suitable for cutting carbon fiber), and the dust during processing is harmful to health, so it is not suitable for use in our laboratory environment.

3. Details of experimental condition design

   For details on the design of the force experiment, see Part 2.

4. Deeper Considerations in Material Selection: Vibration and Natural Frequency

   In addition to static strength and stiffness, we also consider dynamic vibration characteristics.

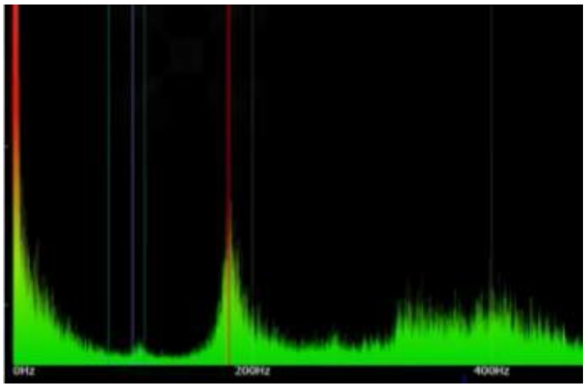


Figure 15. Arm vibration frequency-amplitude diagram

(Spectrum analysis shows that there is obvious vibration energy at about 180 Hz, which is a resonance risk area)

The drone arm is affected by two vibration sources at the same time during flight: the rotational unbalanced force of the motor and propeller, and the blade passing frequency (BPF, Blade Passing Frequency) of the propeller. If the natural frequency of the machine arm is close to these excitation frequencies, resonance will occur, transmitting the vibration energy of the machine arm into the fuselage, seriously interfering with the gyroscope measurement.

The natural frequency of the machine arm , where k is the bending stiffness and m is the mass. The bending rigidity of MDF is about 50% higher than that of PLA, and the mass is about 17% more. After substitution, the natural frequency of the MDF arm is about 27% higher than that of the PLA arm, pushing the resonance point to a higher frequency band, away from the main excitation frequency of the motor.$f_n \propto \sqrt{k/m}$

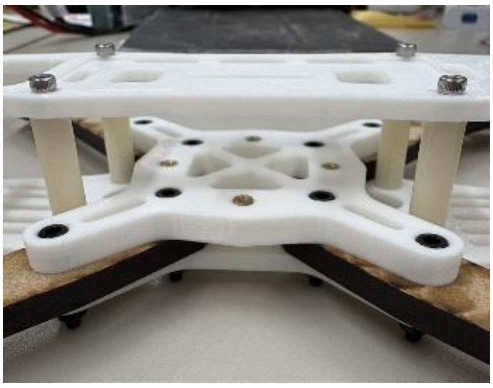


Figure 16: Sandwich connection design between the arm and the fuselage

In addition, our "sandwich" arm installation design (the arm is sandwiched between the upper and lower fuselage panels instead of being directly screwed to one side) allows vibration energy to be converted into heat energy (friction damping) through friction on the contact surface between the arm and the fuselage panel, further reducing the amount of vibration transmitted into the fuselage.

5. Different choices of body materials and arms

   The different material choices for the fuselage frame and arms are a deliberate trade-off:

   Inventor FEA analysis of the fuselage frame shows that under normal flight and reasonable crash conditions, the maximum stress in the fuselage frame is well below the material strength and there is no risk of fracture. Under this condition, "the strongest" is not the selection criterion for the fuselage material, but "the lightest" and "the most flexible design" are. 3D printing PLA outperforms MDF in both dimensions.

Where strength requirements are high, choose the strongest material, and where strength requirements are low, choose materials with better other factors.

1.8 Flight control firmware route conversion: the decision-making process from Betaflight to INAV

1. Symptoms and initial diagnosis of the problem

   After the first test flight of the mounted sub-car system, we found that Z-axis control was the most serious problem: poor altitude stability (it is difficult to maintain a stable altitude with manual throttle), and each landing caused a large impact because the descent speed was difficult to control. This problem directly led to repeated damage to the connectors.

   Preliminary diagnosis: The sub-car increases the moment of inertia of the system, causing the dynamic characteristics of the system to mismatch the original PID parameters. The standard parameter tuning suggestions (increasing the P term for stability, increasing the D term to reduce overshoot) have limited effect on our high-inertia system.

2. In-depth understanding of the process of flight control architecture

   To find the root cause, we spent a day or two digging into Betaflight's control architecture documentation. The key points to understand are:

   Betaflight's Z-axis control is designed to be "auxiliary", not "dominant". Even if the barometer is enabled, Betaflight only uses the barometer reading as a reference signal, and the main control remains with the pilot's throttle input. This is deliberately designed because the pilot of the traversing aircraft does not want the flight control to automatically adjust the throttle to "help" maintain altitude when performing stunts.

   But for our mission, we just need the flight control to "automatically maintain altitude" - we don't have the precision of a shuttle pilot, and we can't use pure manual throttle to accurately control the altitude when approaching the ground.

3. INAV selection basis

   The design goal of INAV (inertial navigation) is autonomous navigation, and NAV ALTHOLD is one of its core functions. In ALTHOLD mode, the barometer is the primary sensor for the Z-axis closed loop, and the pilot's stick input (throttle) corresponds to the target altitude or target climb rate, rather than direct throttle output. This is exactly the control model we need.

   Reasons to choose INAV over ArduPilot: ArduPilot is more comprehensive but more complex to configure, making it too bulky for our task. There is a lot of community information on INAV's configuration on a four-axis multi-rotor, and the difficulty of getting started is reasonable.

   Key settings after INAV switching: Set the hover throttle reference value accurately at 50%, so that the neutral point of the remote control corresponds to the static balance point, so that there is symmetrical control margin in the up and down directions. This setting minimizes the error of the integral term of the height control loop near the equilibrium point, which is an important factor in improving RMSE.

   The cost of switching firmware: All parameter adjustment work completed on Betaflight needs to be redone from scratch on INAV (the parameter interface and logic of the two firmwares are different); INAV's operating logic also needs to be relearned (for example, the control meaning of the joystick in ALTHOLD mode is different from Betaflight). This is one to two days worth of work.

   Benefits of switching firmware: Z-axis RMSE is reduced from 8.66 cm to 1.35 cm, the problem of falling impact is fundamentally improved, and the problem of connector breakage is greatly reduced. This benefit far exceeds the switching costs.

1.9 Systematic Experimentation in Propeller Selection: Comprehensive Comparison of Six Configurations

1. Trigger event for changing paddles

In the early flight test after mounting the sub-car system, obvious low-frequency jitter occurred when using the 5045 three-blade propeller close to the ground (height $H < 1.5$ times the propeller disk radius), and the body would deflect diagonally rearward when landing. After consulting the data, it was confirmed that this is a typical ground effect and unstable phenomenon of the rotor wake at low altitudes - the wake structure of a three-blade propeller at low altitude is more likely to produce asymmetric disturbances than a four-blade propeller.

2. Establishment of thrust test bench
   In order to make a data-supported decision on propeller replacement, we designed and produced a thrust test bench: the test arm was made of the same MDF material as the actual arm, installed with an RS2205 motor, controlled the throttle through SpeedyBee F405 in conjunction with Betaflight's motor test function, and used a precision electronic scale to measure the upward thrust (the test bench design allows the motor to push upward, and the decrease in the scale's reading is the thrust).
   At the same time, with the bidirectional DShot reading real-time RPM data from the flight controller, and the current sensor reading the current, a complete thrust-RPM-current characteristic curve is constructed.
3. Analysis of test results of six propellers
   Each of the six test configurations has its own characteristics (see Section 2 for detailed data):
   5045 Four-Blade: The highest thrust at the same RPM, good hovering efficiency, but strong vibration characteristics and steep slope of the thrust response curve (sensitive to throttle input).
   5040 Four-Blade (Final Choice): Slightly less thrust than the 5045 Four-Blade, but with a more linear thrust response curve, smaller vibration signature, and more predictable behavior at low altitudes.
   5045 three-blade: The thrust is good at high speeds, but the low-altitude vibration problem is the main reason for us to change the propellers, which has been eliminated.
   5040 three-bladed propeller: It also has the problem of low-altitude vibration, and the thrust is not as good as that of a four-bladed propeller with the same

propeller diameter, so it is excluded.

Racing three-leaf: high efficiency at high speeds, but poor hovering efficiency, not suitable for tasks that require long-term hovering, excluded.

5045 two blades: The noise is the smallest, but the thrust is the lowest. It does not meet the standard under our load conditions and is excluded.

4. Select the complete logic of 5040 Yotsuba

   The choice between the 5040 Four Leaf and the 5045 Four Leaf is the final core decision. Purely from the perspective of thrust and efficiency, the 5045 four-blade is slightly better. But we chose 5040 four-leaf, and the complete reasons are as follows:

   The critical operational phase of the mission is the final stage of approach to the platform in area A (height H < 20 cm). At this height, the ground effect is the strongest, and the interference caused by the downwash reflected by the platform is the greatest. The left and right motors receive asymmetric ground effect gains due to different heights. Under this condition, the more sensitive the thrust response is, the higher the magnification factor of the thrust difference between the left and right motors, and the greater the attitude disturbance. The flatter thrust response curve of the 5040 four-blade provides more stable characteristics in this critical low-altitude stage.

   Moreover, we found that the 5040 Four-Blade performs better in the landing strategy of "quickly retracting the throttle and letting the drone sink naturally" - the thrust curve is more linear, making the throttle retraction speed and sinking speed more predictable.

1.10 The design evolution of the ball-retrieving mechanism of the sub-cart

1. The formation of the front fork ball guide concept

   After determining the sub-car system, we need to design a specific ball-retrieving mechanism. During the discussion we ruled out several options:

   Active Gripper (re-implementing the gripper on the trolley): Requires precise alignment, easier on the ground than in the air, but still requires the trolley to be able to stop exactly at the correct position on the ball. In a platform space of less than 20 cm, this accuracy requirement is still very high.

Funnel-type introduction: Design a funnel-shaped opening in front of the car to allow the ball to automatically roll in when the car moves forward. The advantage is that it allows for large lateral deviations (the funnel opening is much wider than the ball), but the disadvantage is that the sub-car needs to be aligned in the general direction of the ball and then pushed forward to let the ball enter the funnel.

Front fork guide ball (final choice): Set two side forks in front of the car to form a "V-shaped" or "plow-shaped" guide ball geometry. When the car is moving forward, even if the ball's lateral position is deviated, the side forks can "guide" the ball to the center position, and then the rear baffle prevents the ball from continuing to roll backwards, and the ball is stored in the cavity between the front fork and the baffle.

The fork design is the most forgiving because alignment errors are passively compensated by the geometry and do not rely on precise positioning.

2. Iteration of fork geometry

   The fork geometry was modified several versions during testing:

   Version 1: The fork angle is small (the V-shape is too narrow), and the ball guide effect is not good. The ball easily slides past the outer edge of the side fork without being introduced.

   Version 2: The fork angle is enlarged, and the length of the front fork is lengthened to increase the guide ball stroke, which improves the effect, but the end of the side fork (outermost) easily hits the edge of the platform when the car turns.

   Version 3 (final version): The fork angle is slightly narrowed based on version 2, and the end of the front fork is bent inward to reduce geometric interference during steering. The ball guiding effect and maneuverability are balanced.

3. Continuously rotating servo driven differential steering

   The movement of the sub-car uses two Continuous Rotation Servos to drive the left and right wheels, and steering is achieved through differential speed (difference in speed between the left and right wheels).

Reasons for choosing a continuously rotating servo instead of a DC motor: The speed of a continuously rotating servo can be directly controlled through a PWM signal (fully compatible with the PWM output of the ESP32) and does not require an additional motor driver chip (H-Bridge); while a DC motor requires a motor driver module, a speed sensor (or estimation), and more complex control logic. In the context of rapid prototyping, the integration complexity of continuous rotation servos is lower.

The control logic of differential steering: left wheel speed = base speed + steering amount, right wheel speed = base speed - steering amount. When the readings of the two joysticks are very different, the steering amount is close to the maximum value, forming an in-situ steering (one round of forward rotation and one round of reverse rotation); when the readings of the two joysticks are the same, the steering amount is zero and the vehicle moves forward in a straight line.

4. Later sub-car improvements: addition of support structures

   A problem was discovered during the ball-pickup test of the first version of the car: when the front fork began to push the ball, the reaction force of the ball made the car tend to lean forward (the center of gravity was already in the forward position, and the resistance of the ball caused the front wheel set to experience an upward moment). In extreme cases, the front end of the sub-car will lift up and lose its ability to move forward.

   Solution: Install a support bracket behind the frame to increase the supporting base area of the sub-car and improve its overturning resistance. Because there are pre-holes in the frame, the bracket can be retrofitted in just 20 minutes without redesigning any existing parts.

1.11 Communication system design decisions: Selection and optimization of ESP-NOW

1. Evaluation of options for sub-vehicle communication solutions

   The sub-vehicle needs to receive control instructions (moving direction, speed, gripper opening and closing) from the operator, while not relying on the drone's flight control system (maintaining control independence). The following communication options were evaluated:

Through the flight control AUX channel: The sub-car control signal is transmitted to the flight control through the AUX channel of the remote controller, and then output from the UART of the flight control to the sub-car MCU. The disadvantage is that the flight control becomes a node in the control link. Once there is a problem with the flight control, the sub-car will also lose control; and the number of AUX channels is limited, and the multiple control axes (front and rear speed, steering, clamping claw) required by the sub-vehicle require multiple channels.

Bluetooth (BLE): Low latency (about 15~30 ms), but Bluetooth is not as reliable as the Wi-Fi series in noisy electromagnetic environments, and the pairing process is inconvenient at the competition site.

Wi-Fi (TCP/UDP): High bandwidth but unstable latency (Wi-Fi latency fluctuates due to routers and competing traffic), and requires the establishment of network infrastructure (AP or router), making the environment at the competition site complex.

ESP-NOW (final choice): A point-to-point low-latency protocol based on Wi-Fi hardware that does not require a router or AP, and devices at both ends communicate directly. Design latency is typically less than 2 ms, suitable for real-time control. The API has complete function library support in the Arduino ecosystem, making it easy to develop.

2. Root cause analysis of communication problems

   ESP-NOW experienced a delay of 1 to 2 seconds during the integration test. For a complete analysis and solution to this problem, see Section 4 (Control Logic and System Reliability).

   The first reaction to the delay problem is "ESP-NOW is too slow, change to Bluetooth or UDP", but the root cause is actually the program architecture. Changing the protocol will not help, but only increases the integration complexity. Do a root cause analysis first, find four specific problems, and then fix them individually. The work hours are much less than changing the agreement, and the results are direct.

1.12 The evolution of connector design: from failure to experimentation to

improvement

1. Failure discovery process

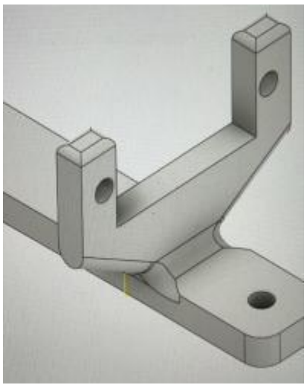


Figure 17. Valve connector design drawing and fracture mode

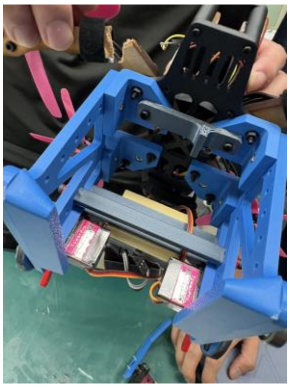


Figure 18. Photo of physical fracture

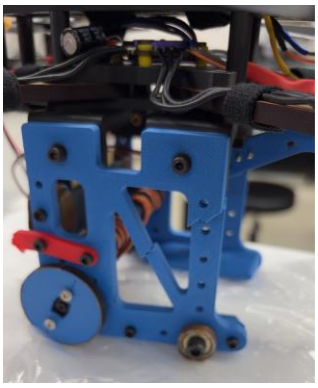


Figure 19. Side view of the sub-wheel set (side impact weak point)

The failure of the sub-car connecting parts was not discovered suddenly, but gradually accumulated: the connecting parts were intact after the first landing test, slight cracks appeared after the third time, and broke after the fifth time. This "cumulative damage" pattern shows that the problem is not just the strength of a single impact, but also fatigue - the downwash airflow continues to vibrate the connectors with lateral forces during flight, and the strength of the connectors has already declined due to fatigue before the landing impact.

We later listed "fatigue resistance" and "impact resistance" as design goals in the design revision, instead of only considering static strength.

2. The need for systematic drop experiments

   After discovering the problem, we made a decision: instead of directly modifying the design, we first conducted systematic experiments to find "where the failure boundary is" and then designed for the boundary.

   The design and results of the drop experiment are detailed in Section 2.

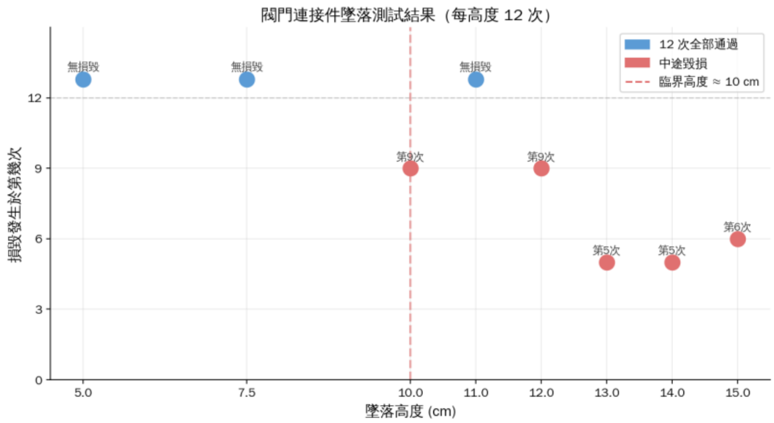


Figure 20. Drop test results of valve connectors

(Height above 10 cm has a high probability of damage, and the critical conditions are clear)

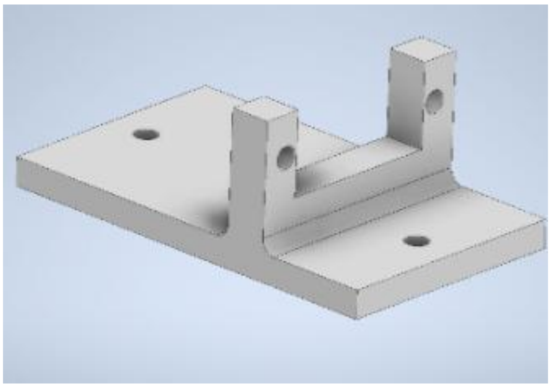


Figure 21. New version of connector CAD

3. The logic of the dual-track improvement strategy

   After confirming the failure boundary, we made improvements in two directions at the same time:

   (1) Firmware speed limit: Limits the maximum descent speed so that the actual landing impact speed is much lower than the failure boundary speed, including a safety factor of 3.0. This measure can be implemented immediately and does

not require any hardware modifications, making it a quick and effective risk mitigation measure.

(2) Geometric design improvement: Redesign the right-angle connection geometry of the connector, widen the contact area, and introduce rounded corners. This improvement addresses the root cause of the stress concentration, allowing the failure boundary itself to be raised rather than just moving the operating conditions further away from the boundary.

The two measures are implemented simultaneously to provide double protection: even if the firmware speed limit fails occasionally (such as an operator error causing a high-speed descent), the stronger connector geometry can provide an additional safety margin.

1.13 Three-ball subsystem design, production and evaluation

1. Design motivation and architectural positioning
   After the mid-term acceptance, we conducted an analysis on the efficiency of ball retrieval. Calculated based on the ball retrieval process of a single-ball machine at that time, B→A flies for about 8.5 seconds, retrieving the ball on the platform is about 10 seconds, and A→B returns for about 8.5 seconds. The total time of a single cycle is about 27 seconds. If you want to get four consecutive perfect scores at the end of the term, each pass must be within 23 seconds, so shortening the number of passes to retrieve the ball is the most direct optimization direction. Under this premise, we began to evaluate the feasibility of "taking multiple balls at once". We set two core constraints when designing the three-ball machine: first, the existing drone body structure should not be changed - the flight control adjustment involved in the airframe design has spent a lot of time to verify, and the risk and cost of re-modifying the airframe are high; second, the new subsystem must be modularly installed at the bottom of the existing frame, retaining the flexibility to replace it with a single-ball machine in the future. Therefore, for the three-ball machine, we focused all our design efforts on the sub-car mechanism, and the flight end part remained unchanged.
2. Center of gravity offset evaluation and installation position offset calculation
   With the three cabins arranged in this shape, the center of gravity is not in the middle. There must be an offset moment when hovering. Therefore, before

drawing the CAD, we first roughly estimated how far the center of gravity would be.

The three ball-retrieval cabins are all simplified into rectangular entities. Cabin 1 and Cabin 2 are the two ball-retrieval cabins in the front, with design dimensions of 54 mm × 76 mm each; Cabin 3 is the rear tail cabin, which needs to accommodate the ESP32-C3 control board at the same time, so its size is 76 mm × 76 mm. Substitute the area of each cabin into the moment balance equation to find the distance x that the drone's locking center point needs to shift forward: $(54 - x)(76 + 76) = x(76 + 76) + (76)(76)$. The solution is $x = 8$ mm.

In other words, if the geometric configuration of the three cabins is maintained, the frame locking point of the drone needs to be moved 8 mm forward relative to the geometric center of the sub-vehicle in order to allow the center of gravity of the entire aircraft to fall back below the thrust center of the propeller. This result directly determines the position of the mounting holes on the sub-roof plate, which has been adjusted accordingly in the early stages of CAD design to avoid continuous pitch bias moments during subsequent flights.

3. Line design optimization

   There are three independent cabins inside the three-ball machine, five MG90S servos, an ESP32-C3 control board and corresponding power wiring. All lines need to be gathered in the extremely limited interior cavity of the car. If the wires are routed one by one according to the original single-ball machine method, repeated disassembly and insertion during assembly and maintenance will easily cause poor contact, and also increase the risk of wires falling off due to vibration during flight.

   Our solution is to connect all signal lines and power lines into 9×2 rows of DuPont pin headers, so that the internal wiring harness can be plugged and unplugged in batches, taking into account compactness and maintainability. This design refers to the lessons learned from past car tests where the servo failed due to a single wire loosening. Although using a pin header requires more welding processes, it shortens the disassembly and assembly time from groping one by one to completing it in one action.

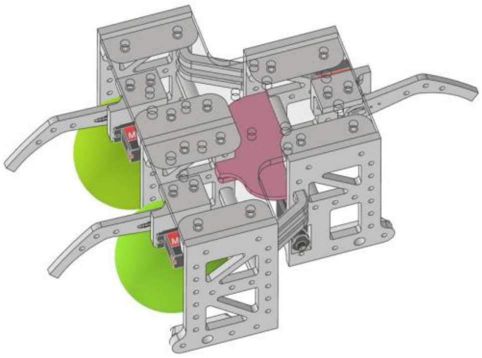


Figure 22. CAD diagram of the internal circuit of the three-ball machine

(The pink mark is the installation position of the control board, and the wiring routing of the five servos can be seen).

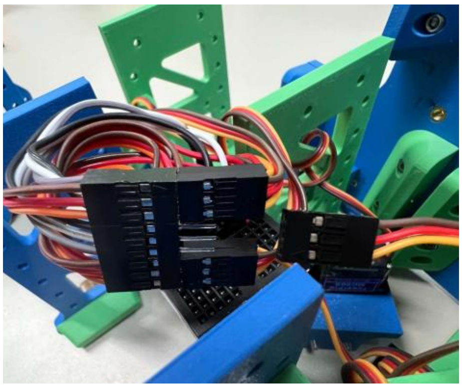


Figure 23: Actual photo of 9×2 DuPont pin header connector

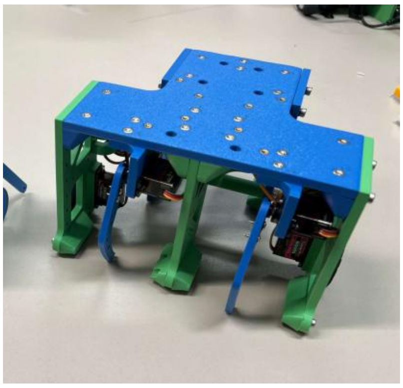


Figure 24: Front view of the three-ball machine

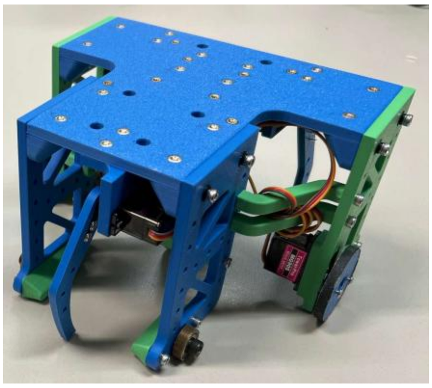


Figure 25: The physical back of the three-ball machine

4. Static analysis comparison: battery life

The key specifications differences between the three-ball machine and the single-ball machine are as follows:

Table 1. Specifications of three-ball machine and single-ball machine

Project	Single ball machine	three ball machine
Total unladen weight (g)	747	948
Total ball weight (g)	805	1121
Servo motor quantity	3	5

The increase in weight directly affects the hovering current and battery life. We use a 2700 mAh, 4S, 25C battery for estimation, and include the following conservative assumptions: the available power is 75% of the total power, the current required for flight is multiplied by a safety factor of 1.15, it is assumed that all servos are running at the same time, and the calculation is based on a lower power conversion efficiency.

Three-ball version (single cycle 60 seconds): 30 seconds of flight (10.52 A without ball, 12.44 A with full load), 30 seconds of stopping to retrieve the ball (a total of 2.5 A for five servos). The system has an average current of 7.15 A and a maximum battery life of about 16.99 minutes. It can complete 16 mission cycles and is expected to transport up to 48 balls.

Single-ball version (single cycle 62 seconds): 44 seconds of flight (8.29 A without ball, 8.94 A with full load), 18 seconds of shutdown (a total of 1.5 A of three servos). The system has an average current of 7.37 A and a maximum battery life of about 16.48 minutes. It can complete 15 mission cycles and is expected to transport up to 45 balls.

In terms of battery life alone, the three-ball machine has a higher proportion of shutdowns (50% vs. 29%) and fewer flights. The average current of the two is similar. In theory, the three-ball machine can deliver 3 more balls. The advantage is not huge, but it is there.

5. Static analysis comparison: mechanism bottom area and landing difficulty

Box A has a usable platform area of 2500 cm^2. The bottom area of the single-ball machine is about 88.39 cm^2, accounting for 3.54% of the platform area; the bottom area of the three-ball machine is about 214.61 cm^2, accounting for 8.58%, which is nearly 2.4 times that of the single-ball machine. The larger the bottom area, the smaller the lateral drift that can be tolerated during landing, and the higher the accuracy requirements for the pilot. This analysis was confirmed in the dynamic test of the three-ball machine - when the platform landed, the stability of the three-ball machine was significantly lower than that of the single-ball machine. Especially after the drone carried a fully loaded three-ball machine, the inertia increased, and the hovering offset was more difficult to control.

6. Dynamic ball retrieval comparison: three main questions

   After actually using a drone to perform a ball-retrieval test on a simulated venue, there were three main problems:

   (1) The large bottom area makes it difficult to move in box A: the three-ball machine has a long body, and the space for rotation in the box is limited. Especially when the ball is blown to the corner by the downwash airflow, the single-ball machine can enter at an angle and be aligned and clamped; the three-ball machine has insufficient operating space at the same oblique angle due to the limitation of the length of the car body, and must spend more time re-aligning, which lengthens the ground operation time for each ball retrieval.

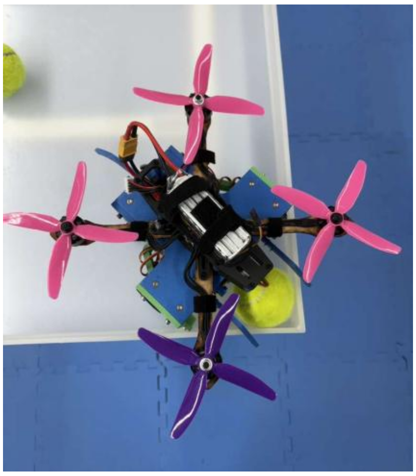


Figure 26: Top view of the three-ball machine taking the corner ball in box A

   (2) The ball inside the car will affect subsequent control: when the ball that has been taken in rolls and contacts the floor or the wall of the mechanism, the friction will make the car body unstable when going straight and turning. This

problem almost does not exist on the single-ball machine, but it occurs repeatedly when the three-ball machine takes the second and third balls, becoming one of the main reasons for the large fluctuations in ground operation time.

Figure 27: Side view of the three-ball machine that has picked up the ball

(3) The operation of the two-way car is not intuitive: the three-ball machine is designed with two valves facing forward and one valve facing backward. When the rear valve needs to be used to get the ball, all control directions are opposite. The operator must switch the psychological sense of direction in real time during execution. This is prone to errors under pressure and further increases the ground time fluctuation of dynamic testing.

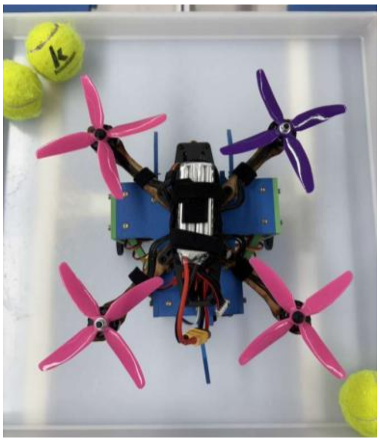


Figure 28: Top view of the three-ball machine in box A

7. Actual performance data comparison

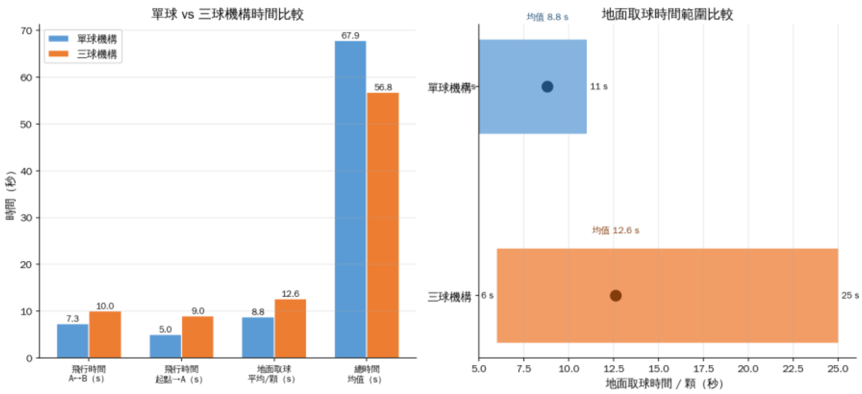


Figure 29. Single-ball vs. three-ball mechanism time comparison (left) and ground ball retrieval time distribution (right)

The average results of three valid tests are as follows:

Table 2. Single-ball vs. three-ball agency time allocation table

	single ball mechanism	**three ball agency**
Total time (average of 3 times, s)	67.9	56.8
Flight time A↔B(s)	7.3	10.0
Flight time starting point→A(s)	5.0	9.0
Average ground work time/ball (s)	8.8	12.6
Ground time range (s)	5–11	6–25

The three-ball machine was only about 16% faster overall, well below expectations of over 50%. The most obvious problem is the fluctuation in ground operation time - the single-ball machine fluctuates between 5-11 seconds per ball retrieval, and the three-ball machine pulls from 6 seconds to 25 seconds, and the standard deviation increases significantly. The instability of the ground time reflects the above three operational problems: the ball scatters to the corner, the ball that has entered the cabin interferes with the progress, and the judgment of the backward ball is incorrect. The median per game was stretched significantly higher on several 25-second pickups, negating the time advantage of fewer flights.

Another point is that the flight stability of the three-ball machine when landing on the platform is significantly lower than that of the single-ball machine. There were observable attitude oscillations in many tests, but the system still maintained an operational state. We currently speculate that the reason is related to the increase in the weight of the three-ball machine, the change in the inertia tensor, and the more asymmetric downwash flow field due to the increase in the bottom area. However, a systematic analysis has not yet been completed and is expected to be supplemented in the final report.

8. Summary: Adopting strategies at the end of the period
   Based on the results of static analysis and dynamic testing, the advantages of the three-ball machine in terms of theoretical battery life and single-mission efficiency are greatly reduced in actual operation by the limitations of the bottom area, ball interference movement, and two-way operation problems. The 16% time advantage corresponds to higher operational uncertainty and flight stability costs. Under the rule that the final score is calculated as the median, a single mistake has a greater impact on the final score.
   Therefore, we decided to use the single-ball machine as the main mechanism at the end of the period, while retaining the three-ball machine as a backup option. If you successfully get full marks in the first few times of the final test, it will depend on the on-site conditions to decide whether to switch to the three-ball machine and try to shorten the total time. In the tense and chaotic environment of a game, the repeatability of getting perfect scores is more important than a strategy that is theoretically faster but prone to mistakes.

1.14 Final preparation

1. Establishment of operational SOPs
   After deciding on the plan, we used the last few exercises to establish a standard operating procedure (SOP).
   The specific contents of the SOP: pre-takeoff system check list (power confirmation, propeller fixation confirmation, sub-car communication confirmation, flight control GPS/barometer status confirmation), standard take-off procedures, path and speed to area A, standard landing operations (throttle retraction timing, attitude adjustment), sub-car ball-retrieval operation sequence (direction of approach to the ball, speed control, judgment criteria to confirm that the ball has been received), return path to area B and landing, and standard recovery procedures after each step fails.

2. backup drone

A week before the exam, we built a backup drone with the exact same design.

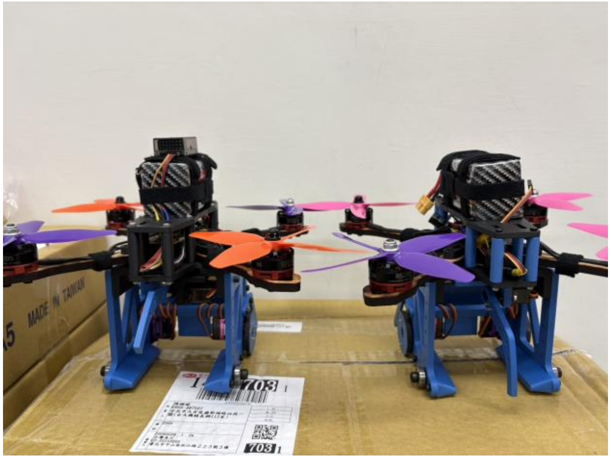


Figure 30: Side-by-side photo of the main unit and backup drone

Risk assessment: Possible failure modes in the competition include: operational errors (can be recovered within the remaining number of attempts), equipment damage at unexpected moments (cannot be recovered during the competition, and the competition cannot continue after the equipment is damaged). Of the two failure modes, the latter is more costly (the entire game ends, irremediable), and the probability of occurrence is not low (we crashed and damaged the drone more than 10 times throughout the development process).

If there is a backup machine, even if the first machine is damaged during the game, the second machine can be played immediately without any conservative operation. The drone was crashed more than 10 times during the entire development process, and the backup was just to take this reality into consideration.

The production strategy of the backup machine: use the exact same design and parameters as the first machine to ensure that the operating feel is consistent and there is no need to readjust. All parameter settings completed on the first station are directly copied to the second station through the backup file of Betaflight/INAV.

3. End-of-period strategy decision tree

   We created a complete game strategy decision tree, covering the optimal subsequent strategy under each "previous attempt score combination":

第一次	第二次	第三次	第四次
0	0	重考，一定要至少一次100	
	40 ~ 90	重考，一定要一次100、一次0	
	100	0	重考
		40 ~ 90	故意拿0重考
		100	只能拿100或0，第四次發現情況不對要硬拿0分
40 ~ 90	0	重考，一定要一次100、一次0	
	絕對不能拿兩次40 ~ 90，因為如果後面要拿0+0才能重考，但這樣八次中位數拿不到100，所以第二次發現情況不對要硬拿0分		
	100	0	故意拿0重考
		只能拿100+100或0+0，第三次發現情況不對要硬拿0分	
		100	只能拿100
100	0	0	重考
		40 ~ 90	故意拿0重考
		100	只能拿100或0，第四次發現情況不對要硬拿0分
	40 ~ 90	0	故意拿0重考
		只能拿100+100或0+0，第三次發現情況不對要硬拿0分	
		100	只能拿100
	100	0	只能拿100或0，第四次發現情況不對要硬拿0分
		40 ~ 90	只能拿100
		100	

Figure 31: End-of-period strategy tree

0 points → 100 points for the second attempt → (if successful) subsequent conservative confirmation; if both the first and second attempts are 0 points → trigger the retake mechanism

40-90 points for the first time → 100 points for the second try (there is still time within the time limit)

100 points for the first time → try the same strategy subsequently (execute

steadily, don't take risks and try faster)

The fourth attempt at any time → the optimal strategy for the remaining possible scores at the moment

This table allows you to think clearly about strategic issues in the game in advance, without having to make decisions while calculating in a tense situation on the spot.

1.15 Overview of design evolution: decision-making process over 16 versions

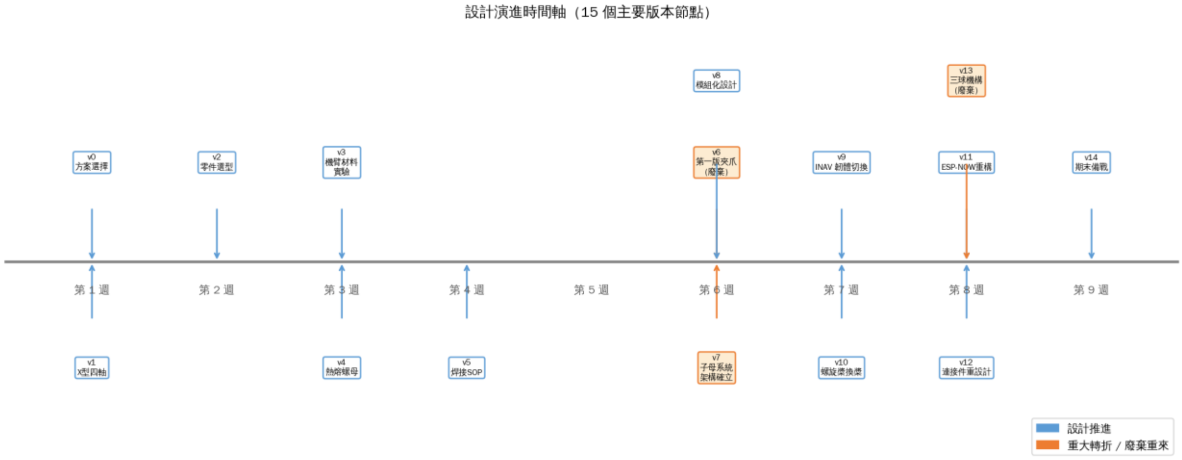


Figure 32: Design evolution time (orange marks major turning points and nodes where abandonment is repeated)

Table 3. Design evolution version and details

version	decision making	Problem trigger	Trade-off	quantifiable results
v0	Establish a five-dimensional assessment framework	Five options need to be compared objectively	Rigor vs. decision speed	Consistent baseline for all subsequent decisions
v1	Select a drone solution	Prioritize fault tolerance and recoverability	The highest difficulty for the highest flexibility	Overall design direction established
v2	Adopt X-type four-axis mature architecture	Resource allocation priority	No innovation for zero uncertainty	A large amount of debugging data can be borrowed directly
v3	Part selection (F405/RS2205/4S-2700mAh)	Thrust-to-weight ratio, current safety, firmware compatibility	Multi-objective joint optimization	The thrust-to-weight ratio is 3.5:1, and the current safety margin is guaranteed.
v4	Production and testing of the first version of the gripper	Verify the feasibility of taking the ball directly	Architectural simplicity vs. operational reliability	Triple failure acknowledgment, ESP32 control link verification
v5	Architecture conversion: child-mother system	There are systematic and unsolvable problems with directly clamping the ball	Weight +350g vs. Fundamental improvement in ball picking stability	Ball recovery improved from close to 0 to acceptable
v6	Comprehensive promotion of modular design	Expect modifications after extensive testing	Design complexity vs. iteration speed	The arms were replaced 20+ times throughout development without a single downtime
v7	MDF arm (experiment-driven material selection)	Vibration versus intensity trade-off	Weight +4g vs. Rigidity +50% + Natural frequency	Gyroscope signal quality significantly

			+27%	improved
v8	Hot melt nuts are fully adopted	3D printed parts are repeatedly disassembled and assembled with sliding teeth	Weight +1g/piece vs. unlimited disassembly and assembly times	Maintenance time reduced from minutes to seconds
v9	Welding quality standardization	Solder joint fell off during test flight	Production time vs. flight reliability	There were no incidents of solder joints falling off during subsequent test flights.
v10	Firmware Betaflight → INAV	Z-axis RMSE 8.66 cm Unacceptable	Firmware learning cost vs. fundamental improvement in control quality	RMSE reduced to 1.35 cm (84% improvement)
v11	Replace the propeller with 5040 four blades (6 types of actual measurements)	Low altitude ground effect jitter	Thrust peak vs. low altitude stability	Landing success rate significantly improved
v12	ESP-NOW communication architecture reconstruction	Control delay 1～2 s	Program refactoring time vs. control immediacy	Delayed to near instantaneous, packet volume -80%
v13	Connector design + INAV speed limit	Falling impact fracture problem	Descent speed vs. structural safety (double rail protection)	No breakage in follow-up tests
v14	Three-ball mechanism production and actual measurement	Reduce total task time	Efficiency +16% vs. reduced control complexity and stability	The total time is 67.9s → 56.8s, but the qualitative problem is serious
v15	Return to single ball + SOP + backup machine + strategy table	End-of-term stability optimization	Speed potential vs. reliable execution	Final exam 100 points

1.16 review

1. Problem framing is more important than technical solutions
   The key to switching from direct ball clamping to a mother-child system is not to change to a better technical solution, but to change the way to ask questions. There is no good answer to "How to make the gripper more accurate in the air"; "Under what conditions is it easiest to retrieve the ball?" The answer is on the ground, without downwash, and you can try again. When the problem changes, the solution naturally changes.

2. The role of numbers in decision-making
   Material selection has a force-displacement curve, firmware switching has RMSE data, propeller selection has a complete curve of the thrust test bench, and the abandonment of the three-ball mechanism has a step-by-step time record. Every important decision has a corresponding quantitative basis.

3. Failed to provide information
   The first version of the gripper and the three-ball mechanism were "failed attempts," but they both provided valuable information. The first version of the gripper identifies three specific reasons for failure of direct ball clamping, providing a basis for decision-making in the parent-child system. The test of the three-ball organization provides a step-by-step time comparison of the two solutions, so that the decision to return to a single ball is not just "it feels like three balls are too troublesome".
   The first version of the gripper and the three-ball mechanism took time, but the specific failure data they gave gave the two subsequent decisions (mother-child system, return to single ball) a basis, not just intuition.

4. Abandonment of the three-ball mechanism
   The three-ball mechanism is the direction in this topic that has the most investment but is not used in the end. The design was completed, the car was printed, and three rounds of actual tests were carried out. The final data showed

that it was not cost-effective: the time was only shortened by 16%, but the control difficulty and variability were significantly worse, and the remaining practice time was not enough to master the three-ball mechanism. We decided to go back to a one-ball institution.

5. The importance of integration testing

   This topic received 100 points at the end of the period, not because one subsystem did a particularly good job, but because all subsystems worked together at the system level. If the best motor is paired with inappropriate flight control settings, the system will be unstable; if the best flight control settings are paired with easily broken connectors, the mission will still fail; if the best mechanical system is paired with communication that is delayed by 2 seconds, the car will not be easy to use.

   After each subsystem passed independent testing, previously unexpected problems still emerged during integration: ESP-NOW delay was triggered only during the integration test, and fatigue damage of the connectors only appeared during the complete test with flight downwash. If these problems are not discovered until the end, there will be no time to fix them. Therefore, we have developed a habit: start integration testing after each subsystem has a preliminary available version, and do not wait until each one is "completed" before integrating them together.

Part 2, Engineering Analysis, Verification and Performance Evaluation

2.1 Basic parameter definition

Table 4. Definition of basic system parameters and geometric dimensions

physical quantity	symbol	numerical value
Drone body quality	m_d	0.6847 kg
Tennis quality	m_b	0.057 kg
Total mass after catching the ball	$m^+ = m_d + m_b$	0.7417 kg
Five-inch paddle diameter	D_p	$5 \text{ in} = 0.127\ m$
Five-inch propeller radius	R_p	$0.0635\ m$
Center distance between adjacent motors	s	$0.169998\ m$
roll/pitch effective force arm	$b = s/2$	$0.084999\ m$
Oblique distance from center to motor	$l = s/\sqrt{2}$	$0.12021\ m$
Lower gripper body height	H_c	$0.095547\ m$
Bottom clamp body width	B_c	$0.096276\ m$
Lower gripper body length	L_c	$0.076933\ m$
height of upper structure	H_u	$0.038\ m$
total height	H_{total}	$0.133859\ m$
Single five-inch paddle disc area	$A_r = \pi R_p^2$	$0.01267\ m^2$
Total propeller disc area of four propellers	$A_{\Sigma} = 4A_r$	$0.05067\ m^2$

gap between adjacent propeller tips	$g_p = s - D_p$	$0.042998\ m$ ≈ 43.0 mm

2.2 Downwash tail watershed discussion

Therefore, the five-inch paddles will not interfere with each other. However, the four rotor wakes will gradually spread and influence each other as they develop downwards. Use wake spreading model:, where is the wake spreading rate, when two adjacent downwash flows come into contact$g_p > 0 R_w(z) = R_p + kz k 2R_w(z) = 2(R_p + kz) = s \Rightarrow z_{\text{merge}} = \frac{s-2R_p}{2k}$

$s - 2R_p = 0.042998\ m$, if:, if:, if:, and the height of the lower gripper body is about:, so near the height of the gripper body, the four rotor wakes are close to interfering with each other, but may not have completely merged.$k = 0.10 z_{\text{merge}} = 0.215\ mk = 0.15 z_{\text{merge}} = 0.143\ mk = 0.20 z_{\text{merge}} = 0.107\ mH_c = 0.0955\ m$

Therefore, the downwash near the gripper is not uniform, but a complex situation including "four local high-speed downwash + central low-speed area + vehicle body obstruction and platform rebound airflow + vortex and turbulence"

2.3 Simplified model of the system

1. UAV = X-shaped quadcopter frame + central structure + rectangular parallelepiped gripper body below

    X-shaped frame: Four motors are located at the four corners of the X-shaped frame, responsible for generating thrust and attitude torque.

    Center stack structure: including flight control, battery, pillars, upper plate, lower plate, wires, etc., regarded as concentrated mass close to the center of the rack

    The rectangular parallelepiped gripper body below: includes gripper, body, steering gear, and bracket. It will cause the center of gravity to shift downward, increase the moment of inertia, block the downwash, and will be subject to the reaction force of the ball when picking up the ball.

2. Coordinate system definition

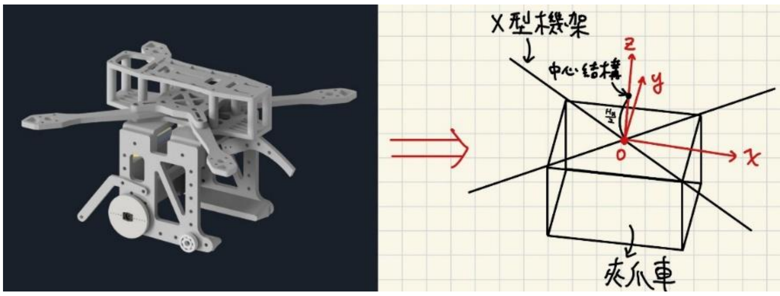


Figure 33. System coordinate definition

x(front of the body), (right side of the body), (upward), origin (center of the$yzOGOxzyzr_1 = \begin{bmatrix} +b \\ +b \\ 0 \end{bmatrix} r_2 = \begin{bmatrix} -b \\ +b \\ 0 \end{bmatrix} r_3 = \begin{bmatrix} -b \\ -b \\ 0 \end{bmatrix} r_4 = \begin{bmatrix} +b \\ -b \\ 0 \end{bmatrix}$ 、$b = \frac{s}{2} =$ $0.084999\ m + z\boldsymbol{F}_i = \begin{bmatrix} 0 \\ 0 \\ T_i \end{bmatrix} i\boldsymbol{M}_i = \boldsymbol{r}_i \times \boldsymbol{F}_i = \begin{bmatrix} y_i T_i \\ -x_i T_i \\ 0 \end{bmatrix}$

That is to say, the roll moment depends on, and the pitch moment depends on$y_i T_i - x_i T_i$

2.4 focus analysis

1. Structural Mass Decomposition

Table 5. Mass breakdown of main UAV structures

structure	quality symbol	numerical value	meaning
X-frame with four motors	m_X	0.0400 kg	Mainly determines the thrust action point and yaw inertia
central structure	m_u	0.3495 kg	Contains upper panels, pillars, flight controls, batteries, and wire concentration areas
Lower rectangular parallelepiped gripper body	m_c	0.2952 kg	Move the center of gravity downward and increase roll/pitch inertia

$\Rightarrow m_d = m_X + m_u + m_c = 0.6847$ kg

2. The center of mass position of the main structure (the total height of the upper structure plus the battery, so the center of mass position is)$2H_u H_u$

$$\boldsymbol{r}_X = \begin{bmatrix} 0 \\ 0 \\ z_X \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} 、 \mathbf{r}_u = \begin{bmatrix} x_u \\ y_u \\ z_u \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0 \\ +H_u \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ +0.038 \end{bmatrix} 、 \boldsymbol{r}_C = \begin{bmatrix} x_C \\ y_C \\ z_C \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0 \\ \frac{-H_C}{2} \end{bmatrix} =$$

$$\begin{bmatrix} 0 \\ 0 \\ -0.04777 \end{bmatrix}$$

The center of mass of the entire drone is:, but the horizontal configuration is approximately symmetrical:$\boldsymbol{r}_G = \frac{m_X \boldsymbol{r}_X + m_u \boldsymbol{r}_u + m_C \boldsymbol{r}_C}{m_X + m_u + m_C} x_G \approx 0 \text{ , } y_G \approx 0,$

and:$z_G = \frac{m_X z_X + m_u z_u + m_C z_C}{m_X + m_u + m_C} = \frac{(0.0400)(0) + (0.3495)(0.0380) + (0.2952)(-0.04777)}{0.6847}$

$$\boxed{z_G = -0.00120 \ m = -1.20 \text{ mm}}$$

3. Changes in center of gravity after catching a tennis ball

Tennis ball quality:, Drone body:, After catching the ball:, New center of gravity after catching the ball:$m_b = 0.0570 \text{ kg} m_d = 0.6847 \text{ kg} m^+ = m_d + m_b = 0.7417 \text{ kg} \boldsymbol{r}_G^+ = \frac{m_d \boldsymbol{r}_G + m_b \boldsymbol{r}_b}{m_d + m_b}$

The tennis ball is clamped under the body, take:, then:$z_b = -0.110 \ m$

$$z_G^+ = \frac{0.6847(-0.00120) + 0.0570(-0.110)}{0.7417} \approx -0.00956 \ m = -9.56 \text{ mm}$$

That is to say, after catching the tennis ball, the center of gravity will move from approximately below the plane of the X-shaped rack to a position approximately approximately below the plane of the X-shaped rack. The displacement of the center of gravity is:$1.20 \text{ mm} 9.56 \text{ mm} \Delta z_G = z_G^+ - z_G \approx -8.36 \text{ mm}$

※If the ball is not directly below, but to the front, for example:, the horizontal center of gravity will shift after catching the ball:, this will cause a pitch trim moment:$x_b = 0.04 \ m x_G^+ = \frac{m_b x_b}{m_d + m_b} = \frac{0.0570(0.04)}{0.7417} = 3.07 \text{ mm} M_y = m^+ g x_G^+ = 0.7417(9.81)(0.00307) = 0.0224 \ N \cdot m$

Corresponding motor differential thrust: , so if the ball is off-center after catching the ball, even if it is only a few centimeters, the motor will need to compensate for the thrust difference of several grams. This is why the clamping jaw should be designed so that the ball is fixed as close to the centerline of the machine body as possible.$\Delta T = \frac{M_y}{4b} = \frac{0.0224}{4(0.084999)} = 0.0658\ N \approx 6.71$ gf

The center of gravity height and mass distribution mainly cause the following effects:

(1) Increasing , makes the roll/pitch angular acceleration smaller.

(2) When the car body below is subject to lateral airflow, a disturbance moment will be generated due to the vertical distance between the point of action and the center of mass.

2.5 3D model analysis

1. Moment of inertia tensor: The inertia tensor relative to the center of mass of the whole machine and selecting three main axes (their eigenvector directions) as the rotation axis is defined as: (principal axis inertia product = 0)$GI\boldsymbol{I}_G = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix} \approx \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$

2. Sum of the inertia tensors of the three main structures

   As mentioned above, it is divided into three main rigid bodies:, where:$k \in \{X, u, c\} G\boldsymbol{I}_G = \boldsymbol{I}_{X,G} + \boldsymbol{I}_{u,G} + \boldsymbol{I}_{c,G} k\boldsymbol{I}_{k,G} = \boldsymbol{R}_k \boldsymbol{I}_k^{(C_k)} \boldsymbol{R}_k^T + m_k(\|\boldsymbol{d}_k\|^2 \boldsymbol{I}_3 - \boldsymbol{d}_k \boldsymbol{d}_k^T) \boldsymbol{d}_k = \boldsymbol{r}_k - \boldsymbol{r}_G.$

Table 6. Symbols and physical meanings of the three-dimensional parallel axis theorem

symbol	meaning
$\boldsymbol{I}_{k,G}$	The inertia tensor of the th component relative to the center of mass of the entire machinekG
$\boldsymbol{I}_k^{(C_k)}$	The inertia tensor of the th component relative to its center of mass and axis of rotationkC_k

$\boldsymbol{R}_k$	Rotation matrix from the local main axis of the th component to the body coordinate axis$k$
m_k	Part quality k
$\boldsymbol{d}_k$	Displacement vector from the center of mass of the component to the center of mass of the complete machine k
$\boldsymbol{I}_3$	3×3Identity matrix

Since the main axes of the three main structures are parallel to the coordinate axis of the body, then:, then:$\boldsymbol{R}_k \approx$

$\boldsymbol{I}_3 \boxed{\boldsymbol{I}_G = \sum_{k=X,u,c}\left[\boldsymbol{I}_k^{(C_k)} + m_k(\|\boldsymbol{r}_k - \boldsymbol{r}_G\|^2\boldsymbol{I}_3 - (\boldsymbol{r}_k - \boldsymbol{r}_G)(\boldsymbol{r}_k - \boldsymbol{r}_G)^T)\right]}$

3. Inertia tensor of X-shaped frame and motor

(1) Model A: Approximation of two crossed thin rods

If the total mass of the upper X-shaped frame is, the distance from the center to the motor is:, and the X-shaped frame is approximated as two cross thin rods, then:,$m_X l = 0.12021\ m I_{xx,X} \approx I_{yy,X} \approx \frac{1}{6} m_X l^2 I_{zz,X} \approx$

$\frac{1}{3} m_X l^2$

This is an underestimation because it assumes that the mass is evenly distributed along the arm and does not notice that the motor, blades and motor mount are concentrated at the end.

(2) Model B: Four-end point lumped mass

Since the cantilever itself is very light and does not affect the moment of inertia, in an actual four-axis system, the mass of the motor, blades, screws, etc. is mostly concentrated at the four end points. Therefore, a more reasonable model is the four-end point lumped mass.

Let the total mass of the X-shaped frame and motor be, the four endpoints are evenly distributed:, the positions of the four motors are:, if horizontally symmetrical and, then the vertical distance relative to the center of mass of the whole machine is. Therefore:$m_X m_R =$

$\frac{m_X}{4} (\pm b, \pm b, 0) x_G \approx 0, y_G \approx 0 z_X - z_G \approx -z_G I_{xx,X}^{(G)} = \sum_{i=1}^{4} m_R\ (y_i^2 +$

$(z_X - z_G)^2) = 4\left(\frac{m_X}{4}\right)(b^2 + z_G^2) = m_X(b^2 + z_G^2)$

Same reason:$I_{yy,X}^{(G)} = m_X(b^2 + z_G^2)$

And the yaw direction:$I_{zz,X}^{(G)} = \sum_{i=1}^{4} m_R\,(x_i^2 + y_i^2) = 4\left(\frac{m_X}{4}\right)(b^2 + b^2) =$

$2m_X b^2$

If the distribution of the four endpoints is completely symmetrical, then:$I_{xy,X} = I_{xz,X} = I_{yz,X} = 0.$

Therefore, under the four-end lumped mass model:

$$\boxed{\boldsymbol{I}_{X,G} = \begin{bmatrix} m_X(b^2 + z_G^2) & 0 & 0 \\ 0 & m_X(b^2 + z_G^2) & 0 \\ 0 & 0 & 2m_X b^2 \end{bmatrix}}$$

4. Inertia tensor of center stack structure

The center stack structure includes flight control, upper panels, pillars, batteries, wire concentration areas and other parts near the center. It is mainly located near the center of the X-shaped frame, and its plane position is approximately:$x_u \approx 0, y_u \approx 0.$

But it has a height along the direction, and its center of mass can be approximated as: Here, the center stack structure and the battery are regarded as an equivalent upper module, and it is assumed that its equivalent total height is approximately; therefore, its equivalent center of mass is located at half of the total height, that is.$zz_u \approx +H_u = +0.0380\ m2H_u z_u = H_u$

If the central stacking structure is approximated as a rectangular parallelepiped, and its equivalent size is , and its mass is , then the moment of inertia at its own center of mass is:$L_u, B_u, H_u m_u I_{xx,u}^{(C)} = \frac{1}{12} m_u (B_u^2 +$

$H_{u,\text{eff}}^2), I_{yy,u}^{(C)} = \frac{1}{12} m_u (L_u^2 + H_{u,\text{eff}}^2), I_{zz,u}^{(C)} = \frac{1}{12} m_u (L_u^2 + B_u^2)$

Relative center of gravity of the whole machine:$\boldsymbol{d}_u = \boldsymbol{r}_u - \boldsymbol{r}_G \approx \begin{bmatrix} 0 \\ 0 \\ z_u - z_G \end{bmatrix}.$

Therefore the parallel axis correction is:$m_u(\|\boldsymbol{d}_u\|^2 \boldsymbol{I}_3 - \boldsymbol{d}_u \boldsymbol{d}_u^T) =$

$$\begin{bmatrix} m_u(z_u - z_G)^2 & 0 & 0 \\ 0 & m_u(z_u - z_G)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore, the moment of inertia of the center stack structure relative to the center of mass of the entire machine is:G

$I_{xx,u}^{(G)} = \frac{1}{12} m_u (B_u^2 + H_{u,\mathrm{eff}}^2) + m_u (z_u - z_G)^2, I_{yy,u}^{(G)} = \frac{1}{12} m_u (L_u^2 + H_{u,\mathrm{eff}}^2) + m_u (z_u - z_G)^2, I_{zz,u}^{(G)} = \frac{1}{12} m_u (L_u^2 + B_u^2)$

5. The inertia tensor of the lower cuboid gripper body

Dimensions:, the mass of the car body is, its own center of mass position is approximately:, its moment of inertia at its own center of mass is:$L_c = 0.076933\ m, B_c = 0.096276\ \text{m}, H_c = 0.095547\ m m_c \boldsymbol{r}_c \approx \begin{bmatrix} 0 \\ 0 \\ -H_c/2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -0.04777 \end{bmatrix} m$

$I_{xx,c}^{(C)} = \frac{1}{12} m_c (B_c^2 + H_c^2), I_{yy,c}^{(C)} = \frac{1}{12} m_c (L_c^2 + H_c^2), I_{zz,c}^{(C)} = \frac{1}{12} m_c (L_c^2 + B_c^2),$

relative to the center of gravity of the whole machine: , therefore:$\boldsymbol{d}_c = \boldsymbol{r}_c - \boldsymbol{r}_G \approx \begin{bmatrix} 0 \\ 0 \\ z_c - z_G \end{bmatrix}$

$I_{xx,c}^{(G)} = \frac{1}{12} m_c (B_c^2 + H_c^2) + m_c (z_c - z_G)^2, I_{yy,c}^{(G)} = \frac{1}{12} m_c (L_c^2 + H_c^2) + m_c (z_c - z_G)^2, I_{zz,c}^{(G)} = \frac{1}{12} m_c (L_c^2 + B_c^2)$, which shows that the lower car body mainly increases the moment of inertia in the roll and pitch directions, that is, and , and the influence on mainly comes from its horizontal size.$I_{xx} I_{yy} I_{zz}$

6. Total moment of inertia formula

$I_{xx} = I_{xx,X}^{(G)} + I_{xx,u}^{(G)} + I_{xx,c}^{(G)}$, that is:

$I_{xx} = m_X(b^2 + z_G^2) + \left[\frac{1}{12}m_u(B_u^2 + H_{u,\mathrm{eff}}^2) + m_u(z_u - z_G)^2\right] +$

$\left[\frac{1}{12}m_c(B_c^2 + H_c^2) + m_c(z_c - z_G)^2\right]$

$I_{yy} = m_X(b^2 + z_G^2) + \left[\frac{1}{12}m_u(L_u^2 + H_{u,\mathrm{eff}}^2) + m_u(z_u - z_G)^2\right] +$

$\left[\frac{1}{12}m_c(L_c^2 + H_c^2) + m_c(z_c - z_G)^2\right]$

$I_{zz} = 2m_X b^2 + \frac{1}{12}m_u(L_u^2 + B_u^2) + \frac{1}{12}m_c(L_c^2 + B_c^2)$(Remaining inertia product = 0)

7. Inertia tensor after grabbing a tennis ball

The total mass after catching the ball is:. If the new center of gravity is regarded as a point mass, its inertia tensor relative to the new center of gravity is$m^+ = m_X + m_u + m_c + m_b\boldsymbol{r}_G^+ = \frac{m_X\boldsymbol{r}_X + m_u\boldsymbol{r}_u + m_c\boldsymbol{r}_c + m_b\boldsymbol{r}_b}{m_X + m_u + m_c + m_b}G^+\boldsymbol{I}_{b,G^+} = m_b(\|\boldsymbol{d}_b\|^2\boldsymbol{I}_3 - \boldsymbol{d}_b\boldsymbol{d}_b^T)$

Among them:, if you want to consider the rotational inertia of the tennis ball itself, the tennis ball can be approximated as a thin spherical shell:, therefore:, the total inertia tensor after catching the ball is:$\boldsymbol{d}_b = \boldsymbol{r}_b - \boldsymbol{r}_G^+\boldsymbol{I}_b^{(C)} = \frac{2}{3}m_b R_b^2\boldsymbol{I}_3\boldsymbol{I}_{b,G^+} = \frac{2}{3}m_b R_b^2\boldsymbol{I}_3 + m_b(\|\boldsymbol{d}_b\|^2\boldsymbol{I}_3 - \boldsymbol{d}_b\boldsymbol{d}_b^T)\boxed{\boldsymbol{I}_{G^+} = \boldsymbol{I}_{X,G^+} + \boldsymbol{I}_{u,G^+} + \boldsymbol{I}_{c,G^+} + \boldsymbol{I}_{b,G^+}}$

8. Numerical estimation and correction of moment of inertia

The equivalent mass of the$L_u = B_u = 0.080\ m$

Table 7. Numerical estimation of the moment of inertia of the whole machine

parts	$\boldsymbol{I}_{xx}$	$\boldsymbol{I}_{yy}$	$\boldsymbol{I}_{zz}$
X rack four point model	0.289×10^{-3}	0.289×10^{-3}	0.578×10^{-3}
center stack structure	0.892×10^{-3}	0.892×10^{-3}	0.373×10^{-3}
Lower gripper body	1.09×10^{-3}	1.01×10^{-3}	0.374×10^{-3}

Sum of three structures	2.27×10^{-3}	2.19×10^{-3}	1.32×10^{-3}

2.6 Quadrotor thrust and torque distribution

1. Single rotor thrust model: , , can also be written as: ()$T_i = k_T \Omega_i^2 Q_i = k_Q \Omega_i^2 Q_i = \gamma T_i \gamma = \frac{k_Q}{k_T}$

   Among them:

   T_i:th rotor thrust$i$

   Q_i : Rotor reaction torque$i$

   Ω_i: Motor angular speed

   k_T:Thrust coefficient

   k_Q: Torque coefficient

2. Hover thrust requirements

   Drone body weight:$W_d = m_d g = 0.6847(9.81) = 6.717\ N$

   On hover:$T_\Sigma = T_1 + T_2 + T_3 + T_4 = W_d$

   Average hovering thrust of a single rotor:$T_h = \frac{W_d}{4} = \frac{6.717}{4} = 1.679\ N$

   Convert to grams:$T_h = \frac{1.679}{9.81} = 0.171\ \text{kgf} \approx 171\ \text{gf}$

   After catching the tennis ball:$W^+ = m^+ g = 0.7417(9.81) = 7.276\ N$

   The hovering thrust of a single rotor after catching the ball:$T_h^+ = \frac{7.276}{4} = 1.819\ N \approx 185\ \text{gf}$

3. Thrust to weight ratio

   Thrust to weight ratio:, where:$\lambda = \frac{T_{max}}{W} T4, max3, max2, max1, max_{max}$

   Theoretically: Just hover. However, it is necessary to actually pick up the ball, resist ground effect, resist platform rebound flow, and resist contact moment. A conservative estimate is that if the ball is not included, the maximum thrust of a single rocket is required to be at least:$\lambda > 1\lambda \approx 1.5 \sim 2.0\lambda = 1.5$

$T\frac{10.075}{4}Ngf_{i,max}$, if requested:$\lambda = 2.0T\frac{2W_d}{4}Ngf_{i,max}$

If you still ask after catching the ball:$\lambda = 1.5$

$T\frac{1.5W^+}{4}\frac{1.5(7.276)}{4}Ngf_{i,max}$, if requested after catching the ball:$\lambda = 2.0$

$$T\frac{2W^+}{4}Ngf_{i,max}$$

Conclusion:
If we only consider empty aircraft, a single five-inch propeller must be at least$257{\sim}342\ \text{gf}$
If you are considering catching tennis balls, a single ball should be at least$57\ g278{\sim}371\ \text{gf}$

4. Quadrotor moment distribution matrix

The four motor positions are:

$$\boldsymbol{r}_1 = (+b,+b,0)^T, \quad \boldsymbol{r}_2 = (-b,+b,0)^T, \quad \boldsymbol{r}_3 = (-b,-b,0)^T,$$

$$\boldsymbol{r}_4 = (+b,-b,0)^T.$$

The thrust direction of each particle is:$\boldsymbol{F}_i = (0,0,T_i)^T \Rightarrow \boldsymbol{M}_i = \boldsymbol{r}_i \times \boldsymbol{F}_i = \begin{bmatrix} y_iT_i \\ -x_iT_i \\ 0 \end{bmatrix}$

The total thrust available is:$T_\Sigma = T_1 + T_2 + T_3 + T_4$

roll moment：$M_x = b(T_1 + T_2 - T_3 - T_4)$

pitch moment：$M_y = b(-T_1 + T_2 + T_3 - T_4)$

The yaw moment comes from the propeller reaction torque. If 1 and 3 are in the same direction and 2 and 4 are in opposite directions, then:$M_z = \gamma(T_1 - T_2 + T_3 - T_4)$

Therefore:$\boxed{\begin{bmatrix} T_\Sigma \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ b & b & -b & -b \\ -b & b & b & -b \\ \gamma & -\gamma & \gamma & -\gamma \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}}$

Among them:$b = 0.084999\ m$

2.7 Three-dimensional Euler equation and attitude perturbation

1. Translational motion

Center of mass translation

equation:$\boxed{m\ddot{\boldsymbol{r}}_G = \boldsymbol{R}\begin{bmatrix} 0 \\ 0 \\ T_\Sigma \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + \boldsymbol{F}_{\text{aero}} + \boldsymbol{F}_{\text{contact}}}$

Among them:

$\boldsymbol{R}$: Rotation matrix from body coordinates to inertial coordinates

$\boldsymbol{F}_{\text{aero}}$: Aerodynamic disturbance force, including downwash obstruction, platform rebound flow and wall jet lateral force

$\boldsymbol{F}_{\text{contact}}$: Contact force when the gripper carriage hits the platform

At a small angle, if the roll angle is and the pitch angle is, then the horizontal component of the thrust vector is approximately:, therefore:, ,, where is the equivalent downward load caused by the lower car body blocking the downwash.$\phi\theta\boldsymbol{R}\begin{bmatrix} 0 \\ 0 \\ T_\Sigma \end{bmatrix} \approx \begin{bmatrix} T_\Sigma\theta \\ -T_\Sigma\phi \\ T_\Sigma \end{bmatrix} m\ddot{x} \approx T_\Sigma\theta + F_x m\ddot{y} \approx -T_\Sigma\phi + F_y m\ddot{z} \approx T_\Sigma - mg - F_{\text{block}}F_{\text{block}}$

2. Euler rotation equation

The rigid body rotation equation is:$\boldsymbol{I}_G\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\boldsymbol{I}_G\boldsymbol{\omega}) = \boldsymbol{M}$

Among them: roll rate, pitch rate, yaw rate respectively. That means p is the positive angular velocity around the x-axis, q is the positive angular velocity around the y-axis, and r is the positive angular velocity around the z-axis.$\boldsymbol{\omega} = \begin{bmatrix} p \\ q \\ r \end{bmatrix}$

The total moment can be decomposed into:$\boldsymbol{M} = \boldsymbol{M}_{\text{rotor}} + \boldsymbol{M}_{\text{aero}} + \boldsymbol{M}_{\text{contact}} + \boldsymbol{M}_{\text{gyro}}$

Among them:

$\boldsymbol{M}_{\text{rotor}}$: The control torque caused by the thrust difference and reaction torque of the four rotors

$\boldsymbol{M}_{\text{aero}}$: Aerodynamic disturbance torque caused by downwash, ground effect, wall jet, and vehicle body obstruction.

$\boldsymbol{M}_{\text{contact}}$: Contact moment when the clamping jaw contacts the platform.

$\boldsymbol{M}_{\text{gyro}}$: The gyroscopic moment caused by the angular momentum of the motor and blade rotor.

※When the symmetry of the inertia tensor is good enough and the inertia product is much smaller than the diagonal inertia, it can be simplified to a diagonal tensor, then the Euler equation is expanded to:$\boldsymbol{I}_G \approx \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$

$$I_{xx}\dot{p} + (I_{zz} - I_{yy})qr = M_x$$
$$I_{yy}\dot{q} + (I_{xx} - I_{zz})pr = M_y$$
$$I_{zz}\dot{r} + (I_{yy} - I_{xx})pq = M_z$$

When taking the ball, if the attitude angular velocity is not large: , it can be linearized: , at a small angle: , so: , this is the equation that connects "fluid disturbance" to "attitude shaking"$p, q, r \approx 0 \dot{p} \approx \frac{M_x}{I_{xx}}, \dot{q} \approx \frac{M_y}{I_{yy}}, \dot{r} \approx \frac{M_z}{I_{zz}} p \approx \dot{\phi}, q \approx \dot{\theta} \boxed{\ddot{\phi} \approx \frac{M_x}{I_{xx}},\ \ddot{\theta} \approx \frac{M_y}{I_{yy}}}$

3. Attitude angular acceleration caused by thrust difference

Assume the total thrust difference between the left and right sides:$\Delta T_{LR} = 0.2\ N$

Then roll moment:, using the moment of inertia after the summation of the three structures:, then:$M_x = b\Delta T_{LR} = 0.084999(0.2) = 0.0170\ N \cdot m I_{xx} = 2.27 \times 10^{-3}\ \mathrm{kg} \cdot m^2 \ddot{\phi} = \frac{0.0170}{2.27\times10^{-3}} = 7.48\ \mathrm{rad/s^2}$

If the thrust difference increases to:,$\Delta T_{LR} = 0.5\ N M_x = 0.0425\ N \cdot m \ddot{\phi} = 18.7\ \mathrm{rad/s^2}$

Therefore, as long as local downwash, ground effect or platform rebound airflow causes a thrust difference of tens of grams of force, it is enough to cause obvious attitude shaking.

4. fluid disturbance

The thrust disturbance of a single rotor can be written as:, where:, respectively represent:$T_i = T_{i,0} + \delta T_i \delta T_i = \delta T_{i,\Omega} + \delta T_{i,h} + \delta T_{i,\mathrm{turb}} + \delta T_{i,\mathrm{block}}$

Thrust disturbance caused by changes in motor speed:$\delta T_{i,\Omega} = 2k_T \Omega_i \delta\Omega_i$
Thrust disturbance caused by ground effect height change:$\delta T_{i,h} = T_{i,0} G'(h_i)\delta h_i$
Thrust disturbance caused by turbulent velocity disturbance:$\delta T_{i,\mathrm{turb}} \sim \rho A_i (2\bar{w}_i \delta w_i + \delta w_i^2)$
Damage caused by obstruction by the vehicle body or supports below:$\delta T_{i,\mathrm{block}} \sim -\frac{1}{2}\rho C_D A_{\mathrm{block}} U^2$

Disturbance torque:,$\delta M_x = b(\delta T_1 + \delta T_2 - \delta T_3 - \delta T_4)\delta M_y = b(-\delta T_1 + \delta T_2 + \delta T_3 - \delta T_4)$

Substitute into Euler equation:,$\boxed{\delta\ddot{\phi} = \frac{\delta M_x}{I_{xx}} \delta\ddot{\theta} = \frac{\delta M_y}{I_{yy}}}$

The full impact link is

below: $\boxed{\delta w \to \delta T_i \to \delta M \to \delta\ddot{\phi}, \delta\ddot{\theta} \to flight\ oscillation}$

2.8 Actuator Disk Model analyzes thrust and downwash flow

1. Concept:BEMT = ADM + BET = Actuator Disk Model + Blade Element Theory

   ADM regards the entire bladed rotor as a very thin disk, and only cares about the momentum and energy changes of the entire fluid before and after passing through the rotor. Therefore, it can answer: How much does the rotor change the fluid momentum in total? Therefore, it can be estimated what the total thrust is.

   However, ADM cannot see blade geometry, such as blade chord length, pitch angle, airfoil lift coefficient, and drag coefficient. BET, on the contrary, cuts the blade into many small segments, and each segment is regarded as a two-dimensional airfoil. It can answer: How much lift, drag, thrust and moment does a certain small segment of blade produce under the local relative wind speed?

   The core of BEMT is to let the annular momentum change given by ADM = the force on the blade element given by BET

   Therefore, ADM starts from the "overall momentum of the fluid", BET starts from the "local airfoil force of the blade", and BEMT requires the two to be consistent in each annulus.

2. Basic assumptions of ADM and control volume perspective

   Standard ADM basic assumptions:

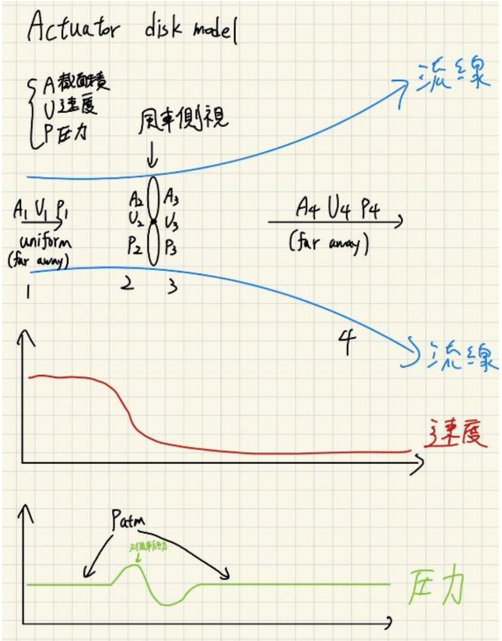
      Steady-state flow field: The average state of the flow field does not change rapidly with time.

      Incompressible flow: The air near a low-speed propeller can be approximated by. ρ = constant

The action disk is very thin: the axial speed in front of the disk and behind the disk is continuous.$U_2 = U_3$

Far upstream and far downstream pressure returns to atmospheric pressure:.$p_1 = p_4 = p_\infty$

The radial velocity, tip loss and strong three-dimensional vortex effect are



first ignored.

Figure 34: Actuator Disk Model control volume, velocity distribution and pressure changes

Take a flow tube control volume that surrounds the action disk. The far upstream is section 1, the front of the disk is section 2, the back of the disk is section 3, and the far downstream is section 4. The steady-state incompressible continuity equation is:$\dot{m} = \rho A_1 U_1 = \rho A_2 U_2 = \rho A_3 U_3 = \rho A_4 U_4$.

If the action disk is very thin, then:$A_2 = A_3 = A, U_2 = U_3$.

Bernoulli's equation cannot be used directly across the disk because the disk does work on or from the fluid; therefore it must be used separately with:$1 \rightarrow 23 \rightarrow 4$

$$p_1 + \frac{1}{2}\rho U_1^2 = p_2 + \frac{1}{2}\rho U_2^2, \quad p_3 + \frac{1}{2}\rho U_3^2 = p_4 + \frac{1}{2}\rho U_4^2.$$

Because: The relationship between the disk pressure jump and the speed change can be obtained.$p_1 = p_4 = p_\infty, \quad U_2 = U_3,$

3. The difference between wind turbine ADM form and propeller form

Generally, the derivation of ADM mainly uses the wind turbine mode, that is, the rotor takes out energy from the flow field, and the speed of the fluid decreases after passing through the rotor. Define the axial induction factor:, can be obtained from Bernoulli eq: Therefore:, the mass flow rate is:, for the wind turbine form, the momentum equation thrust is written as:$a = \frac{U_1 - U_2}{U_1}$, $U_2 = (1-a)U_1 U_2 = U_3 = \frac{U_1+U_4}{2}, U_4 = (1-2a)U_1 \dot{m} = \rho A U_2 = \rho A(1-a)U_1 T = \dot{m}(U_1 - U_4)$,

$\boxed{T = \rho A(1-a)U_1[U_1 - (1-2a)U_1] = 2\rho A U_1^2 a(1-a)}$, since the flow

pressure is dimensionless, the thrust coefficient is:, if the power taken out by the rotor from the flow field is:, then the power coefficient is:, for differentiation, the maximum value can be obtained at: and the Betz limit is obtained: $\boxed{C_T = \frac{T}{\frac{1}{2}\rho A U_1^2} = 4a(1-a)}$ $P = TU_2$, $\boxed{P = 2\rho A U_1^3 a(1-a)^2}$ $\boxed{C_P = \frac{P}{\frac{1}{2}\rho A U_1^3} = 4a(1-a)^2}$ $C_P a a = \frac{1}{3}, C\frac{16}{27}_{P,max}$

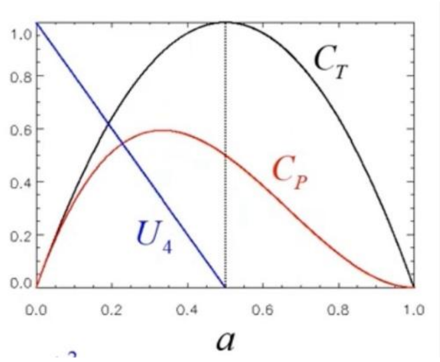


Figure 35. Relationship between thrust coefficient, power coefficient and axial induction factor in wind turbine ADM

But the object we want to analyze is not a wind turbine, but a propeller. The propeller does not draw energy from the flow field, but inputs power from the motor, causing the air to gain downward momentum. The physical direction is therefore opposite to that of the wind turbine:

Wind turbine: The fluid does work on the rotor and the flow rate decreases.

Propeller: The motor does work on the fluid, increasing the flow rate.

Subsequent analysis of the four-axis five-inch propeller was performed in the form of propeller hovering. In the hovering problem, the far upstream incoming flow is approximately zero, the propeller induces velocity on the disk surface, and the far downstream wake velocity is approximately:v_i

$V_w \approx 2v_i$.From the momentum theory, the hovering thrust can be obtained:, after the four propellers are combined:, this is the ADM form that we will subsequently use to estimate the induced velocity, disk loading and ideal power of the quadcopter.$T = 2\rho A v_i^2 T_\Sigma = 2\rho A_\Sigma v_i^2$

4. Momentum theory: The hovering thrust of the five-inch propeller quadcopter of this aircraft

Think of the four propellers as four actuator disks, when hovering:$T_\Sigma = 2\rho A_\Sigma v_i^2$

T_Σ:Total thrust
ρ: air density
A_Σ: Total propeller disc area of four propellers
v_i: induced velocity at the propeller disk

Therefore:$v_i = \sqrt{\frac{T_\Sigma}{2\rho A_\Sigma}}$

When the aircraft is hovering:, take:,$T_\Sigma = W_d = 6.717\ N \rho = 1.2$ kg/m$^3 A_\Sigma = 0.05067\ m^2$

Then: $\boxed{v_i = \sqrt{\frac{6.717}{2(1.2)(0.05067)}} = 7.43 \text{ m/s}}$

Far downstream speed: $\boxed{V_w \approx 2v_i = 14.86 \text{ m/s}}$

And after catching the tennis ball:,$T_\Sigma = W^+ = 7.276\ N$

$$\boxed{v_i^+ = \sqrt{\frac{7.276}{2(1.2)(0.05067)}} = 7.74 \text{ m/s}}\ \boxed{V_w^+ \approx 15.47 \text{ m/s}}$$

5. Disk loading: $DL = \frac{W}{A_\Sigma}$

Empty aircraft:$DL_d = \frac{6.717}{0.05067} = 132.6 \text{ N/m}^2$

After catching the ball:$DL^+ = \frac{7.276}{0.05067} = 143.6 \text{ N/m}^2$

Disk loading is also equal to the average propeller disc pressure difference:, so when the empty aircraft is hovering:, this pressure difference may not seem large, but when it acts on the entire propeller disc area, it forms sufficient lift.$\Delta p \approx \frac{T}{A}$ $\boxed{\Delta p \approx 133 \text{ Pa}}$

6. mass flow rate

The mass flow rate through the paddle disk is:, empty aircraft:, after catching the ball:, this means that when the drone is hovering, it accelerates the air downwards approximately per second; after catching the tennis ball, it is approximately.$\dot{m} = \rho A_\Sigma v_i \dot{m} = 1.2(0.05067)(7.43) = 0.452 \text{ kg/s} \dot{m}^+ = 1.2(0.05067)(7.74) = 0.470 \text{ kg/s} 0.45 \text{ kg} 0.47 \text{ kg/s}$

7. Ideal hover power

Ideal induced power:, empty machine:$P_i = T v_i P_i = 6.717(7.43) = 49.9\ W$

After catching the ball:$P_i^+ = 7.276(7.74) = 56.3\ W$

The induced power increase ratio after catching the ball:$\frac{P_i^+}{P_i} = \left(\frac{m_d+m_b}{m_d}\right)^{3/2} = \left(\frac{0.7417}{0.6847}\right)^{3/2} = 1.127$

So after grabbing the tennis ball, the ideal hover power increases by approximately:57 g12.7%

2.9 BEMT：Blade Element Momentum Theory

$$\text{BEMT} = \text{ADM} + \text{BET}$$

ADM explains overall downwash flow, induced velocity, disk loading and ideal power

BET explains why blade pitch, chord length, airfoil, and rotational speed affect thrust and torque.

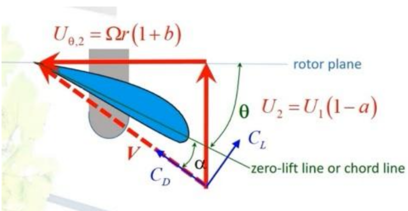
BEMT explains that propeller thrust is the result of "fluid momentum change" and "blade local lift and drag".

After obtaining the , , and of the five-inch propeller, you can use BEMT to further predict the , and under different RPMs.$c(r)\theta(r)C_L(\alpha)C_D(\alpha)TQP$

Table 8. BEMT blade element analysis parameters and symbol definitions

symbol	meaning
B	Number of leaves. If the five-inch propeller is a two-blade propeller, then B = 2; if it is a three-blade propeller, then B = 3
r	Local radius position, 0 < r < Rp
$c(r)$	local chord length
$\theta(r)$	local pitch angle
$\varphi(r)$	local inflow angle / flow angle
$\alpha(r)$	local angle of attack
$V(r)$	The relative wind speed seen by the blade element
C_L, C_D	Airfoil lift coefficient and drag coefficient are functions of α and Reynolds number
σ	local solidityσ= Bc / (2πr)
a	axial inducible factor
a_θ	tangential inducible factor

1. Rotation Effect and Velocity Triangle



←The rotor can be cut into many annular elements

Figure 36: Velocity triangle and force direction of rotating blade elements

A real propeller not only changes the axial speed, but also exerts a tangential moment on the fluid, causing the wake to rotate.

$U_x = U_1(1-a)$, $U_\theta = \Omega r(1+a_\theta)$, for a general rotor, the relative wind speed seen by the blade element can be written as:, the inflow angle satisfies:, approximated by propeller hovering, the axial velocity can be regarded as the local induced velocity:, the average induced velocity obtained by ADM can be used as the local axial velocity. The angle of attack definition must be consistent with the geometry. Adopt propeller convention:.$V(r) = \sqrt{U_x^2 + U_\theta^2}\ \phi = tan\left(\frac{U_x}{U_\theta}\right)^{-1}\ U_x \approx v_i(r) U_x \approx v_i = 7.43$ m/s$\alpha(r) = \theta(r) - \phi(r)$

2. Blade element local velocity

   For blade tiny elements at radius:$r U(r) = \sqrt{(\Omega r)^2 + (V_\infty + v_i)^2}$

   When hovering:, so:, inflow angle:, local angle of attack:$V_\infty = 0 U(r) = \sqrt{(\Omega r)^2 + v_i^2}\ \phi(r) = \tan^{-1}\left(\frac{V_\infty + v_i}{\Omega r}\right) \alpha(r) = \theta(r) - \phi(r)$

3. local lift and drag

   Cut the blade into blade elements with radius and thickness. This small section of blade is regarded as a two-dimensional airfoil section, and the tiny lift and drag are$r dr dL = \frac{1}{2}\rho U(r)^2 c(r) C_L(\alpha) dr,\ dD = \frac{1}{2}\rho U(r)^2 c(r) C_D(\alpha) dr$

   If the number of propeller blades is$B$

   $$dT = B(dL\ cos\ \phi - dD\ sin\ \phi), dQ = Br(dL\ sin\ \phi + dD\ cos\ \phi)$$

   Substitute:$dL, dD\ \frac{dT}{dr} = \frac{1}{2} B\rho U^2 c(C_L\ cos\ \phi - C_D\ sin\ \phi), \frac{dQ}{dr} = \frac{1}{2} B\rho U^2 c(C_L\ sin\ \phi + C_D\ cos\ \phi) r$

   Total thrust:, Total torque:, Shaft power:$\boxed{T = \int_0^{R_p} d\,T}\ \boxed{Q = \int_0^{R_p} d\,Q}\ P_{\text{shaft}} = Q\Omega$

4. Momentum annulus equation

   Radius, thickness:$r dr dA = 2\pi r dr$.

   The annular thrust can be written as:, if wake rotation and tangential induction are added, the angular momentum flux can be derived as the moment formula:$\frac{dT}{dr} = 4\pi r \rho U_1^2 a(1-a)\ \frac{dQ}{dr} = 4\pi r^3 \rho U_1 \Omega (1-a) a_\theta$

   For the propeller hovering problem, if the local induced velocity is used to represent the axial velocity, the approximation can be used: , where is the

Prandtl tip/root loss factor; if the tip and root losses are ignored first, then: , the core of BEMT is: and: From this, we can find , , and then integrate to get , .$v_i(r)\frac{dT}{dr} \approx 4\pi F\rho r v_i(r)^2 F F \approx 1 \left(\frac{dT}{dr}\right)_{\text{BET}} = \left(\frac{dT}{dr}\right)_{\text{ADM}} \left(\frac{dQ}{dr}\right)_{\text{BET}} = \left(\frac{dQ}{dr}\right)_{\text{ADM}} v_i(r) a(r) a_\theta(r) TQP$

Definition: And: Let the thrust equations of ADM and BET be equal, and the axial induction factor equation can be obtained:, and the moment equations of ADM and BET be equal, and the tangential induction factor equation can be obtained:$\sigma = \frac{Bc}{2\pi r}, \quad J = \frac{U_1}{nD}, \quad \lambda_r = \frac{\Omega r}{U_1}, V = \frac{U_1(1-a)}{\sin\phi},$

$$\boxed{a = \frac{\sigma(1-a)}{4\sin^2\phi}(C_L\cos\phi + C_D\sin\phi)}$$

$$\boxed{a_\theta = \frac{\sigma(1-a)}{4\sin^2\phi}\frac{JD}{2\pi r}(C_L\sin\phi - C_D\cos\phi) = \frac{\sigma(1-a)}{4\lambda_r\sin^2\phi}(C_L\sin\phi - C_D\cos\phi)}$$

5. Actual calculation of BEMT

Static thrust and electric power experimental data

Based on the empty weight of the machine and the weight after catching the ball, the following table serves as the static thrust data when four rotors are output simultaneously. This data allows the aircraft to hover near the approximate throttle, which can support the hovering requirement after grabbing a tennis ball, and the throttle above can provide a close thrust-to-weight ratio.$W_d = 6.717\ N W^+ = 7.276\ N, 52\%55\%75\%2$

Table 9. Thrust, speed, thrust-to-weight ratio and electric power under different throttle ratios

Throttle ratio (%)	**Average RPM(rpm)**	**Total thrust TΣ(N)**	**Single thrust (gf)**	**Thrust-to-weight ratio TΣ/Wd**	**Electric powerPelec(W)**
45	9000	5.15	131	0.77	70
50	9800	5.95	152	0.89	82
52	10400	6.72	171	1.00	95
55	11200	8.10	206	1.21	122
60	12000	9.65	246	1.44	154

65	12800	11.50	293	1.71	195
70	13600	13.20	337	1.97	245
75	14400	15.20	388	2.26	300

It can be seen from the table that the hovering point of the empty aircraft can be taken as At this time, the equivalent hovering efficiency can be estimated by. From the aforementioned ADM, the ideal induced power of the empty machine can be obtained, so$T_{\Sigma} \approx 6.72\ N, \quad P_{elec} \approx 95\ W. \eta_{hover} = \frac{P_i}{P_{elec}} P_i \approx 49.9\ W, \eta_{hover} \approx \frac{49.9}{95} = 0.525.$

The measurement values of the data required by BEMT are as follows:

$$B = 3, \quad D_p = 0.127\ m, \quad R_p = 0.0635\ m,$$

RPM = 10400, $n = 173.3\ s^{-1}$, $\Omega = 2\pi n = 1089$ rad/s, $v_i = 7.43$ m/s.

This rotational speed point is close to the hovering operating point of the empty machine. For each blade element, use the velocity triangle and propeller convention to estimate the local thrust and moment:$\phi(r) = tan\left(\frac{v_i}{\Omega r}\right)^{-1}, \quad V(r) = \frac{v_i}{sin\,\phi},$

$\frac{dT}{dr} = \frac{1}{2} B\rho V^2 c(C_L\, cos\,\phi - C_D\, sin\,\phi), \ \frac{dQ}{dr} = \frac{1}{2} B\rho V^2 cr(C_L\, sin\,\phi + C_D\, cos\,\phi).$

Input and local output of BEMT blade elements (single rotor):

Table 10. BEMT blade element input parameters and local thrust calculation

r/R	**r(m)**	**c(mm)**	**φ(°)**	**θ(°)**	**α(°)**	**V(m/s)**	**C_L**	**C_D**	**dT/dr(N/m)**
0.30	0.0191	24	19.7	37	17.3	22.0	1.00	0.065	12.9
0.50	0.0318	20	12.1	30	17.9	35.4	1.05	0.060	30.4
0.70	0.0445	17	8.7	24	15.3	49.0	1.05	0.060	50.3
0.90	0.0572	13	6.8	19	12.2	62.7	0.95	0.065	57.3

Using trapezoidal integration and making the blade root and blade tip point loads approach zero, the result of a single rotor can be obtained:
$T_{BEMT} \approx 1.69\ N, \quad Q_{BEMT} \approx 1.63 \times 10^{-2}\ N \cdot m, P_{shaft} = \Omega Q_{BEMT} \approx 17.8\ W.$

The total weight of the four rotors is: , which is quite close to the weight of the empty machine.$4T_{BEMT} \approx 6.78\ NW_d = 6.717\ N$

Therefore, this set of BEMT data is compatible with the ADM hover thrust requirements. Assuming that the efficiency of the motor and ESC is approximately , the electrical power of the four rotors is approximately the same as the hovering point in the table.$\eta_{motor+ESC} \approx 0.75, P_{elec} \approx \frac{4P_{shaft}}{0.75} = 94.9\ W, P_{elec} \approx 95\ W$

6. Model limitations and corrections

   Blade tip and root loss: The limited number of blades will cause tip vortex, which will reduce the effective load. Add the Prandtl tip/root loss factorF correction.

   Misaccuracy in the high induction region: When the induction factor is too large, the simple momentum theory may fail and high induction correction is required.

   Airfoil stall: If the angle of attack is too large, it is sensitive to poles, and the reliability of the model depends on the quality of the airfoil data.$C_L C_D \alpha$

   Three-dimensional and unsteady effects: When the five-inch paddle is close to the platform, the gripper body and the tennis ball, radial flow, backflow, tip vortex interaction, wall jet and unsteady turbulence will occur. These cannot be fully described by basic BEMT.

2.10 Propeller coefficient: propeller performance analysis

If the energy is measured in RPM, let:$n = \frac{\text{RPM}}{60}$

Thrust coefficient:, power coefficient:, torque coefficient: and:, so:$C_T = \frac{T}{\rho n^2 D_p^4} C_P = \frac{P}{\rho n^3 D_p^5} C_Q = \frac{Q}{\rho n^2 D_p^5} P = Q\Omega, \Omega = 2\pi n C_P = 2\pi C_Q$

If there is forward speed: , where is advance ratio. When hovering:, the hover efficiency can be used as a figure of merit:, if only electric power can be obtained:, then the equivalent hover efficiency: is defined, which can be used to analyze the efficiency of the entire propulsion system.$J = \frac{V_\infty}{nD_p} J J = 0 FM = \frac{P_i}{P_{\text{shaft}}} P_{\text{elec}} = VI \boxed{\eta_{\text{hover}} = \frac{T\sqrt{T/(2\rho A_\Sigma)}}{VI}}$

2.11 Finite control volume method: reverse thrust from wind speed field

1. Governing Volume Momentum Equation

   Reynolds transport theorem：$\sum F = \frac{\partial}{\partial t}\int_{CV} \rho\, V dV + \int_{CS} \rho\, V(V \cdot n) dA$

   Take the vertical direction, let it be the downward velocity, and average the steady-state time:$w \sum F_z \approx \int_{CS} \rho\, w(V \cdot n) dA$

   If the control surface is selected below the downwash flow and the main outflow direction is downward, then:$T_{\mathrm{CV}} \approx \rho \int_A w^2\, dA$

   Complete ME:$T_{\mathrm{CV}} = \rho \int_{A_b} w^2\, dA + \rho \int_{A_s} w\,(V \cdot n) dA + \int_{CS}(p - p_\infty)\, n_z dA + F_{\mathrm{viscous}}$

   The first term: vertical momentum flux on the lower control surface

   Item 2: Correction caused by side outflow

   The third item: pressure correction

   Item 4: Adhesion correction

   The simplified formula can only be used if the control surface is far enough away from the propeller disc, the pressure is close to atmospheric pressure, and the vertical component of the lateral outflow is small.

2. differential discretization

   Cut the lower measuring plane into a grid:$\Delta A_{ij} = \Delta x \Delta y$

   Measured at each point:, Time average:, Disturbance:$w_{ij}(t) \bar{w}_{ij} = \frac{1}{N_t}\sum_{k=1}^{N_t} w_{ij}\,(t_k) w'_{ij}(t) = w_{ij}(t) - \bar{w}_{ij}$

   Standard deviation:, because thrust is related to velocity squared:$\sigma_{w,ij} = \sqrt{\frac{1}{N_t - 1}\sum_{k=1}^{N_t}\left[w_{ij}(t_k) - \bar{w}_{ij}\right]^2}\; T \propto w^2$

   So you can't just use:, but you should use:, that is:$\bar{w}^2 \overline{w^2} = \bar{w}^2 + \overline{w'^2} \overline{w^2} = \bar{w}^2 + \sigma_w^2$

   Therefore the thrust is estimated as:$T_{\mathrm{CV}} \approx \rho \sum_{i,j}\left(\bar{w}_{ij}^2 + \sigma_{w,ij}^2\right) \Delta A_{ij}$

   ※Why can’t we just use average wind speed?

   If you only use the plane mean velocity: , then write: you will usually underestimate or misestimate the thrust. There are three reasons:$\bar{w}_A = \frac{1}{A}\int_A w\, dA T \approx \rho A \bar{w}_A^2$

First, the momentum flux is related to the square of the velocity:, Second, the velocity field is not uniform. The downwash flow of the quadcopter has four high-speed zones, a central low-speed zone, a boundary shear layer and a vehicle body obstruction. Third, turbulent fluctuations will also contribute:, so the momentum correction factor is defined:, in discrete form:$\int_A w^2\, dA \neq A\left(\frac{1}{A}\int_A w\, dA\right)^2 \overline{w^2} = \bar{w}^2 + \sigma_w^2 \beta = \frac{\int_A w^2 dA}{A\bar{w}_A^2} \beta = \frac{\sum_{i,j} w_{ij}^2 \Delta A_{ij}}{A\left[\frac{1}{A}\sum_{i,j} w_{ij}\Delta A_{ij}\right]^2}$

Therefore:, if the velocity field is uniform:, if the velocity field is not uniform:.$\boxed{T_{\mathrm{CV}} = \rho A \beta \bar{w}_A^2}\beta = 1\beta > 1$

3. Estimating the thrust action line and disturbance moment from the flow field

It can be estimated whether the thrust line of action is offset and the thrust center is defined: discrete form:$x_T = \frac{\int_A x w^2 dA}{\int_A w^2 dA},\ y_T = \frac{\int_A y w^2 dA}{\int_A w^2 dA} x_T = \frac{\sum x_{ij} w_{ij}^2 \Delta A_{ij}}{\sum w_{ij}^2 \Delta A_{ij}},\ y_T = \frac{\sum y_{ij} w_{ij}^2 \Delta A_{ij}}{\sum w_{ij}^2 \Delta A_{ij}}$

If the line of action of the thrust deviates from the center of mass, a moment will be caused:$M_x \approx y_T T,\ M_y \approx -x_T T$

4. Actual data calculation

Take the measurement plane approximately below the propeller disk, and use the grid coordinates to remove the table and measure the time-averaged vertical downwash velocity with the anemometer. This is defined as the magnitude of the downward velocity, so it is a positive value.$z_m = 0.15\ m \Delta x = \Delta y = 0.05\ m, \Delta A = 0.0025\ m^2. x, y \in \{-0.10, -0.05, 0, 0.05, 0.10\}\ m. \bar{w} w$

Table 11. Average vertical velocity distribution of downwash flow measurement plane

y\x	-0.10	-0.05	0	+0.05	+0.10
+0.10	5.8	8.9	6.6	8.6	5.6
+0.05	9.1	14.0	11.0	13.7	8.9

0	6.4	10.8	7.8	10.6	6.3
-0.05	8.9	13.6	10.7	13.4	8.8
-0.10	5.5	8.5	6.2	8.4	5.3

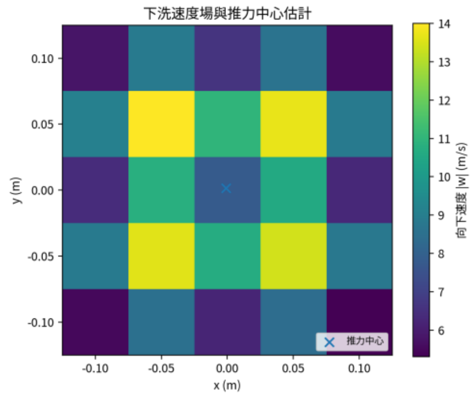


Figure 37. Downwash velocity field heat map and thrust center estimation

Considering the turbulent velocity disturbance, the standard deviation of each point is approximately , then the control volume thrust is estimated to be , which can be obtained by substituting into the table: This value is consistent with the weight of the empty aircraft, indicating that this set of downwash velocity fields can represent the hovering state.$\sigma_w \approx 0.16\bar{w} + 0.15 T_{CV} = \rho \sum_{i,j} \left(\bar{w}_{ij}^2 + \sigma_{w,ij}^2 \right) \Delta A \boxed{T_{CV} \approx 6.72\ N.} W_d = 6.717\ N$

If the turbulence disturbance is ignored and only used, then , is approximately lower than the estimate containing the turbulence term, which supports the aforementioned conclusion: when the downwash flow is a high Reynolds number turbulent flow, the velocity disturbance term should not be completely ignored.$\bar{w}^2 T_{CV,mean} \approx 6.52\ N \frac{6.72-6.52}{6.72} = 3.0\% \sigma_w^2$

The plane average velocity of this velocity field is and the momentum correction factor is$\bar{w}_A = 8.94$ m/s,

$\beta = 1.09$, so if only the uniform average velocity is used for estimation, the momentum flux correction caused by the non-uniform downwash flow will be ignored.$T \approx \rho A \bar{w}_A^2$

In addition, the thrust center can be estimated from the velocity field:

$x_T = \frac{\sum x_{ij} (\bar{w}_{ij}^2 + \sigma_{w,ij}^2) \Delta A}{\sum (\bar{w}_{ij}^2 + \sigma_{w,ij}^2) \Delta A}$, $y_T = \frac{\sum y_{ij} (\bar{w}_{ij}^2 + \sigma_{w,ij}^2) \Delta A}{\sum (\bar{w}_{ij}^2 + \sigma_{w,ij}^2) \Delta A}$.Substituting it in, we can get that the corresponding disturbance torque is about. If it is substituted, the roll direction angular acceleration is. This magnitude shows that although the downwash asymmetry only has a millimeter-level thrust center offset, it may still cause considerable attitude disturbance.$x_T = -1.1$ mm, $y_T = -1.7$ mm$M_x \approx y_T T_{CV} = (-0.0017)(6.72) = -0.011\ N \cdot m,\ M_y \approx -x_T T_{CV} = 0.007\ N \cdot m I_{xx} = 2.27 \times 10^{-3}$ kg $\cdot\, m^2 \boxed{\ddot{\phi} \approx \frac{M_x}{I_{xx}} \approx -4.9 \text{ rad/s}^2}$

2.12 Static lift experimental verification and propeller selection

1. Experimental purpose, propeller number and test structure

The purpose of the lift experiment is to convert the aforementioned thrust demand and propeller model into actual and verifiable throttle, current, RPM and single thrust curves. The propeller number abcd represents the diameter D = a.b in and the pitch P = c.d in. For example, 5040 represents a diameter of 5.0 inches and a pitch of 4.0 inches.

Pitch is usually defined by a blade section with an average radius of approximately 3/4R. The larger the pitch, the thrust usually rises faster with throttle and RPM, but the resistance and load are also greater. We used 5045 four-blade propellers for the initial configuration, but the vibration and noise were loud during actual flight, and there was also obvious lateral deflection when descending. Therefore, we later switched to 5040 four-blade propellers, which have a milder response and a more stable control feel.

Regarding the problem of lateral deviation of the drone when it descends, based on the ground effect analysis conclusion later in our report, we speculate that the lift of the 5045 is more sensitive to the throttle (or RPM) response. When the throttle is slightly changed, the thrust change will be more obvious than that of the 5040. Therefore, if the ground effect is asymmetric, even if 5045 only produces a little more thrust difference than 5040, it can still cause considerable and difficult-to-control angular acceleration.

Therefore, we want to observe the thrust curves of different propeller blades (mainly 5045 and 5040) and analyze the thrust/current ratio of different propellers under different throttle inputs to determine the efficiency of the propeller.

Experimental architecture:

Figure 38. Static lift experimental test frame and torque transmission mechanism

The lifting table refers to the design of the motor test frame and is adjusted according to the size of the machine. The L-shaped test stand was chosen because it allows the front and rear space of the propeller to move closer to free air without excessive interference from the table or the ground. At the same time, the actual drone arm is used as the motor holder to make the flow field closer to the on-board operating situation. The thrust of the motor transmits torque through the bearings, keeping the wooden bars in contact with the scale horizontal and transmitting force vertically. The distance from the motor center to the bearing and the distance from the scale center to the bearing are both 20.5 cm, so the torque arm is the same, and the scale reading can directly correspond to the motor thrust.

The motor control and readings use the SpeedyBee flight control board, the firmware uses Betaflight, and bidirectional DShot is enabled to monitor the motor RPM. The power supply uses a power supply, and the current limit is set to 15.1 V and 3.5 A to avoid battery voltage sag or excessive discharge affecting the results.

2. Experimental process

Six sets of propellers were tested: 5040 four-blade propeller, 5045 four-blade propeller, 5040 three-blade propeller, 5045 three-blade propeller, racing three-blade propeller and 5045 two-blade propeller. Each group was measured from 5% to 60% throttle at 5% throttle intervals, and the current, motor speed and scale thrust readings were recorded at the same time.

3. Throttle-thrust curve and hover throttle verification

When using the 5040 four-blade propeller in the mission, the actual flight hovering throttle is about 36% (the throttle reading on the remote control is about 1360), which corresponds to a total thrust of the four motors of about 751 g. The lift table measured the linear interpolation throttle of a single 5040 four-blade propeller at 187.75 g (= 751/4) thrust as 36.50%. The difference between the actual flight measurement and the test stand interpolation is about 0.5 percentage points, which shows that the lift table results can reasonably represent the static thrust characteristics on the aircraft.

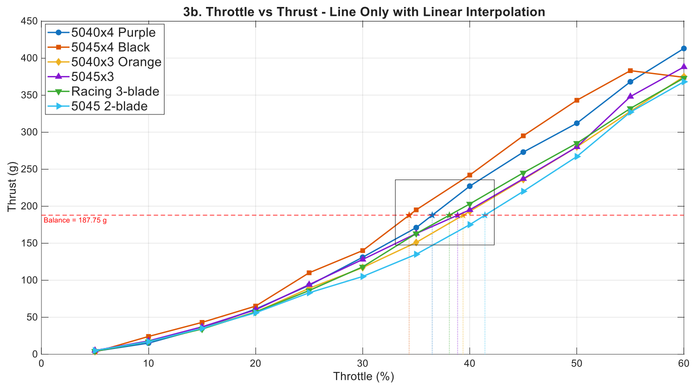


Figure 39: Thrust to Throttle

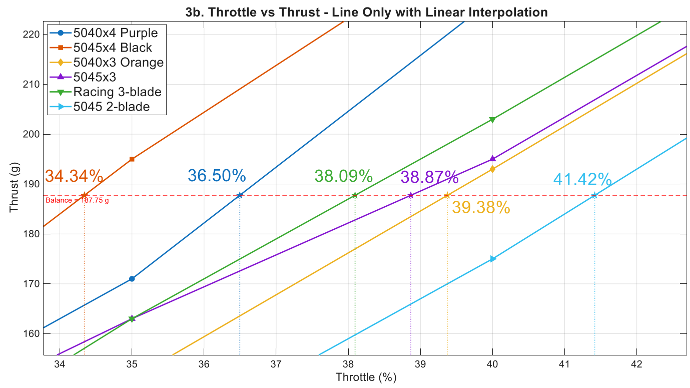


Figure 40, Thrust to Throttle (magnification)

Table 12. Comparison of throttle, RPM, current and efficiency when each propeller reaches hovering thrust

propeller	Up to 187.75 gf throttle (%)	Corresponds to RPM	Corresponding current (A)	Thrust/current near hover (gf/A)
5040x4 Purple	36.50	10070	1.071	175.4
5045x4 Black	34.34	9226	1.046	179.4
5040x3 Orange	39.38	11227	1.101	170.5
5045x3	38.87	10605	1.234	152.2
Racing 3-blade	38.09	10319	1.169	160.6
5045 2-blade	41.42	11268	1.412	133.0

As can be seen from the table, the 5045 four-blade propeller can achieve hovering thrust at about 34.34% of the throttle due to its larger pitch; the 5040 four-blade propeller is 36.50%. This is consistent with the expectation that the larger the pitch, the higher the thrust at the same throttle/speed. However, the more sensitive thrust curve also means that when the ground effect is asymmetric at low altitude, a small throttle or RPM disturbance may be amplified into a larger thrust difference and attitude disturbance.

4. RPM-thrust curve, thrust coefficient and theoretical verification

From the basic propeller thrust model, it can be expected that thrust and rotational speed have an approximately quadratic relationship. After plotting the thrust of the six propellers against RPM, all curves show a trend close to T ∝ RPM², and the fitting R² is all high, verifying that the aforementioned thrust model can be used as a reasonable approximation within the test range.$T = k_T \Omega^2$

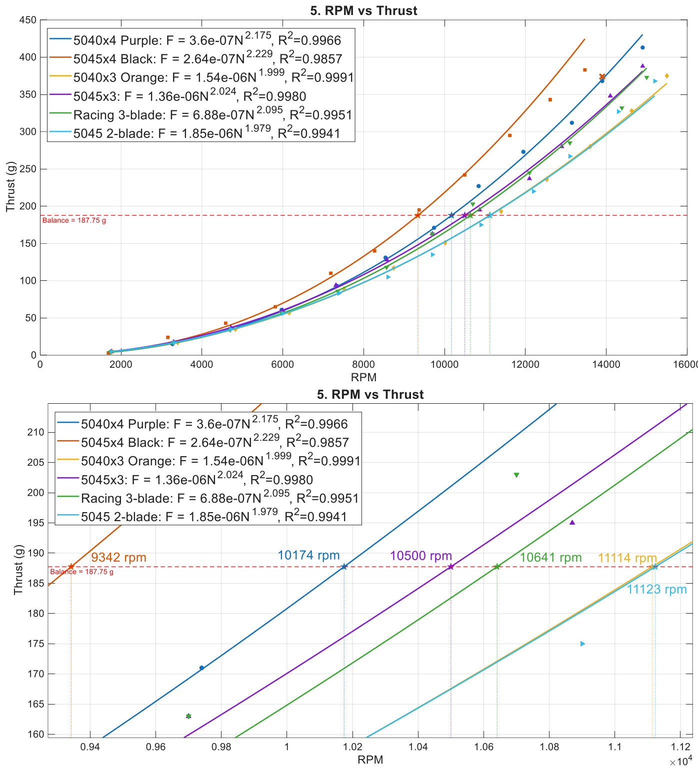


Figure 41. Thrust to RPM

Calculate the average thrust coefficient in the 30%~60% throttle range, and eliminate the 60% abnormal pressure drop point of the 5045 four-blade propeller, and the following table can be obtained. The average value of the 5045 four-blade propeller is higher than 0.20590 of the 5040 four-blade propeller; this means that the 5045 four-blade propeller provides higher thrust at the same speed, but it is also more likely to amplify low-altitude ground effect asymmetry and control input disturbance.$C_T = 0.23847$

Table 13. Comparison of average thrust coefficients of each propeller

propeller	**Average thrust coefficient C_T**

5040x4 Purple	0.20590
5045x4 Black	0.23847
5040x3 Orange	0.16826
5045x3	0.18870
Racing 3-blade	0.18532
5045 2-blade	0.16702

This result has experimentally verified the original theoretical estimation: if the 5045 propeller is more likely to produce larger thrust at the same RPM, then near the low-altitude platform, a small rotational speed difference on either side of the rotor caused by ground effects, downwash rebound or vibration will be converted into a larger thrust difference; and then the roll/pitch angular acceleration is caused by M = bΔT. This is the reason why the lateral deflection and low-altitude oscillation of 5045 during descent were more obvious during the dynamic test flight.

5. Current-thrust efficiency and propeller selection trade-off

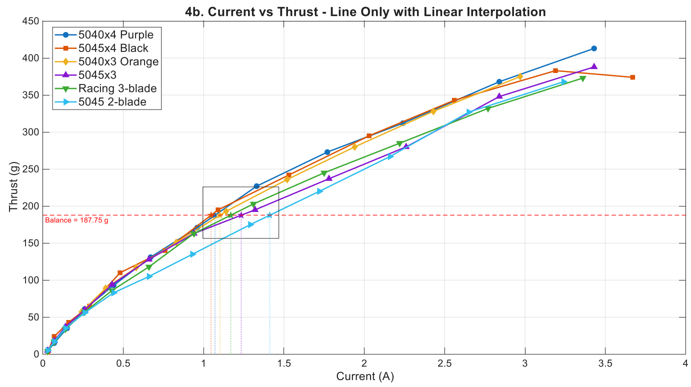


Figure 42. Thrust to Current

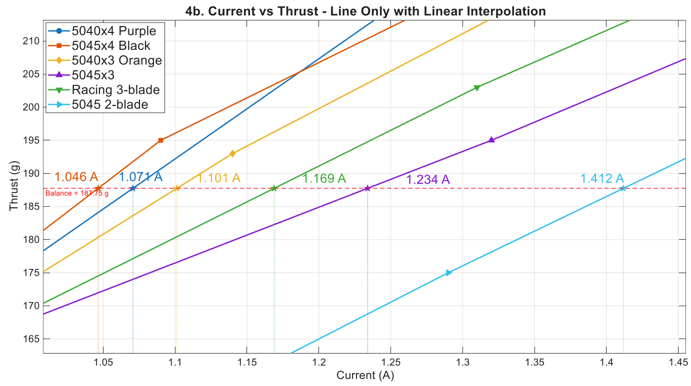


Figure 43. Thrust to Current (magnification)

Using the thrust/current near hovering as a simplified efficiency indicator, the 5045 four-blade propeller is slightly higher than the 5040 four-blade propeller, but the difference is not large, and the measurement itself may be affected by the scale reading, the current limit of the power supply, and the flow field of the single test rack. In comparison, the vibration and lateral deflection of the 5045 series blades during the dynamic test flight were significantly larger, and the cost of control stability was higher than the small efficiency difference.

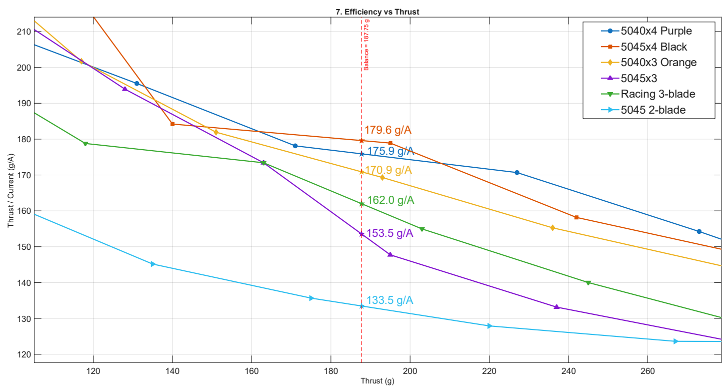


Figure 44. Thrust/current to Throttle

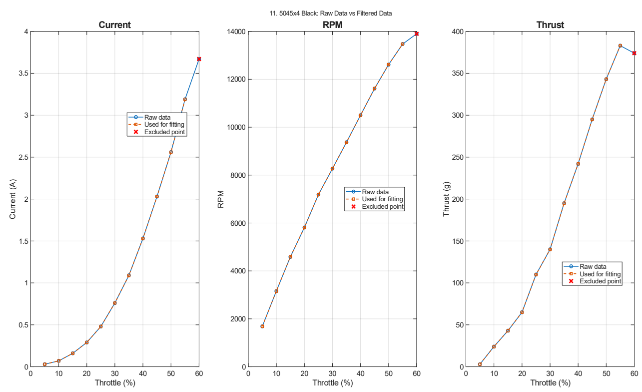


Figure 45. 5045 Original data and fitting results of four-blade propeller

current, RPM and thrust

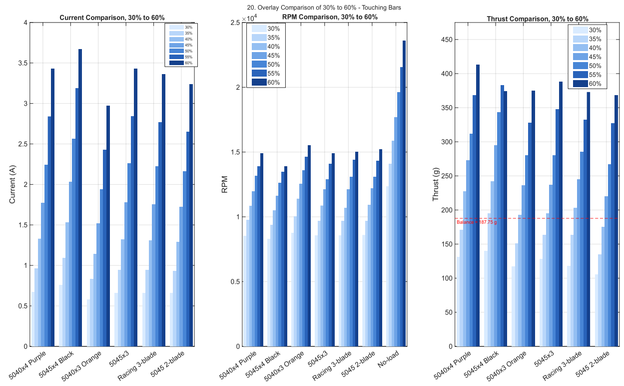


Figure 46. Comparison 30% throttle to 60% throttle

Therefore, the 5040 four-blade propeller was finally selected for the following reasons:

The interpolated hovering throttle of the 5040 four-blade propeller lift platform is 36.50%, which is highly consistent with the actual flight hovering throttle of about 36%. The model is consistent with the actual measurement.

The 5045 four-blade propeller has a higher C_T and is more sensitive to thrust response, making it easier to amplify attitude disturbances in asymmetric ground effect and platform backflow environments.

During dynamic flight, the vibration and noise of the 5045 series blades are significantly larger, which may increase the flight control IMU measurement noise and make the controller more difficult to stabilize.

Although the efficiency of the 5040 four-blade propeller is slightly lower than that of the 5045 four-blade propeller, it has better control predictability; and the team’s spare parts are mainly 5040 four-blade propeller, which has lower maintenance and replacement costs.

2.13 Downwash, Reynolds number and turbulence analysis

Downwash flow can be laminar or turbulent. Because of the high speed, blade rotation, and strong shear layer of the five-inch propeller four-axis, it is usually a turbulent downwash with a high Reynolds number.

1. Reynolds number

The following definitions of downstream speed and propeller diameter are as follows:$V_w = 14.86$ m/s$D_p = 0.127\ m Re = \frac{\rho V_w D_p}{\mu}$

Take: then:$\rho = 1.2$ kg/m^3, $\mu = 1.8 \times 10^{-5}$ Pa $\cdot\, s Re = \frac{1.2(14.86)(0.127)}{1.8\times10^{-5}} \approx$

1.26×10^5

This means that the downwash flow is a high Reynolds number flow and will be affected by: shear layer, tip vortices, wake interaction, recirculation, and turbulent fluctuation effects.

2. Reynolds decomposition

Speed breakdown:

$u = \bar{u} + u', v = \bar{v} + v', w = \bar{w} + w'$Where: , turbulence intensity: where: , if it means that the speed disturbance is approximately that of the average speed.$\bar{u}' = \bar{v}' = \bar{w}' = 0$ $\boxed{I_w = \frac{\sigma_w}{|\bar{w}|}}$, $\sigma_w = \sqrt{\overline{w'^2}} I_w = 0.220\%$

Since, the small disturbance approximation is:, if, then the local thrust fluctuation may reach. Although the four rotors and the overall control will lose part of the balance, it is still enough to cause posture shaking.$T \propto$

$w^2 \frac{\delta T}{T} \approx 2 \frac{\delta w}{w} \frac{\delta w}{w} = 0.2 \frac{\delta T}{T} \approx 0.440\%$

3. RANS and Reynolds stress

Navier--Stokes equation:, do Reynolds decomposition and take time average:$\rho\left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right) = -\frac{\partial p}{\partial x_i} + \mu \frac{\partial^2 u_i}{\partial x_j \partial x_j} \rho\left(\bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j}\right) = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} -$

$\rho \frac{\partial \overline{u_{i'} u_{j'}}}{\partial x_j}$

Where: called Reynolds stress. For vertical direction:$\boxed{-\rho \overline{u_{i'} u_{j'}} w}$

$$\rho\left(\bar{u}\frac{\partial \bar{w}}{\partial x} + \bar{v}\frac{\partial \bar{w}}{\partial y} + \bar{w}\frac{\partial \bar{w}}{\partial z}\right) = -\frac{\partial \bar{p}}{\partial z} + \mu \nabla^2 \bar{w} - \rho\left[\frac{\partial \overline{u'w'}}{\partial x} + \frac{\partial \overline{v'w'}}{\partial y} + \frac{\partial \overline{w'w'}}{\partial z}\right]$$

After the downwash flow hits the platform, a strong shear layer and backflow are formed. Reynolds stress changes the average flow field and pressure field, causing unsteady thrust and attitude disturbance.

4. Turbulent kinetic energy

If the three-axis velocity can be measured, define turbulent kinetic energy:. If only the vertical velocity can be measured, use: as an approximation.$k = \frac{1}{2}\left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2}\right) k \sim \frac{3}{2}\overline{w'^2}$

This can be used to draw or diagram showing where the turbulence is strongest.$k(x, y) I_w(x, y)$

2.14 Ground effect analysis

1. average ground effect correction

The ground effect can be modified in a simplified form:$G(h) = \frac{T_{\text{IGE}}}{T_{\text{OGE}}} = \frac{1}{1-\left(\frac{R_p}{4h}\right)^2}$

T_{IGE} ： in-ground-effect thrust

T_{OGE} ： out-of-ground-effect thrust

h: Distance from paddle disc to platform

The height of the lower gripper body is:,, substituted into:, so the average thrust gain is about$h \approx 0.0955\ m \frac{h}{R_p} = \frac{0.0955}{0.0635} = 1.50 \boxed{G(h) = \frac{1}{1-\left(\frac{0.0635}{4(0.0955)}\right)^2} = 1.028}$ 2.8%

Conclusion: At , the average ground effect gain is only a few percent; the real causes of instability are ground effect asymmetry, wall jet, backflow and turbulence.$h/R_p \approx 1.5$

2. Tilt causes ground effect asymmetry

When the body has a roll angle, the rotor heights on the left and right sides are different. Small angle approximation:$\phi \delta h_i \approx -y_i \phi$

Thrust of each rotor:, linearization:, by:, where:, derivative: roll, ground effect disturbance moment magnitude:, where, in, can be estimated as:, using angular acceleration:, fairly controllable.$T_i(h_i) = T_{i,0} G(h_i) \delta T_i \approx T_{i,0} G'(h_0) \delta h_i G(h) = \frac{1}{1-c/h^2} c = \frac{R_p^2}{16} G'(h) = -\frac{2c/h^3}{(1-c/h^2)^2} M_{x,\text{ground}} \approx 4b^2 T_0 |G'(h_0)| \phi T_0 \approx \frac{W_d}{4} = 1.679\ N h_0 = 0.0955\ \text{m},\ \ \phi = 5^\circ = 0.0873\ \text{rad} M_{x,\text{ground}} \approx 0.00259\ N \cdot m I_{xx} = 2.27 \times 10^{-3}\ \text{kg} \cdot m^2 \boxed{\ddot{\phi}_{\text{ground}} = \frac{M_{x,\text{ground}}}{I_{xx}} = \frac{0.00259}{2.27 \times 10^{-3}} \approx 1.14\ \text{rad/s}^2}$

However, if the paddle is closer to the platform, for example:, the ground effect sensitivity will increase significantly. It is estimated that:$h = 0.05\ m \boxed{\ddot{\phi}_{\text{ground}} \approx 9.3\ \text{rad/s}^2}$

Therefore, the design conclusion: try to keep the height of the paddle plate at$h/R_p \gtrsim 1.3$~1.5

2.15 Impinging jet and wall jet

1. Downwash impingement platform

When the downwash flows downward and hits the platform, it will form:

impinging jet → stagnation region → radial wall jet

There is stagnation pressure near the center of the platform:, if:, if:, this pressure field will make the flow field near the platform non-uniform and may push the ball away.$p_s - p_\infty \approx \frac{1}{2} \rho U_j^2 U_j = 10\ \text{m/s} \boxed{p_s - p_\infty = \frac{1}{2}(1.2)(10)^2 = 60\ \text{Pa}} U_j = 15\ \text{m/s} \boxed{p_s - p_\infty = 135\ \text{Pa}}$

2. Wall jet speed estimate

Downwash mass flow rate:, after impacting the platform, a radial wall jet of thickness is formed at the radius:, if only a part of the downwash flow acts near the ball, let the ratio be:, then:. This model explains:$\dot{m} = \rho \int_A w \, dA r \delta \dot{m}_j \approx \rho(2\pi r\delta)U_r \eta_j \dot{m}_j = \eta_j \dot{m} \boxed{U_r \approx \frac{\eta_j \dot{m}}{\rho 2\pi r\delta}}$

- rThe smaller, the closer the ball is to the center of impact, the larger it is.U_r
- δThe thinner, the faster the wall jet.
- η_jThe bigger it is, the easier it is to blow the ball away.

3. The influence of the lower gripper body on the downwash flow

Horizontal projected area of the lower car body:$A_{\text{block}} = L_c B_c = 0.076933(0.096276) = 0.00741 \ m^2$

The total area of the four paddles is:, the ratio is:, so the lower gripper body is not an object that can be ignored, it may significantly block the downwash.$A_\Sigma = 0.05067 \ m^2 \boxed{\frac{A_{\text{block}}}{A_\Sigma} = \frac{0.00741}{0.05067} = 14.6\%}$

4. Downwash blocking power

The car body below is subject to the aerodynamic resistance of the downwash:$F_{\text{block}} = \frac{1}{2}\rho C_D \chi A_{\text{block}} U^2$

- C_D: Vehicle body resistance coefficient, you can first take 1.0~1.2
- χ: Effective masking ratio
- U: Partial downwash speed

If:$C_D = 1.1, \quad \chi = 1, \quad U = 10$ m/s

Then: $\boxed{F_{\text{block}} = \frac{1}{2}(1.2)(1.1)(0.00741)(10)^2 = 0.489\ N}$

Equivalent to the weight ratio of the empty machine:, the weight after catching the ball is larger, the same blocking force is equivalent to:, if the local downwash speed is taken as the far downstream speed of the empty machine, then, is approximately the weight of the empty machine $\boxed{\frac{F_{\text{block}}}{W_d} = \frac{0.489}{6.717} = 7.3\%}$ $\boxed{\frac{0.489}{7.276} = 6.7\%}$ $U = 14.86\ \text{m/s} F_{\text{block}} \approx 1.08\ N 16.0\%$

Conclusion: The lower gripper body may cause an equivalent thrust loss of several to more than ten percent.

5. Wall jet blows to the side of the car body

Car body side area:, lateral aerodynamic force: $A_{\text{side}} = B_c H_c = 0.096276(0.095547) = 0.00920\ m^2$

$F_{\text{side}} = \frac{1}{2}\rho C_D A_{\text{side}} U_r^2$, if, then: $U_r = 5$ m/s, $\quad C_D = 1.1 F_{\text{side}} =$

$\frac{1}{2}(1.2)(1.1)(0.00920)(5)^2 = 0.152\ N$

Calculated from the center of gravity of the three structures: $z_G = -0.00120\ m, \quad z_c = -0.04777\ m.$

Therefore, the magnitude of the vertical distance between the lateral stress point of the lower vehicle body and the center of gravity is desirable, then the disturbance torque is:, corresponding to the motor differential thrust: $d \approx |z_c - z_G| =$

$0.04657\ m$ $\boxed{M_{\text{side}} = dF_{\text{side}} = 0.04657(0.152) = 0.00707\ N \cdot m}$ $\boxed{\Delta T = \frac{M_{\text{side}}}{4b} = \frac{0.00707}{4(0.084999)} = 0.02}$

If, the force and moment will become 4 times. $U_r = 10$ m/s $\Delta T \approx 8.5$ gf

Therefore, the lateral blow of the platform wall jet to the car body is enough to cause the need for control correction. Since the center of gravity is approximately approximately below the plane of the X-shaped frame, and the moment arm relative to the center of gravity of the lower car body is approximately approximately, the lateral wall jet will still cause measurable disturbance to the roll/pitch.$1.20\ \text{mm}46.57\ \text{mm}$

2.16 Will the downwash blow away the tennis balls?

Tennis quality:$m_b = 0.057\ \text{kg}$

1. ball force model

Horizontal wall jet speed near the platform, windward area of tennis ball:, air resistance of ball:, ball Reynolds number:,$U_r A_b = \frac{\pi d_b^2}{4} = \frac{\pi(0.067)^2}{4} = 3.53 \times 10^{-3}\ m^2 F_D = \frac{1}{2}\rho C_D A_b U_r^2 Re_b = \frac{\rho U_r d_b}{\mu}$

If:$U_r = 5\ \text{m/s} Re_b = \frac{1.2(5)(0.067)}{1.8\times10^{-5}} \approx 2.2 \times 10^4$

This is a high Reynolds number sphere outflow, and the drag coefficient is taken as an estimate.$C_D = 0.5$

2. rolling critical wind speed

If using rolling resistance:$F_{\text{resist}} = C_{rr} m_b g$

Critical wind speed:, if, then:, if then:$U_{r,\text{crit,roll}} = \sqrt{\frac{2C_{rr} m_b g}{\rho C_D A_b}} C_{rr} = 0.02 \boxed{U_{r,\text{crit,roll}} = \sqrt{\frac{2(0.02)(0.057)(9.81)}{1.2(0.5)(3.53\times10^{-3})}} \approx 3.25\ \text{m/s}} C_{rr} = 0.01 \boxed{U_{r,\text{crit,roll}} \approx 2.30\ \text{m/s}}$

Conclusion: The tennis ball rolls freely and about a wall jet is possible to make it move.$57\ g 2\sim4\ \text{m/s}$

2.17 Battery, power and endurance analysis

1. ideal power

Ideal for empty machines induced power$P_i = 49.9\ W$

After catching the ball:$P_i^+ = 56.3\ W$

Actual electric power:, where, if:, empty aircraft hovering electric power:$P_{\text{elec}} = \frac{P_i}{\eta_{\text{total}}} + P_{\text{control}} + P_{\text{servo}} + P_{\text{loss}} \eta_{\text{total}} = \eta_{\text{prop}} \eta_{\text{motor}} \eta_{\text{ESC}} \eta_{\text{battery}} \eta_{\text{total}} = 0.45 \sim 0.60$ $\boxed{P_{\text{elec}} \approx \frac{49.9}{0.60} \sim \frac{49.9}{0.45} \approx 83 \sim 111\ W}$

Electric power for hovering after catching the ball: Taking into account downwash obstruction, ground effect disturbance, control correction, and gripper servo, the average mission power can be estimated:$\boxed{P_{\text{elec}}^+ \approx \frac{56.3}{0.60} \sim \frac{56.3}{0.45} \approx 94 \sim 125\ W}$ $\boxed{P_{\text{avg}} \approx 105 \sim 155\ W}$

2. Measured voltage and current integral

Actual measurement:, Total task energy consumption:, Discretization:, Average power:$P(t) = V(t) I(t) E = \int_0^t P(t) dt$ $\boxed{E \approx \sum_{k=1}^{N} V_k I_k \Delta t}$ $\boxed{P_{\text{avg}} = \frac{1}{t} \sum_{k=1}^{N} V_k I_k \Delta t}$

3. Battery power and flight time

The battery is a series LiPo:, capacity:$N_s V_{\text{nom}} = 3.7 N_s C$ Ah

Nominal Energy:, Consider not fully discharging, Available Energy:, Flight Time:$E_{\text{nom}} = V_{\text{nom}} C \quad [\text{Wh}] DOD = 0.8 E_{\text{use}} = DOD \cdot V_{\text{nom}} C$ $\boxed{t_{\text{min}} = 60 \frac{DOD \cdot V_{\text{nom}} C}{P_{\text{avg}}}}$

4. actual calculation

The battery we use is therefore: , the nominal voltage is used to estimate energy here; although the full-charge voltage of 4S LiPo is approximately, it is not appropriate to use the full-charge voltage to overestimate the battery life and energy margin analysis. The nominal energy is: If the available discharge ratio is:, then the available energy is: If:, then:2700 mAh, $4S$, $14.8\ V$, $35C N_s = 4,\ V_{\text{nom}} = 14.8\ \text{V},\ C_{\text{batt}} =$ $2700\ \text{mAh} = 2.7\ \text{Ah} 14.8\ V 16.8\ V$

$$\boxed{E_{\text{nom}} = V_{\text{nom}} C_{\text{batt}} = 14.8(2.7) = 39.96\ \text{Wh}}$$

$$DOD = 0.8$$
$$E_{\text{use}} = 0.8(14.8)(2.7) = 31.97\ \text{Wh}$$
$$P_{\text{avg}} = 110\ W t_{\min} = 60 \frac{31.97}{110} = 17.4\ \text{min}$$

If:, then$P_{\text{avg}} = 155\ W t_{\min} = 60 \frac{31.97}{155} = 12.4\ \text{min}$

If: , then: Therefore, the actual battery used has significantly higher available energy. Even if the average task power increases to , the theoretical battery life can still exceed the 5-minute test time.$P_{\text{avg}} = 180\ W t_{\min} = 60 \frac{31.97}{180} =$ $10.7\ \text{min}$

$$4S\ 2700\ \text{mAh} 155 \sim 180\ W$$

5. Five-minute task required energy test time:, task energy consumption

$$t = 5\ \text{min} = \frac{5}{60}\ h E_{5\min} = P_{\text{avg}} \frac{5}{60}$$

If:, then:, the available energy of the actually used battery is, then the remaining energy margin is:, margin ratio: If the average power is increased to:, then the 5-minute task consumption energy is:, the remaining energy margin is:, margin ratio: Therefore, from the perspective of energy capacity, the actually used battery is enough to support the 5-minute mid-term ball fetching task.$P_{\text{avg}} = 155\ W E_{5\min} = 155 \frac{5}{60} = 12.9\ \text{Wh} 31.97\ \text{Wh} 31.97 -$ $12.9 = 19.07\ \text{Wh} \frac{19.07}{31.97} = 59.7\%$

$$P_{\text{avg}} = 180\ W E_{5\min} = 180 \frac{5}{60} = 15.0\ \text{Wh} 31.97 - 15.0 = 16.97\ \text{Wh} \frac{16.97}{31.97} = 53.1\%$$

$4S\ 2700\ \text{mAh}$

6. Battery current and C-rate average current:, based on the actual battery nominal voltage:, if the average task power range is:, then the average current is approximately: If replaced by the representative value:, then: The battery capacity is:, so the average task discharge rate is:

$$I_{\text{avg}} = \frac{P_{\text{avg}}}{V_{\text{nom}}} V_{\text{nom}} = 14.8\ V P_{\text{avg}} = 105{\sim}155\ W I_{\text{avg}} = \frac{105}{14.8} \sim \frac{155}{14.8}$$

$$\approx 7.1{\sim}10.5\ A P_{\text{avg}} = 125\ W I_{\text{avg}} = \frac{125}{14.8} = 8.45\ A$$

$$C_{\text{batt}} = 2.7\ \text{Ah} C_{\text{rate}} = \frac{I_{\text{avg}}}{C_{\text{batt}}}$$

Taking for example:, if the task average power range is used, then:, if the average power is increased to:, the actual nominal battery discharge rate is, so the theoretical maximum continuous discharge current is: The corresponding nominal maximum discharge power is about: However, it is the nominal upper limit of the battery's discharge capacity, which does not mean that the system should operate at this current for a long time. The actual available current will still be limited by the battery's internal resistance, connectors, wires, ESC, motor temperature rise and heat dissipation conditions. The internal resistance of the battery causes voltage drop: internal resistance loss. If the current is too large, it will cause: voltage sag, reduction in motor thrust, shortened battery life, and unstable control. $P_{\text{avg}} = 125\ W C_{\text{rate}} = \frac{8.45}{2.7} = 3.13 C 105{\sim}155\ W C_{\text{rate}} = \frac{7.1}{2.7} \sim \frac{10.5}{2.7} \approx 2.6C{\sim}3.9C 180\ W I_{\text{avg}} = \frac{180}{14.8} = 12.16\ A,\ \ C_{\text{rate}} = \frac{12.16}{2.7} = 4.50C 35C I_{\text{max,cont}} = 35 C_{\text{batt}} = 35(2.7) = 94.5\ A$

$$P_{\text{max,nom}} = V_{\text{nom}} I_{\text{max,cont}} = 14.8(94.5) \approx 1.40 \times 10^3\ W$$

$$35C$$

$$V = V_{OC}(SOC) - I R_{\text{int}} P_{\text{loss}} = I^2 R_{\text{int}}$$

2.18 Summary of kinetic analysis

This system is a five-inch propeller X-type four-rotor drone, equipped with a rectangular parallelepiped gripper body below. The center distance between adjacent motors is, the roll/pitch effective force arm is, and the total propeller disc area of the four propellers is. Using the right-hand body coordinate system, define the axis pointing to the front of the body, the axis pointing to the left of the body, and the axis pointing above the body,

so.169.998 mm84.999 mm0.05067 $m^2 xyz\boldsymbol{e}_x \times \boldsymbol{e}_y = \boldsymbol{e}_z$

A rigid body model is established based on the measured mass of the mechanism: the central stacked structure containing the battery is , the X-shaped frame is , and the rectangular parallelepiped gripper body below is , so the empty machine mass is . The center stack structure plus battery is regarded as the upper module of the equivalent total height, so its center of mass is located at half height, and the equivalent center of mass height of the center stack is. After the weighting of the three structures, the height of the center of gravity of the empty machine is, which means that the center of gravity of the system is almost on the X-shaped frame plane and slightly lower than this plane; after grabbing the tennis ball, the total mass increases to, if the tennis ball is located, the new center of gravity is approximately.$m_u = 0.3495$ kg$m_X = 0.0400$ kg$m_c = 0.2952$ kg$m_d = 0.6847$ kg$2H_u z_u = H_u = 0.0380$ $m z_G = -1.20$ mm0.057 kg0.7417 kg$z_b = -0.110$ $m z_G^+ = -9.56$ mm

The total thrust requirement for hovering when empty is, and the average thrust required by a single rotor is; after grabbing a tennis ball, the total thrust requirement for hovering is, and the hovering thrust requirement for a single rotor increases to. If the ball retrieval, platform rebound flow and attitude control margin are taken into account, the maximum thrust of a single rotor should be reached when empty and after catching the ball, in order to maintain a thrust-to-weight ratio.6.717 N171 gf7.276 N185 gf257~342 gf278~371 gf1.5~2.0

From actuator disk theory, it can be estimated that the induced velocity of empty machine hovering is, the far downstream downwash speed is about, and the ideal hovering power is; after catching the ball, the induced velocity increases to, and the ideal power increases to, which increases to about. The actual electric power needs to take into account the propeller, motor, ESC, battery internal resistance, the obstruction of the lower car body and the power consumption of the steering gear. The estimated average mission power is approximately.7.43 m/s14.86 m/s49.9 W7.74 m/s56.3 W12.7%105~155 W

The thrust is also verified using the finite control volume method, measured at the downwash plane, and the momentum flux is estimated, which is used to include the contribution of turbulent velocity fluctuations to the thrust. Compared with the single-point average wind speed, this method can reflect the downwash non-uniformity and turbulence fluctuation, and can further calculate the thrust center offset to estimate the roll/pitch disturbance moment caused by the asymmetry of the downwash.$w(x,y,z,t)T_{\mathrm{CV}} = \rho \sum(\bar{w}_i^2 + \sigma_{w,i}^2)\, \Delta A_i \sigma_{w,i}^2 (x_T, y_T)$

In terms of rigid body dynamics, the system is divided into three main rigid bodies: the X-shaped frame and rotor units, the center stack structure, and the lower cuboid gripper body. The overall inertia tensor is the sum of the inertia tensors of the three relative to the center of gravity of the entire machine: after catching the ball, the tennis ball inertia tensor needs to be added: under the estimation of the equivalent horizontal size and equivalent vertical height of the center stack, we can get,,. Rotational motion can be described by the Euler equation. If the total thrust difference between the left and right is, it can result in a roll angular acceleration of approximately, showing that the local thrust difference caused by downwash, ground effect and platform rebound will directly affect attitude shake.$\boldsymbol{I}_G = \boldsymbol{I}_{X,G} + \boldsymbol{I}_{u,G} + \boldsymbol{I}_{c,G}.\boldsymbol{I}_{G^+} = \boldsymbol{I}_{X,G^+} + \boldsymbol{I}_{u,G^+} + \boldsymbol{I}_{c,G^+} + \boldsymbol{I}_{b,G^+}.L_u = B_u = 80\ \mathrm{mm}H_{u,\mathrm{eff}} = 2H_u = 76\ \mathrm{mm}I_{xx} = 2.27 \times 10^{-3} I_{yy} = 2.19 \times 10^{-3} I_{zz} = 1.32 \times 10^{-3}\ \mathrm{kg} \cdot m^2 \boldsymbol{I}_G \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\boldsymbol{I}_G \boldsymbol{\omega}) = \boldsymbol{M} 0.2\ N 7.48\ \mathrm{rad/s}^2$

When picking up the ball, the height from the paddle to the platform is about 1,000 feet, and the average ground effect gain is about 10,000 feet. However, what really causes instability is not the average thrust gain, but the asymmetry of the ground effect, the transformation of impinging jet into radial wall jet, the effect of lateral airflow on the lower body, and the contact moment caused by the contact between the clamping jaws. The horizontal projected area of the lower car body is approximately 1,000 square meters, accounting for the total propeller disc area of the four propellers, so it may cause downwash obstruction and additional aerodynamic load. If the platform wall jet blows to the side of the car body below, it can cause a disturbance torque of approximately; if it rises to 100, the required motor differential compensation can reach approximately.$h/R_p \approx 1.5 2.8\% 0.00741\ m^2 14.6\% U_r = 5\ \mathrm{m/s} 0.00707\ N \cdot m U_r 10\ \mathrm{m/s} 8.5\ \mathrm{gf}$

The main mechanism of the tennis ball being blown is that the downwash flow hits the platform and forms a horizontal wall jet, which creates resistance to the ball. For a tennis ball with mass and diameter of approximately, if estimated by rolling resistance, the critical wind speed is approximately. Therefore, if wall jet reaches near the platform, it may still cause displacement of several centimeters to ten centimeters in a short period of time. $F_D = \frac{1}{2}\rho C_D A_b U_r^2$ 0.057 kg67 mm2~4 m/s5~10 m/s

2.19 Material experiments, structural material selection and verification

The foregoing analysis is mainly based on system-level models such as thrust demand, center of gravity position, moment of inertia, downwash and attitude disturbance, etc., to illustrate the main mechanical limitations faced by this system in flight and ball retrieval tasks. However, in actual engineering systems, theoretically having sufficient thrust and control margin does not mean that the system can complete the task stably, because the rigidity, strength, vibration resistance and failure mode of the structural material itself will directly affect the flight control sensing quality, attitude stability and reliability during repeated tests. Therefore, we further proceed from material experiments to verify the judgment of the material configuration of the arm and fuselage.

In the system, the arm is the structural component that most directly bears the motor thrust, propeller vibration, landing impact and crash impact. If the arm rigidity is insufficient, periodic vibrations during motor operation will cause elastic deformation of the arm, which will be further transmitted to the flight control board, causing the IMU to measure unreal attitude disturbances, causing the controller to produce unnecessary correction outputs. If the strength of the machine arm is insufficient, the fixed end will easily bend and break during landing or collision, causing the overall system to fail. Therefore, the selection of machine arm materials cannot only be based on weight minimization, but must also consider rigidity, strength, processing speed, cost and maintainability.

1. Experimental purpose and basic assumptions

   Compare two candidate arm materials: 5 mm 3D printing PLA and 5.5 mm dense board MDF. The purpose of the experiment is to quantify the deformation and damage load-bearing capacity of the two under approximate arm stress conditions, and to serve as a basis for selecting the arm material.

This experiment uses the following assumptions:

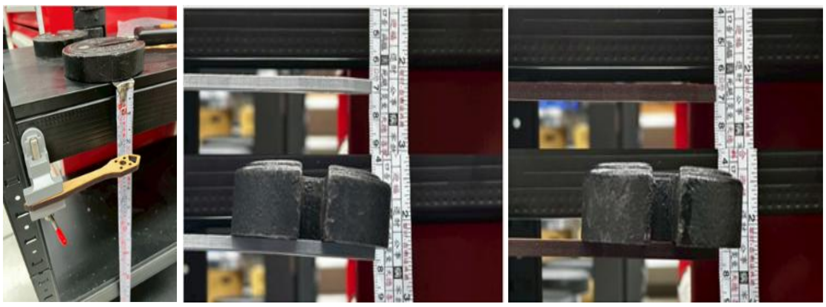
The machine arm can be approximated as a cantilever beam with one end fixed and one end stressed.

The main effect of motor thrust and landing impact on the arm is bending deformation, so endpoint concentrated force is used as the main test mode for material comparison.

The same group of material test pieces are consistent in geometric size, fixation method and stress position, making material differences the main comparison variable.

2. Experimental methods

Material experiments are divided into two groups. The first group is a small load deformation test. A 100 g weight is placed at the end of the machine arm to simulate elastic deformation caused by motor thrust or small external force, and measure the end point displacement. The second group is a damage test. A tensile gauge is used to gradually increase the downward pulling force on the



tail end of the machine arm, and the deformation amount and fracture mode at the moment of damage are observed with slow-motion video. The first set of experiments mainly reflects the material rigidity, and the second set of experiments mainly reflects the material strength and failure mode.

Figure 47. Arm load-bearing experimental setup and actual results

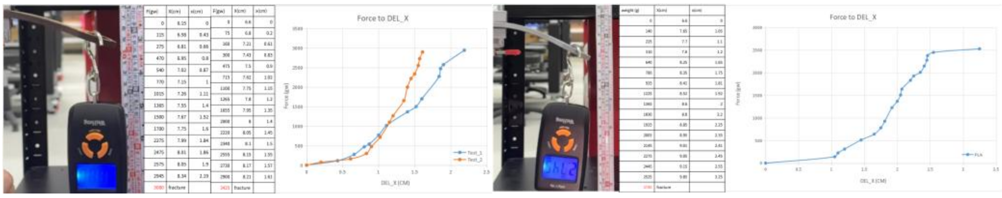
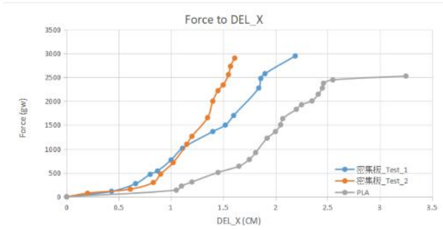


Figure 48: Actual results of the machine arm tension test



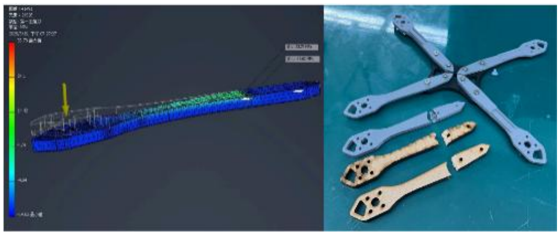


Figure 49: Simulated and actual broken machine arm

The results obtained from the experiment are as follows:

Table 14. Comparison of experimental results between MDF and PLA arm materials

Material	**100 g load deformation (cm)**	**Equivalent end point rigidity (N/m)**	**Force at break (gw)**	**Force at break (N)**	**Deformation amount at break (cm)**	**Single weight (g)**

5.5 mm MDF	0.32	About 307	3080–3425	30.2–33.6	1.61–2.19	7
5 mm 3D printing PLA	0.48	About 204	2745	26.9	3.25	6

The stiffness ratio between the two is, that is, under the same external force, the bending stiffness of the MDF arm is about 1.5 times that of the PLA arm. If compared with the breaking load, the breaking load of MDF is 3080-3425 gw, while that of PLA is about 2745 gw. As an average estimate, the failure-bearing capacity of MDF is approximately 18-20% higher than that of PLA. Although the weight of a single MDF arm is about 7 g, 1 g more than PLA, the total weight of the four arms only increases by about 4 g, accounting for about 684.7 g of the empty machine mass. This weight cost is far lower than the increase in rigidity and strength it brings. Therefore, in terms of flight stability and structural safety, MDF is a more reasonable choice of arm material. $\frac{k_{\mathrm{MDF}}}{k_{\mathrm{PLA}}} = \frac{307}{204} \approx 1.50$ $\frac{4}{684.7} \times 100\% \approx 0.58\%$

3. Effect of material selection on flight stability

The rigidity of the arm material not only affects whether the structure is broken, but also affects the quality of flight control. For a machine arm that can be approximated as an elastic structure, its first-order natural frequency can be expressed as, where is the equivalent rigidity and is the equivalent vibration mass. Assuming that the equivalent mass is approximately the same, since the rigidity of MDF is approximately 1.5 times that of PLA, its natural frequency can be increased by approximately 22.5%. Even considering the slightly higher weight of MDF itself, based on the weight of a single arm, MDF still has a natural frequency advantage of about 13%. This means that the MDF arm is less likely to produce large-amplitude resonance near the operating frequencies of the motor and propeller, which helps reduce high-frequency vibration and false attitude errors measured by the flight control IMU.$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m_{\text{eff}}}} k m_{\text{eff}} \sqrt{1.5} \approx 1.225 k/m$

In addition, the fuselage of this set does not adopt an integrated rigid structure, but uses multi-layer plates to clamp the arms. This structure can provide a certain degree of friction damping at the connection between the arm and the fuselage, so that part of the vibration energy is converted into friction heat at the contact surface of the parts, instead of being completely transmitted to the fuselage and flight control board. In other words, the high rigidity of the arm material and the friction damping at the fuselage connection jointly form a structure that "increases the natural frequency and reduces vibration transmission."

2.20 Finite element analysis and practical experiments

1. Finite element analysis of the impact of machine arm falling

In actual test flights, the most common failure mode of the arm occurred near the fixed end. This is because the machine arm is similar to a cantilever beam. When external force acts on the motor end or the machine arm end, the maximum bending moment will appear at the root fixed end. Therefore, taking the vertical concentrated force at the end of the machine arm as an extreme situation, Inventor was used to conduct finite element analysis to observe the relative improvement effects of stress concentration locations and different structural designs.

The analysis uses the following assumptions:

The total mass of the unmanned aerial vehicle is$m_d = 0.6847$ kg

The drone falls freely from a height of 2 m, and the instantaneous speed when it touches the ground is estimated by the free fall formula.$v = \sqrt{2gh} = \sqrt{2(9.81)(2)} = 6.264\ \text{m/s}$

Assuming that the velocity decelerates from to 0 during the impact, the average impact force is estimated by the momentum theorem as$\Delta t =$

$$0.01\ svF = \frac{m\Delta v}{\Delta t} = \frac{0.6847(6.264)}{0.01} \approx 428.9\ \mathrm{N}$$

Therefore, in the finite element analysis, a vertical force of about 430 N acts on the tail end of the machine arm, and the fixed end of the machine arm is regarded as a fixed boundary condition. Although this scenario is extreme, it can be used to determine the location of stress concentrations under the most adverse conditions and the relative improvement effects of different reinforcement designs.

Table 15. Comparison of finite element analysis results between the original version and the enhanced version of the machine arm

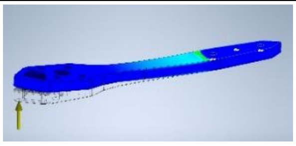
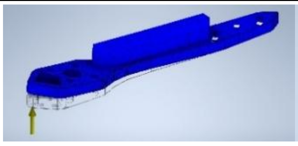
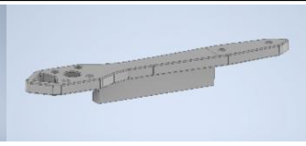
	 
First version of the machine arm	New version of machine arm

Table 16. Comparison of finite element analysis numerical values between the original and enhanced versions of the machine arm.

version	boundary conditions	Endpoint force	maximum equivalent stress	main stress concentration location	relative improvement
Original MDF flat arm	Fixed end fully constrained	430 N	1738.37 MPa	Near the fixed end of the machine arm	None
Spinal type reinforced machine arm	Fixed end fully constrained	430 N	973.29 MPa	The junction between fixed end and reinforcement	Approximately 44.0% reduction

The new version of the reinforcement design adds a vertical 5.5 mm MDF board under the arm, turning the originally thin plate-like arm section into a T-shaped or spine-type section. This design can increase the second moment of section in the vertical direction and reduce the bending normal stress. From the finite element analysis, it can be seen that the maximum stress after reinforcement is reduced from 1738.37 MPa to 973.29 MPa, with a decrease of$I\sigma = \frac{Mc}{I}\frac{1738.37-973.29}{1738.37} \times 100\% \approx 44.0\%$

2. Impact test of sub-car floor connecting parts

The lower ball-retrieval car and the valve connector are also structures that are prone to failure of this system when the platform lands. Since sub-cars are mostly made of 3D printed PLA, their strength is affected by the printing direction, interlayer adhesion, filling rate, holes and local stress concentration, making it difficult to accurately predict only by a homogeneous material model. Therefore, this group conducted drop experiments on the connectors that are actually most likely to break, directly simulating the ground impact when the platform lands.

The experimental method is as follows:

Release the drone and sub-vehicle from different heights in an approximately vertical manner so that the sub-vehicle connector can withstand the impact.

The operating variable is the drop height, and the test heights include 5 cm, 7.5 cm, 10 cm, 11 cm, 12 cm, 13 cm, 14 cm and 15 cm.

Each height is evaluated based on whether the connector can withstand multiple falling impacts, and the number of times when it breaks is recorded.

Table 17. Drop test results of sub-car connectors

Drop height (cm)$\boldsymbol{h}$	**Terminal velocity of free fall (m/s)$\boldsymbol{v = \sqrt{2gh}}$**	**Pre-impact potential energy (J)$\boldsymbol{E = m_d gh}$**	**Experimental results**

5	0.99	0.34	No damage
7.5	1.21	0.50	No damage
10	1.40	0.67	9th damage
11	1.47	0.74	No damage
12	1.53	0.81	9th damage
13	1.60	0.87	5th damage
14	1.66	0.94	5th damage
15	1.72	1.01	6th damage

It can be seen from the results that 5 cm and 7.5 cm falling impacts have little impact on the connector; when the height reaches more than 10 cm, even if the connector does not necessarily break at the first impact, fatigue accumulation and brittle fracture will occur after multiple impacts. The landing speed corresponding to a drop height of 10 cm is. This speed can be regarded as the critical speed level at which the sub-car connector begins to have obvious risk of failure. In order to avoid damage to the sub-car or connecting parts due to high-speed descent during the final test, the team limited the maximum Z-axis ascending and descending speeds in the INAV firmware to approximately. If compared with kinetic energy, the kinetic energy of a 10 cm free falling object before impact is approximately, and when the limited descent speed is 0.50 m/s, the kinetic energy before landing is approximately. Comparing the two,. It means that after limiting the descent speed, the kinetic energy before landing is reduced to approximately 12.7% of the 10 cm free fall situation, which means that the impact energy is reduced by approximately 87.3%. If the impact force is roughly estimated with the same contact time, a free fall of 10 cm corresponds to an average impact force of approximately, and a falling speed of 0.50 m/s corresponds to an average impact force of approximately. Therefore, the Z-axis speed limit in the firmware is not simply an adjustment of the operating feel, but a protective design parameter based on structural impact experiments.$v = \sqrt{2(9.81)(0.10)} \approx 1.401\ \mathrm{m/s} = 140.1\ \mathrm{cm/s} v_z = 50\ \mathrm{cm/s} = 0.50\ \mathrm{m/s} E_{10\mathrm{cm}} = m_d g h = 0.6847(9.81)(0.10) \approx$

$0.672\ \mathrm{J} E_{0.50} = \frac{1}{2} m_d v^2 = \frac{1}{2}(0.6847)(0.50)^2 \approx 0.0856\ \mathrm{J} \frac{E_{0.50}}{E_{10\mathrm{cm}}} = \frac{0.0856}{0.672} \approx$

$0.127 \Delta t = 0.01\ \mathrm{s} F_{10\mathrm{cm}} = \frac{0.6847(1.401)}{0.01} \approx 96.0\ \mathrm{N} F_{0.50} = \frac{0.6847(0.50)}{0.01} \approx 34.2\ \mathrm{N}$

2.21 System dynamic experiments and crash cause records

After completing the basic functional verification, we started a large number of test flight plans, and carried out basic height setting and positioning verification in sequence; then tested the landing on the platform; and finally optimized the anti-interference ability of the system. The following will start from the system failure record and explain our preliminary discussion on the cause of the incident. Subsequent chapters will cover solutions to more complex problems:

1. Height setting and positioning verification stage
   (1) Lateral dynamic hypersensitivity and no-input drift: the body moves sideways too fast, and even small operations hit the boundary of the field; in addition, the body continues to drift to the left and front when the joystick is centered. Lateral allergies are caused by the high thrust-to-weight ratio of the system and the preset RC Rate being too large. No input drift may be caused by the superposition of two factors:
      a. There is a hardware offset in the neutral value of the remote control channel and it is not accurately aligned with the standard value of 1500
      b. The ESP32 sub-car module or battery configuration is asymmetrical, causing the center of gravity to shift and produce a fixed bias moment.

   Improvement plan: Recalibrate the neutral values of the flight control level and remote control channel, and add Deadband settings; also adjust the body weight to ensure that the physical center of gravity coincides with the thrust geometric center.
2. Platform landing function test phase
   (1) Propulsion close to the ground and loss of control after overturning after retrieving the ball:
      a. When the propeller power was used for ground-to-ground propulsion testing on the platform, once the sub-wheel set hooked

onto the edge of the platform, the body instantly overturned forward.

b. If the ball is not completely retracted into the fuselage, the center of gravity will shift and greater rotational torque will be generated during maneuvers, making operation difficult.

Improvement plan: Adopt a "dynamic and static separation" operation strategy. After the aircraft reaches the platform, the rotor power is turned off, and the ESP32 drives the sub-car to move in the plane. This fundamentally eliminates the risk of overturning caused by the thrust vector, and accurately collects the ball into the aircraft body when picking up the ball.

(2) High-speed landing caused subsystem structural damage

When the platform was being landed, the sub-car structure was broken and damaged due to the excessive descent terminal speed.

The vertical kinetic energy during landing is released in a very short contact time, producing a huge impact force. 3D printed PLA parts have low impact toughness, and brittle fracture occurs when the instantaneous stress exceeds the yield strength of the material.

Improvement plan: In addition to optimizing the throttle curve, redesign the connection between the sub-car and the frame, increase the stress area and introduce a stress-relief fillet design.

3. Anti-interference test stage

(1) Improper propeller matching causes low-altitude oscillations

Although the original 5045 three-blade propeller was used to fly, it would vibrate violently at low altitudes and would deflect diagonally to the rear when landing. After inquiries, we learned that the jitter and deflection of three-blade propellers at low altitudes are typical ground effects, so we purchased 5040 and 5045 four-blade propellers for experiments. We found that the 5040 did have worse response than the 5045 as expected, but provided smoother flight control.

Improvement plan: Use 5040 four-blade propellers to ensure stability when landing on the platform; adopt a quick power cutoff strategy for low-altitude landings to avoid prolonged hovering in the ground effect area.

2.22 System altitude-fixed flight optimization

In the above flight test, we found that the most serious problem is the difficulty in controlling the Z-axis (flying height), because it not only increases the time and difficulty it takes to land on the platform, but also often causes the fuselage to be damaged due to high-speed descent, so we will focus on solving this problem before the mid-term:

1. Essential limitations of Betaflight architecture

   To address the above issues, we're trying to enable height accessibility in Betaflight. Although some versions support basic height restriction mode, search data found that the effect is not as good as other firmwares (INAV, ArduPilot), and there is a height drop when performing Roll/Pitch lateral movement.

   Betaflight's core design philosophy was developed for FPV traversing and fancy flying applications, emphasizing the instant response characteristics of "absolutely obeying the pilot's original input". Under this architecture, Z-axis control has a lower priority: system resources are mainly allocated to high-frequency operations of the attitude loop. Even if the barometer is enabled, its weight in the control loop is deliberately lowered to avoid interfering with the pilot's direct control intention. The Z-axis controller lacks sufficient integral gain to eliminate steady-state errors and cannot effectively combat the disturbance caused by the high inertia characteristics of this system.

2. INAV firmware migration and its barometer closed loop

   After recognizing the problems with Betaflight, the team believed that the architectural limitations of Betaflight were not suitable for altitude determination tasks, and decided to migrate to INAV firmware designed for autonomous navigation.

   According to the queried data, in the NAV ALTHOLD (altitude hold) mode of INAV, the barometer is no longer just a reference sensor, but the main feedback signal of the Z-axis PID closed loop. The controller architecture can be conceptualized as: $\boldsymbol{\delta_{throttle} = \delta_{hover} + K_p(h_{target} - h_{baro}) +}$

$$K_i\int (h_{target} - h_{baro})\, dt + K_d \frac{d(h_{baro})}{dt}$$

where is the air pressure measurement altitude, and is the hovering throttle reference value automatically learned by the system.$h_{baro}\delta_{hover}$

3. Quantitative analysis of INAV barometer altitude introduction and hover stability

After the firmware conversion, we made key dynamics matching settings: through the measured total body weight and thrust curve, the INAV's baseline hover throttle was accurately set at 50%. This setting makes the neutral point of the remote control exactly correspond to the static equilibrium point of the system, giving the altitude control loop a symmetrical control margin in the climbing and descending directions, effectively reducing the burden of the cumulative error of the integral term.

We use image analysis (Tracker) to quantify hover stability. A reference grid with a 5-centimeter spacing scale was set on the background of the test site, and a fixed 60fps camera was used to record the hovering process of the drone. Afterwards, the displacement history of the body relative to the target height within the test interval is captured. To ensure data reliability, each set of conditions was tested three times independently, with 10 seconds of steady-state hovering data captured each time, and the root mean square error was used as a quantitative indicator of stability.

Table 18. Betaflight pure manual throttle control table and INAV barometer closed-loop height setting

Number of tests	**maximum height**	**minimum height**	**Number of tests**		**maximum height**	**minimum height**	**peak to peak**	**RMSE**
1	52.8	47.5	1	5.3	63.5 / 1.42	34.2	29.3	8.74
2	53.2	48.1	2	5.1	61.8 / 1.38	36.5	25.3	7.92
3	52.5	47.8	3	4.7	65.2 / 1.26	32.8	32.4	9.31
average	52.8	47.8	average	5.0	63.5 / 1.35	34.5	29.0	8.66

It can be seen from the data that the peak-to-peak height drift under manual control is close to 30 centimeters, which is an unacceptable error range for tennis ball picking tasks that require precise alignment. After introducing the barometer height determination, the Z-axis RMSE dropped from 8.66 cm to 1.35 cm. In addition, the data variability of the three tests of the experimental group (standard deviation 0.08 cm) is much lower than that of the control group (standard deviation 0.70 cm), showing that closed-loop control not only improves the absolute accuracy, but also greatly improves the repeatability and predictability of the system.

The remaining displacement in the experimental group (approximately ±2.5 cm) may come from two main factors:

(1) Sensor hardware limitations: measurement noise and resolution limitations of the barometer itself

(2) Aerodynamic disturbance: the pressure field disturbance produced by the propeller downwash as it approaches the ground

These residual errors are within the acceptable range of the mission, confirming that the decision-making of firmware conversion and hover throttle matching effectively provides the high degree of stability required for precise pick-and-place missions across altitudes.

2.23 Task execution time evaluation and trajectory optimization strategy

We also used the standard assessment at the end of the term in the test and found that the average process of picking up a ball takes about 27 seconds (B → A: 8.5 seconds; picking up the ball: 10 seconds; A → B: 8.5 seconds). However, in order to achieve four perfect scores at the end of the semester, it is safer to keep the ball retrieval time below 23 seconds. Therefore, we propose the following plan as an upgrade direction after the mid-term.

1. Car function upgrade

   (1) Speed up the movement of the car: Use a high-speed servo motor to improve the linear movement of the car on the platform and shorten the ball-retrieval time by 10 seconds.

   (2) Accelerate ball retrieval: Optimize the guide ball geometry of the sub-car opening to achieve fault-tolerant grasping that locks on touch, eliminating the fine-tuning time required for fine alignment

   (3) Increase the number of balls that can be carried in one trip: Expand the internal cavity design of the sub-car and try to increase the carrying

capacity in a single trip to 2

2. Increase descending platform speed
   (1) Reduce the difficulty of precise landing: Relax the landing accuracy requirements of the X-Y axis of the drone, as long as it lands within the safe range of the platform
   (2) Improve hard landing tolerance: According to the impulse theorem, high-damping sponges and reinforced locking points are installed at the bottom of the machine, and the descent strategy is changed from slow descent to limited free fall to save time on the Z axis.
   (3) Optimize the trajectory approaching the platform: Abandon the traditional right-angle trajectory of "level flight, arrival and vertical landing" and adopt a smooth "parabolic glide path" to maintain system kinetic energy and touch down early.→

Part 3. Control logic and system reliability

3.1 Mission requirements and control design goals

The topic of this course, Aero-Carrier, requires the team to design an electromechanical integrated system that can complete cross-height carrying tasks. The system must be able to start from the starting point, go to a designated platform to pick up objects, and then place the objects on another platform. This task not only includes mechanism design and manufacturing, but also includes many engineering integration issues such as flight control, ground movement control, ball pick-and-place mechanism control, wireless communication and failure protection. The course syllabus also clearly states that the final report needs to present system design, production, electronic control, measurement, analysis, debugging, optimization and verification, etc. Therefore, this chapter takes "whether the control logic is reasonable" and "whether the system is stable and reliable" as the main axis of analysis.

This group of systems consists of two main control levels. The first part is the main four-axis flight system, which is responsible for transporting the vehicle to the target platform and maintaining a stable flight attitude. The second part is the auxiliary ball retrieval and dribbling subsystem, which is responsible for ground movement on the platform, catching the ball, and releasing the ball. Although the functions of the two are different, the control design concept is the same: the operator does not directly control each actuator, but inputs the task intention, and the system converts it into executable motor or servo motor commands through the controller, sensing feedback and safety logic.

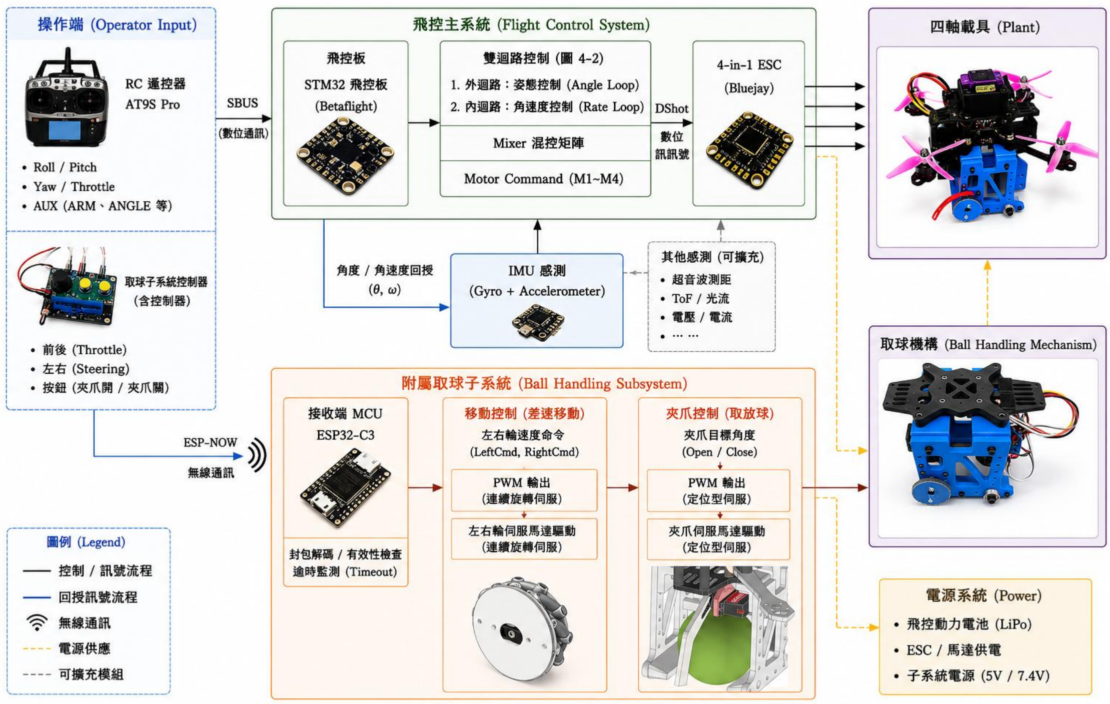


Figure 50: Overall system control architecture diagram

3.2 Overall control architecture and signal flow

The control architecture of this system can be divided into two signal streams: "flight control main system" and "auxiliary ground ball retrieval subsystem".

In the main flight control system, the operator inputs Roll, Pitch, Yaw, Throttle and AUX mode switching signals through the AT9S Pro remote controller. After the signals are received by the R9DS receiver, they are transmitted to the SpeedyBee F405 flight control board through SBUS digital communication. After the flight control board receives the operator's input, it does not directly regard it as the output command of a single motor, but converts it into the system target state, such as the desired attitude angle, desired angular velocity or throttle command. Then, the flight control board combines the actual attitude and angular velocity measured by the IMU, calculates the correction amount through the closed-loop controller, and finally distributes the control output to the four motors through the Mixer, and then transmits it to the ESC through the DShot digital signal to drive the brushless motor to generate thrust. This design allows the operator to input "flight intention" rather than directly controlling a single motor, which can greatly improve the stability and maneuverability of the four-axis vehicle.

The auxiliary ground ball retrieval subsystem adopts a dual microcontroller architecture. The sender uses ESP32, which is responsible for reading the two physical buttons and the HW-504 joystick input; the receiver uses ESP32-C3, which is responsible for driving two continuous rotation servo motors and a positioning gripper servo. ESP32 converts the joystick input into left and right wheel differential commands, and converts the button input into gripper opening or clamping angle, and then transmits it to ESP32-C3 through ESP-NOW. After ESP32-C3 receives the packet, it converts the left and right wheel commands into the servo motor PWM pulse width, and converts the gripper angle command into the positioning servo angle. This architecture separates the functions of the main flight system and the ball-retrieval subsystem, reducing the complexity of a single controller undertaking both flight and ball-retrieval tasks.

3.3 Closed-loop control logic of flight control main system

The four-axis flight system is a highly coupled and dynamically changing nonlinear system. If the operator directly controls the speed of each motor, it will be difficult to instantly compensate for body tilt, load deflection and external disturbances. Therefore, this group uses the closed-loop attitude control architecture built into the flight control board, uses the IMU to feed back the actual attitude, and calculates the motor corrections in real time through the PID controller.

In ANGLE mode, the operator's Roll and Pitch inputs mainly correspond to the desired attitude angle. When the operator releases the direction stick and expects the attitude angle to return to level, the flight control board will calculate the angle error based on the actual attitude measured by the IMU and actively return the aircraft to level. This mode can reduce the manual operation burden during initial flight, and is especially suitable for systems of this type that have bottom loading and carrying mission requirements.

The flight control architecture can be understood as hierarchical closed-loop control. The outer loop is an angle control loop, responsible for converting the error between the desired angle and the actual angle into the desired angular velocity; the inner loop is the angular velocity control loop, responsible for comparing the desired angular velocity with the actual angular velocity measured by the gyroscope, and generating the three-axis control corrections of Roll, Pitch, and Yaw through the PID controller. The inner loop handles angular velocity and has a fast reaction speed, so it is the core of flight stability; the outer loop is responsible for returning the body to the attitude expected by the operator, so that the system has better steady-state performance.

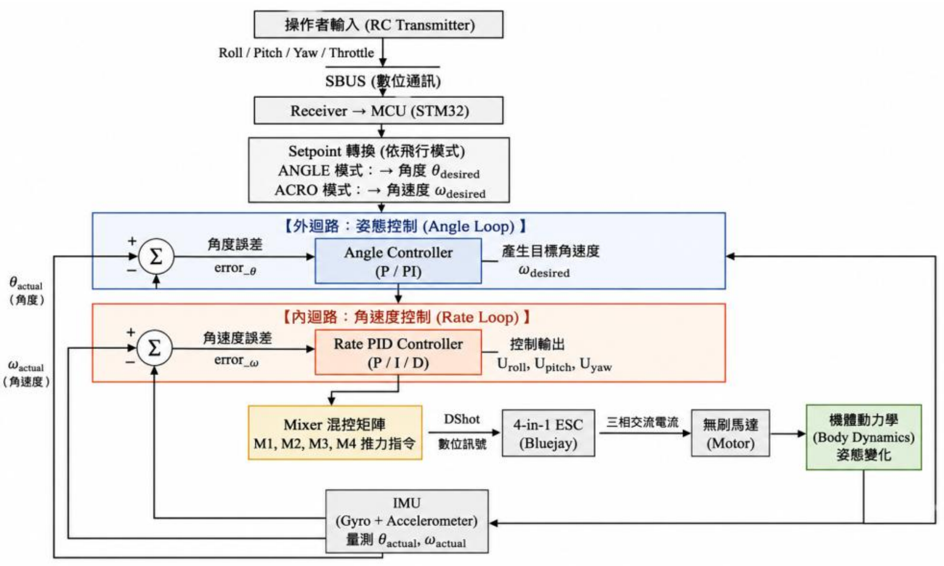


Figure 51. Flight control dual-loop control flow chart

Figure 4-2 shows that the flight control of this system adopts hierarchical dual-loop control (the simple process is RC input → Setpoint → Angle Loop → Rate Loop → PID Output → Mixer → ESC → Motor → Body Dynamics → IMU Feedback → Return to the controller). The operator inputs Roll, Pitch, Yaw and Throttle through the remote control. After being transmitted to the flight control MCU via SBUS, it is converted to the desired angle or desired angular velocity according to the flight mode. In ANGLE mode, the outer loop first calculates the angular error based on the desired attitude and the IMU feedback attitude, and generates the target angular velocity; the inner loop then calculates the angular velocity error based on the target angular velocity and the gyroscope measured angular velocity, and generates three-axis control output through Rate PID. The control output is converted into four motor thrust commands by Mixer and sent to ESC through DShot to drive the brushless motor to change the attitude of the body. Finally, the IMU continuously measures the actual angle and angular velocity and feeds them back to the controller, forming a complete closed-loop control.

This team discovered after the mid-term that although the aircraft body had sufficient thrust, it suffered from slow attitude response, low-frequency swing and drift after load integration. This means that the preset control parameters originally suitable for general lightweight models cannot fully adapt to the dynamic characteristics of this system of "high thrust, large inertia, and bottom mounting." Therefore, this team will not only adjust the parameters, but also re-examine the signal flow, sensor feedback and closed-loop control logic of the flight control to avoid relying solely on experience to adjust parameters and ignore the essence of the system.

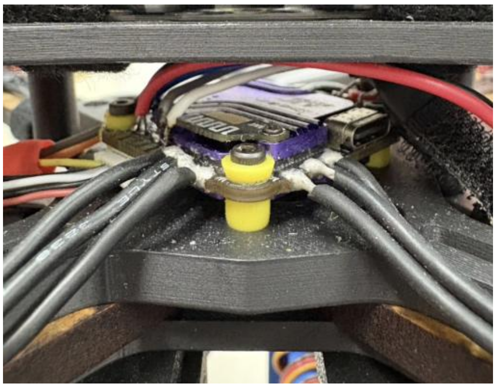


Figure 52: Flight control completed with motor welding

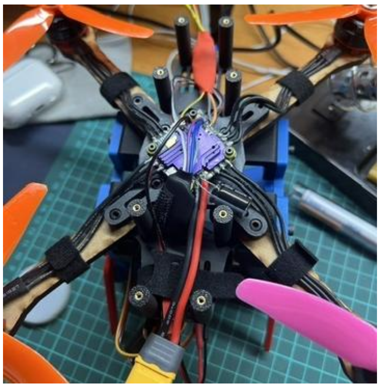


Figure 53: Overview of flight control and motor wiring

3.4 Affiliated ball-taking subsystem control logic

The task of the attached ball retrieval subsystem is to complete finer ground movement and ball retrieval actions on the platform. Since the main four-axis flight system is responsible for movement across heights, and the ball-retrieval task on the platform requires lower-speed, controllable, and close-to-the-ground operations, our team also designed an ESP32/ESP32-C3 subsystem to separate ball-retrieval control from flight control.

The subsystem sending end ESP32 reads the two-axis analog input of the HW-504 joystick, one of which serves as the forward/reverse command, and the other axis serves as the left and right steering command. After the program mixes throttle and steer, it calculates leftCmd and rightCmd, respectively, as the speed commands for the left and right continuous rotation servo motors. This control method is equivalent to differential drive. When the left and right wheel commands are the same, the vehicle will move forward or backward in a straight line; when the left and right wheel commands are different, the vehicle will turn. The two buttons correspond to the gripper release and clamping commands respectively, and switch the gripper servo to the preset angle.

After the receiving end ESP32-C3 obtains the ESP-NOW packet, it converts the left and right wheel commands into the PWM pulse width of the continuous rotation servo motor, and converts the gripper angle command into the PWM output of the positioning servo. In order to avoid the impact of the mechanism caused by the sudden and large movement of the clamping claw, the program also adopts the method of gradually updating the angle, so that the clamping claw servo can approach the target angle smoothly. In the initial design, the clamping jaw angle was determined as RELEASE_ANGLE = 55 and CLAMP_ANGLE = 10 after many tests to comply with the actual ball size and clamping jaw mechanism stroke.

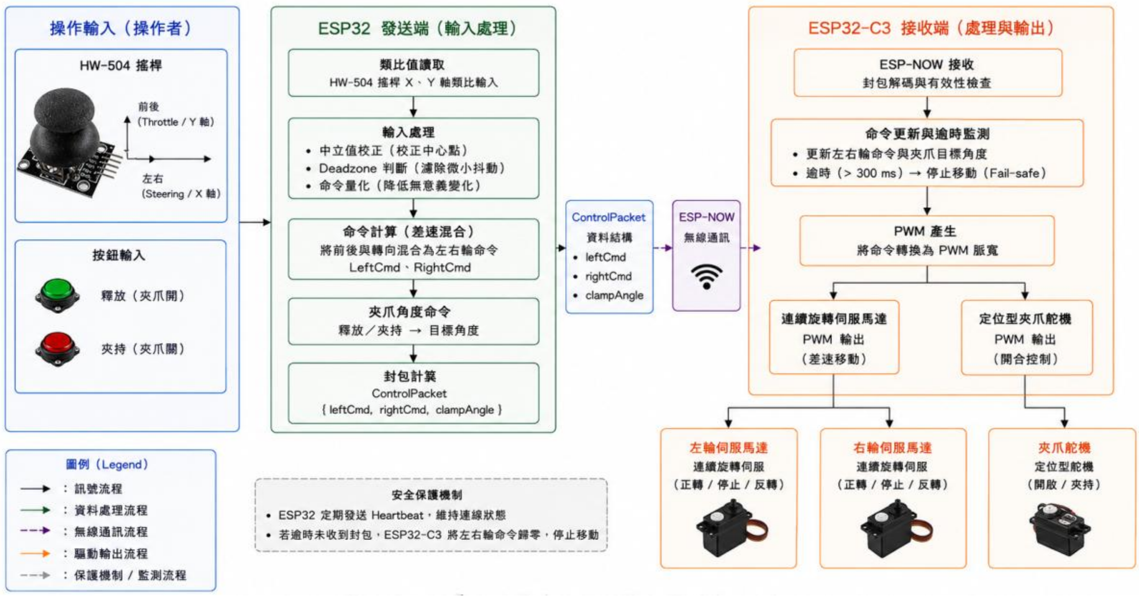


Figure 54. Control flow chart of the accessory ball retrieval subsystem

Figure 4-5 shows the control flow of the attached ball retrieval subsystem (the simple flow is Joystick / Buttons → ESP32 input processing → ControlPacket → ESP-NOW → ESP32-C3 → PWM → Wheel Servos / Clamp Servo). This subsystem is mainly responsible for low-speed movement on the platform, ball retrieval and ball release actions, and is separated from the main flight control system to reduce the complexity caused by integrating flight control and ball retrieval control into a single controller. The operator inputs the front and rear and left and right direction commands through the HW-504 joystick. The ESP32 transmitter first reads the analog value of the joystick. After neutral value correction, deadzone judgment and command quantification, the front and rear commands and steering commands are mixed into the left and right wheel differential control quantities. When the left and right wheel commands are the same, the vehicle can move forward or backward in a straight line; when the left and right wheel commands are different, the vehicle can turn. On the other hand, the operator can input the gripper opening or gripper closing command through two buttons, and the ESP32 converts this command into the gripper target angle. The above left and right wheel commands and gripper angle commands will be packaged into ControlPacket and sent to the ESP32-C3 receiver through ESP-NOW wireless communication. After receiving the packet, ESP32-C3 first performs data decoding and validity check, then converts the left and right wheel commands into PWM pulse width outputs of two continuous rotation servo motors, and at the same time converts the target angle of the gripper into a PWM control signal for the positioning servo. Since the wheel servo is responsible for the differential movement of the vehicle, and the gripper servo is responsible for fixing and releasing the ball, this subsystem can complete the process of moving, aligning, clamping and releasing the ball on the platform. In order to improve system reliability, our team added packet throttling and heartbeat mechanisms to the communication logic. The first version of the program uses a fixed period to continuously send packets, which causes analog joystick jitter, frequent Serial.print, and failure to wait for the completion of sending the previous packet, which together cause control delays. The subsequent modification is to only send a new packet when there is an obvious change in the command, and confirm the completion of the previous packet through send callback before sending the next packet; if there is no operation change in a short period of time, heartbeat will be sent periodically to let the receiving end confirm that the connection still exists. The receiving end sets a timeout protection. If no valid packet is received after the set time, the left and right wheel commands will be reset to zero, causing the vehicle to stop moving to avoid losing control due to the last movement command being maintained after the wireless communication is interrupted. Therefore, the control focus represented by Figure 4-3 is not the high-frequency attitude closed loop, but the stable conversion of operator input into servo motor output, and improved operation immediacy and safety through deadzone, quantification, packet validity check, heartbeat and timeout fail-safe. This design enables the accessory ball-retrieval subsystem to complete the ball-retrieval and dribbling tasks on the platform in a relatively low-speed, predictable and protective manner.

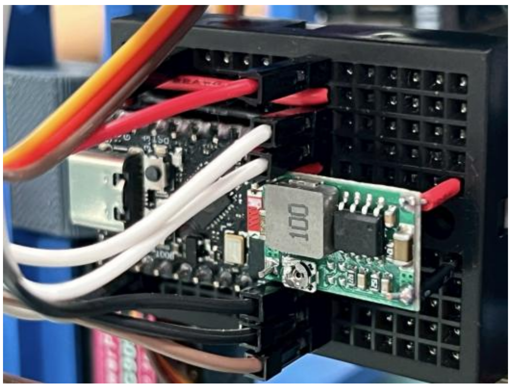


Figure 55. ESP32-C3 behind the subsystem

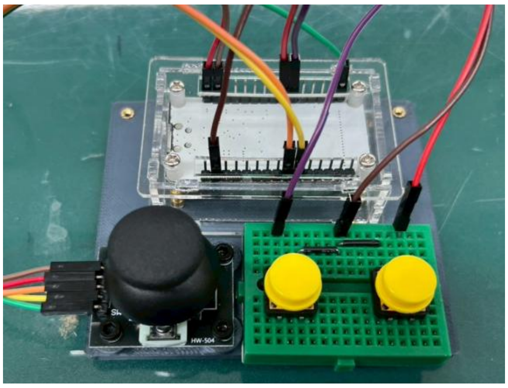


Figure 56: Ball retrieval subsystem controller

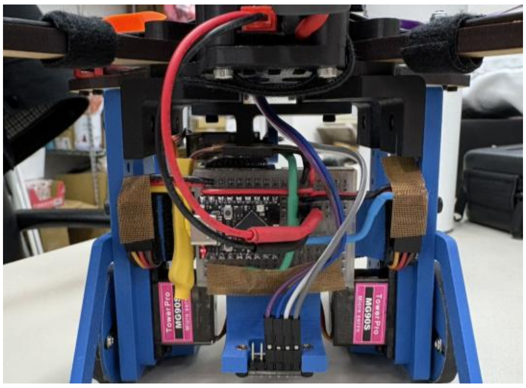


Figure 57: Overview of the wiring of the ball retrieval subsystem

3.5 System state management and State Machine

In order to make the overall operation process predictable, this group decomposes system tasks into several states. Although the main flight control system and the ESP32 subsystem are executed by different controllers, from the perspective of the overall task, the system can be regarded as a finite state machine. Switching between each state is triggered by operator instructions, communication status, platform contact status and task requirements.

Table 19. System task state transition and security control design table

Status	Trigger condition	Main control output	Reliability and safety design
INIT initializatio	System power on	Flight control startup, receiver connection,	Do not take off or retrieve the ball

Status	Trigger condition	Main control output	Reliability and safety design
n		ESP32/ESP32-C3 initialization, servo reset	before initialization is completed
STANDBY Standby	The remote control is connected normally, but not ARM yet	The motor is locked and the jaws maintain the preset release angle.	Check channel neutral, receiver input and servo response
ARM / TAKEOFF take off	Operator engages ARM and advances throttle	The flight control enters the output state, and the four motors generate thrust.	Use ANGLE mode to assist posture stabilization
FLIGHT flight transport	Operator controls Roll / Pitch / Yaw / Throttle	Flight control closed loop to adjust motor output	IMU continuously feeds back posture, and PID is corrected in real time
LAND / DOCK platform landing	Close to target platform	Reduce throttle and control attitude	Avoid high-speed contact with the platform to reduce the risk of overturning
GROUND_ MOVE ground dribbling	Subsystem startup, joystick input	Two-wheel differential servo motor output	ESP-NOW heartbeat maintains the connection and stops after timeout
CLAMP	Press the	The gripper servo moves	Angle limitation

Status	Trigger condition	Main control output	Reliability and safety design
clip ball	clamping button	to the clamping angle	avoids excessive clamping or mechanism jamming
RELEASE release the ball	Press the release button	The gripper servo moves to the release angle	It is recommended to perform this operation when the car body is stopped to avoid the ball deviation.
FAIL_SAFE failsafe	Remote control signal or ESP-NOW interruption	The flight control end stops the motor; the subsystem wheels return to zero.	Prioritize predictable, low-risk behaviors

The purpose of this state management is to prevent all executors from being triggered arbitrarily at any time. For example, the flight phase mainly relies on the flight control attitude closed loop; the platform ground operation phase mainly relies on the ESP32 subsystem differential control. By distinguishing mission status, mutual interference between flight control, ball retrieval control and abnormality protection can be avoided.

3.6 Fail-safe design and safety verification

The fail-safe design of this system is divided into two parts: the main flight control system and the auxiliary subsystem.

1. Regarding the main flight control system, this group considers that wireless communications may encounter interference or interruption in actual fields, so a fail-safe protection must be set. Since this system is not equipped with a spatial

positioning sensor that is sufficient for accurate autonomous landing, if the system is required to automatically stabilize the landing when the signal is interrupted, it may cause unpredictable behavior due to misjudgment of altitude, position or attitude. Therefore, this group chose a more conservative but predictable strategy: when it detects that the remote control signal is lost, the motor slows down and stops output. Although this strategy cannot guarantee a complete landing of the vehicle, it can reduce the risk of the system's continued crash or uncontrolled flight based on the existing sensing capabilities. This protection logic has also been verified by turning off the power of the remote control in a safe state where the propeller is not installed, and it is confirmed that the motor can stop according to the set logic.

2. In terms of the attached ball-retrieval subsystem, the ESP32-C3 receiver sets a receive timeout mechanism. When the receiving end does not receive a new ESP-NOW packet for more than 300 ms, the system immediately resets the left and right wheel commands to zero and stops the wheels. This design can prevent the subsystem from maintaining the last speed command and continuing to move after the wireless signal is interrupted. Since the platform ball retrieval task is usually located in a limited space, if it continues to move forward after losing control, it may cause the mechanism to fall off the platform or hit the ball. Therefore, stopping the wheels after the reception timeout is a safer and more reasonable way to deal with it.

   In addition, the optimized ESP32 sender retains the heartbeat mechanism. In other words, even if there are no new input changes from the operator, the host will still periodically transmit the current status at a lower frequency to let the receiving end know that the connection still exists. This design takes into account two requirements at the same time: on the one hand, it avoids communication congestion caused by fixed high-frequency packets, and on the other hand, it prevents the receiving end from misjudging communication interruption and triggering a stop.

3.7 Control delays, communication congestion, and real-time corrections

During the integration test of the accessory ball-retrieval subsystem, this group encountered significant control delays. When the operator continues to deflect the joystick to control the differential movement of the left and right wheels, if the gripper button is pressed at the same time, the ESP32-C3 receiving end will often delay action for about 1 to 2 seconds. This problem has a great impact on the ball retrieval task, because the allowable error in the position of the ball on the platform is limited. If the clamping command is delayed, the ball may have deviated from the original clamping position, causing the ball retrieval to fail.

The team first eliminated the possibility of simply insufficient power supply, and then decomposed the problem into four levels: control logic, communication frequency, sensing noise, and program blocking. The first version of the ESP32 sender continuously sends packets at a fixed 40 ms cycle. Even if the status of the joystick or button does not change significantly, the system will still repeatedly send the same or similar data. On the other hand, the HW-504 joystick is an analog potentiometer input, and analogRead() itself will have slight jitter; if the program treats a command change of ±1 as a valid input, the system will continue to generate meaningless packets. When these packets continue to enter the ESP-NOW transmission process, and the first version of the program does not wait for the previous packet to complete the callback before sending the next packet, it is easy to cause packet stacking and queue congestion.

In addition, the first version of the program uses Serial.print() extensively on both the sending and receiving ends to output real-time status. This is helpful for observing packet content in the early stages of debugging, but when executing at high frequencies in a control loop, the serial port output speed may not be able to keep up with the main loop, causing the program to wait for the output to complete, causing blockage. At this time, what the operator feels is not a simple wireless communication delay, but the entire control loop is slowed down by the debug output.

In order to solve this problem, our team changed the sending end to an architecture of "event triggering as the main component and heartbeat supplementary sending as the auxiliary". First, the left and right wheel commands are quantified to filter out the tiny jitter of the analog joystick; second, the packet to be sent is only updated when the content of the packet has changed significantly compared to the last valid state; third, before actually calling esp_now_send(), it must be confirmed that the previous packet has been reported through send callback to avoid unlimited stacking of packets;

fourth, even if there are no new operations for a long time, the current status is regularly supplemented in the heartbeat mode, so that the receiving end can confirm that the connection still exists. The receiving end avoids using blocking delay to allow the gripper to rotate slowly at once, and instead adopts non-blocking update: the gripper gradually approaches the target angle, and the left and right wheels continue to refresh the speed command in each main loop.

After the above corrections, the subsystem no longer continues to increase packets due to joystick noise, and also prevents the gripper command from being overwhelmed by a large number of invalid wheel speed packets. The overall operating experience has become significantly smoother, and the response delay when performing multiple operations simultaneously is also significantly reduced. This case shows that the real-time control system not only depends on the wireless protocol itself, but also depends on the data update strategy, whether the program is blocked, whether the sensing noise is processed, and whether the packets have a reasonable throttling mechanism.

3.8 Flight control main system debugging and cross-domain failure analysis

The debugging process of the main flight control system showed that unstable flight or motor abnormalities may not necessarily come from PID parameters, but may also come from signal interpretation methods, communication port settings, flight modes, welding reliability or mechanical assembly problems. Therefore, this team adopts a hierarchical investigation method and divides possible causes into communication layer, software setting layer, control parameter layer, power supply layer and institutional hardware layer.

1. The first case is motor jamming and overload protection. At the beginning of the test, the motor rotated unevenly, stuck three times and then stopped. If you only judge from the software perspective, you may mistakenly think that it is a problem with DShot, ESC settings or PID parameters. However, after cross-reference and hardware inspection, the team found that the real cause was that the screws fixing the motor were too long, and after being locked in, they contacted or even damaged the internal coils of the motor. Damage to the coil will cause abnormal

current, which will trigger the overload protection of the flight control or ESC, causing the motor to "try to rotate several times and then stop." Further observation revealed that some motors could still rotate on the surface, but in fact it was just the screws that temporarily pressed the damaged coils, causing the circuit to conduct temporarily; once the screws were removed, the motors would no longer work properly. Therefore, this group ultimately concluded that such motors do not have long-term reliability and should be replaced instead of left to chance. This case illustrates that mechanical assembly details may directly cause electromagnetic system failure, and all motor abnormalities cannot be attributed to control parameters.

2. The second case is the difference in interpretation of SBUS and PWM signals. In the early days, in order to confirm the R9DS receiver output, our team tried to use a general three-purpose electric meter to measure the receiver pins. However, SBUS is not a PWM type pulse signal, but a high-speed digital stream. A single signal line contains encoded information of Roll, Pitch, Yaw, Throttle and multiple AUX channels. Therefore, ordinary electricity meters cannot effectively interpret SBUS signals. Subsequently, the team switched to using the Betaflight receiver page and oscilloscope as the main judgment tools to avoid misjudgments caused by measurement methods that did not conform to the nature of the signal. This also confirms that we must first understand the signal pattern when debugging sensing and communication problems. We cannot use analog voltage intuition to judge digital communication.
3. The third case is the UART port setting error. There has been a situation in the system where the receiver light signal is normal and the remote control and receiver are connected, but Betaflight cannot read the input from the remote control at all. The team initially suspected that there might be a problem with the receiver, remote control, firmware and wiring. Later, through layered troubleshooting, it was confirmed that the real root cause was that the hardware SBUS signal was connected to UART2 of the flight control board, but UART6

was opened by mistake on the Betaflight software side. This causes the flight control board to continue waiting for data on the wrong communication port, so even if the hardware signal is present, the software cannot read the input at all. After correcting to UART2, the remote control signal immediately returned to normal. This case highlights the importance of software and hardware mapping consistency in electromechanical integration.

4. The fourth case is the issue of AirMode and the neutral value of the receiving channel. Theoretically, Roll, Pitch, and Yaw should be close to 1500 when the joystick is not operated, and the lowest value of the throttle should be close to 1000. If the neutral value deviates, the aircraft may yaw, move laterally or drift even when the operator does not input instructions. On the other hand, AirMode will still maintain attitude control authority at low throttle, resulting in problems such as unintuitive throttle retraction and unpredictable motor behavior during the test phase. Therefore, at this stage, this team chooses to correct the neutral value of the channel and turn off AirMode to make the relationship between throttle and motor output more direct, giving priority to ensuring initial test flight safety and operational predictability.
5. The fifth case is the failure of SBUS solder joints caused by landing impact. After many successful flights, the system suddenly lost contact after landing. The motor could not enter the standby state when ARM was turned on again, and the Roll, Pitch, Yaw, Throttle and AUX channels in Betaflight were unresponsive. After checking the remote control, receiver, Dupont cable, oscilloscope and flight control board layer by layer, the team confirmed that the failure point was located at the SBUS solder joint on the flight control board. Since landing at this stage may still produce a large impact, the solder joints may suffer from fatigue fracture or poor contact after being subjected to repeated vibrations. This case shows that welding reliability itself is part of the system performance; the reliability of electronic systems not only depends on circuit design, but also depends on the mechanical vibration environment, solder joint fixation method and structural shock absorption design.

3.9 The distinction between control problems, sensing problems and institutional problems

In order to avoid blind modifications during debugging, this team divided the sources of anomalies into three categories: control problems, sensing/communication problems, and organizational problems, and adopted different troubleshooting methods for different types.

1. Control problems usually manifest as a system that is able to receive commands, but responds erratically or not as expected. For example, if a four-axis flight controller exhibits low-frequency swings when mounted on the bottom and with large inertia, it may indicate that the PID or filter settings are not suitable for the current aircraft; if the ESP32 subsystem delays amplification when multiple operations are input simultaneously, it may indicate insufficient packet throttling, send callback, or program non-blocking design. This type of problem should be analyzed from the perspective of control cycle, command mapping, controller parameters, data update frequency and algorithm immediacy.
2. Sensing and communication issues often manifest as the controller not getting the correct input. For example, SBUS cannot be read with a general electric meter and must be confirmed through Betaflight or an oscilloscope; when the receiver light is normal but there is no input from the flight control at all, you should first check the receiver output mode, SBUS light, UART correspondence and Betaflight receiver settings; if there is a delay in the ESP-NOW subsystem, you need to check whether packets are stacked, whether the send callback is waiting, whether Serial.print is blocking, and whether the receiving end has a timeout stop mechanism. The core of this type of problem is not "the output does not move", but "whether the controller really receives the correct data."
3. Mechanical problems often appear to be control or electronic control failures, but are actually caused by geometric assembly, part interference, solder joint stress, screw length, structural rigidity, or center of gravity configuration. For example, the motor card stopped after three times, which appeared to be an ESC or control parameter error, but in fact the motor screw was too long and damaged the coil; the remote control lost contact after landing, which appeared to be an error in the receiver or flight control settings, but was actually due to poor contact of the

SBUS solder joints due to impact; when the ball was clamped, the car body turned backwards, which was not a simple servo angle problem, but a stability issue caused by the force application position of the ball clamping mechanism and the center of gravity of the entire machine. These cases all illustrate that electromechanical integrated systems cannot be thought of in a single domain, but must consider control, sensing, communication, mechanism and manufacturing errors at the same time.

Through this classification, the team can first determine the nature of the problem and then select the corresponding testing method during subsequent debugging. For example, if there is no channel input in Betaflight, check the SBUS, UART and receiver modes first; if the channel input is normal but the motor does not move, check the ARM, failsafe, Betaflight connection status and motor test page; if the motor can rotate but not smoothly, further check the screw length, coil status, ESC and motor cross test. This layered debugging approach can reduce unnecessary attempts and avoid changing too many variables at the same time without confirming the root cause.

Table 20. Summary of major abnormal phenomena, root cause analysis and corrective measures

abnormal phenomena	**initial suspicion**	**ultimate root cause**	**Corrective measures**	**Verification method**
The motor freezes, rotates a few times and	ESC / DShot / PID setting error	The motor screw is too long and damages the coil, triggering the overcurrent protection.	Replace the motor and confirm the screw length	The motor can rotate smoothly and continuously

abnormal phenomena	initial suspicion	ultimate root cause	Corrective measures	Verification method
then stops.				
Betaflight cannot read remote control input	Receiver or remote control failure	SBUS entity is connected to UART2, but the software mistakenly opens UART6	Fix Betaflight Ports settings	The Receiver page can read the channel value
Motor behavior is unintuitive after reducing throttle	PID or throttle setting problem	AirMode still maintains attitude control authority at low throttle	Turn off AirMode during testing phase	Throttle control is more predictable
Subsystem gripper delay 1-2 seconds	Insufficient power supply or unstable ESP-NOW	Joystick shaking, high-frequency packet delivery, Serial.print blocking, waiting for callback	Command quantification, change only, send callback, heartbeat	Latency is significantly reduced during multiple operations
Lost contact suddenly after landing	Receiver or flight control failure	SBUS solder joints suffer from poor contact due to falling impact	Improve welding, shock absorption and wire fixing	Check channel input after reconnecting and shaking

Table 4-2 shows that the debugging process of this group does not directly replace parts or blindly adjust parameters. Instead, it first proposes a preliminary hypothesis based on the abnormal phenomenon, then confirms the root cause through cross-testing, software setting inspection and hardware measurement, and finally confirms whether the correction is effective in a verifiable manner.

3.10 Ball clamping mechanism and control reliability design

The ball-retrieval task of this system does not only rely on the servo motor to clamp the ball, but also incorporates the reliability of the mechanism into the control design. If the initial design idea relies entirely on servo motor torque clamping, although it may be enough to catch the ball under static testing, the ball will be affected by gravity, inertia, vibration, attitude changes and downwash during flight, take-off and landing. If the last line of defense only has servo torque, the ball may still fall off once the servo angle error, mechanism shaking or instantaneous impact occurs.

Therefore, our team changed the design idea from "clamping by motor" to "supporting by rigid structure". Finally, a design concept similar to that of a plug-in car was adopted, with fixed rods set under the mechanism so that the ball can enter the support range formed by the two rods. When the vehicle takes off, the main weight of the ball is borne by the fixed structure rather than the continuous output torque of the servo motor. The servo drive component is responsible for the horizontal positioning and anti-sway of the ball, making it difficult for the ball to slide forward, backward, left, and right during flight. This design separates the two functions of "load-bearing" and "positioning" to reduce the risk of the ball falling due to a single servo failure.

In the process of releasing the ball, this team also tried its best to use geometry and relative motion to simplify control. When the vehicle reaches the target platform, the clamping claw fixing member opens, and the vehicle body retreats backward, allowing the ball to naturally stay on the platform. This method is easier to control than complex active throwing or multi-degree-of-freedom grippers, and is more in line with the reliability requirements of the current system.

During the actual test, this team also found that the body may tip backwards when clamping the ball due to the position of the force application and the configuration of the center of gravity. Therefore, a bracket was added behind the ball clamping mechanism to improve the support conditions and center of gravity stability. This improvement makes the ball clamping process more stable and allows the mechanism to approach the ball from more angles. This case once again shows that the success rate of ball retrieval does not only depend on the control program, but also depends on whether the mechanism geometry, center of gravity configuration and structural support are reasonable.

3.11 Operational standardization, SOPs and reproducibility

In addition to correcting single faults, this team also organizes debugging experiences into reproducible operating procedures to reduce human errors in subsequent tests. In terms of flight control settings, the team has established basic SOPs: Before re-burning or significantly modifying settings, back up the current configuration with diff all; after burning the flight control firmware, reconnect power and restart; enter the Ports page to confirm UART correspondence; turn off AirMode; set failsafe; load a Preset suitable for the aircraft size; and finally confirm on the Receiver page that the neutral values of Roll, Pitch, and Yaw are close to 1500, and the minimum throttle value is close to 1000. This process can avoid unpredictable flight behavior caused by missing settings.

In terms of pre-takeoff inspection, this team confirmed that the physical position of the motor must be consistent with the definition of the flight control software, the motor steering and propeller directions must be correct, the length of the screw must not contact the motor coil, the battery voltage must be higher than the test threshold, and failsafe must be tested before the propeller is installed. In terms of remote control operation, you need to confirm that the output mode is SBUS, the receiver light signal is SBUS mode, AUX6 corresponds to ARM, and AUX5 corresponds to ANGLE. Before taking off, the throttle needs to be pulled all the way, and ARM and ANGLE must be turned on in sequence, and then slowly increase the throttle.

The team also noticed a discrepancy between the Betaflight wiring state and the actual motor control state. When the flight control board is connected to Betaflight via USB or Wi-Fi, the operator can see the value changes of the remote control channel in the software interface. However, the remote control input in this state is usually only used for data display and setting confirmation, and does not mean that the motor will actually operate. If you don't understand this, it is easy to misjudge that the receiver is normal but the motor is faulty. Therefore, it is necessary to clearly distinguish between "configuration software monitoring status" and "actual flight control status" when debugging to avoid misjudgments due to the safety locking mechanism.

Through SOPs and checklists, this team transformed the original fragmented debugging experience into a replicable engineering process. This not only facilitates subsequent testing of this group, but also allows different team members to follow the same logical operations when handing over or rebuilding the system. For mechanical engineering practice projects, reproducibility itself is part of system maturity; if the system can only be operated by specific members based on experience and cannot be documented and standardized, its reliability is still insufficient. Therefore, this group regards operational standardization as a part of system reliability design, rather than just ancillary recording work.

In addition to subjective operating experience, this group also defines reproducibility as: multiple vehicles can show consistent ARM, ANGLE, throttle response, motor steering and basic hover performance under the same wiring, the same Betaflight settings, the same remote control channel configuration and the same pre-takeoff inspection process. This definition makes "reproducible" not just an increase in the number of spare machines, but means that the same engineering process can obtain consistent results on multiple physical systems.

In addition, in terms of reproducibility, this group did not just stop at file organization, but actually produced more than three backup machines with the same architecture. Each backup aircraft uses the same flight control configuration, receiver settings, motor and ESC drive logic, and similar mechanical configurations. After assembly, setting and testing, the flight and control performance of these vehicles were basically consistent and met the team's expectations for thrust, control response and mission execution.

This result is of great significance to this project. If the system only operates successfully on a single machine, it may just be an accidental result of specific parts, specific assembly errors, or the experience of a specific operator; but when multiple machines can be assembled and set up with the same process and show nearly consistent control performance, it means that the design of this group has achieved a certain degree of standardization and replicability. This also means that the aforementioned SOP, wiring specifications, Betaflight setting process, remote control channel configuration, motor direction confirmation, failsafe test and pre-takeoff checklist are not just written records, but engineering methods that can actually support multi-machine replication and team handover.

Therefore, this group considers "backup mechanism production" as part of the system reliability verification. The existence of multiple backup machines not only reduces the risk of mission interruption caused by a single vehicle failure, but also proves that the control architecture and assembly process of this system are reproducible. For mechanical engineering practice projects, this means that the results of this group have been upgraded from a single prototype to a replicable, maintainable, and replaceable engineering system.

Figure 58: Reproducibility verification of multiple backup vehicles

3.12 System Reliability Summary

Based on the above design and debugging results, the reliability of this system does not come from a single controller, a single parameter or a single mechanism, but from multi-level engineering integration. The main flight control system uses IMU feedback, hierarchical closed-loop control, Mixer thrust distribution and failsafe settings to enable the four-axis vehicle to maintain basic attitude control capabilities under load changes. The auxiliary ball-taking subsystem uses ESP-NOW communication, differential servo control, send callback, command quantization, heartbeat and receive timeout to make the movement and ball-clipping actions on the platform more real-time and predictable.

During the implementation process, this group also confirmed that control issues, sensing issues and organizational issues are not independent of each other. A stuck motor may appear to be an abnormality in the electronic control, but the actual root cause may be damage to the coil caused by overlong screws; a lost connection in the remote control may appear to be a receiver problem, but may actually be due to poor contact of the SBUS solder joints after being impacted by landing; flight drift may appear to be a poor PID, or it may be due to neutral value deviation, incorrect Pitch direction setting, AirMode behavior not suitable for the test phase, or incorrect propeller direction. Therefore, the team's debugging strategy is not to directly adjust parameters or replace parts, but to first determine the problem attributes in layers, and then check the signals, settings, power supply, mechanism and control logic in sequence.

The core principle of the final control design of this group is: during normal operation, the operator only needs to input the mission intention, and the system completes attitude stabilization, differential movement and ball-clipping action by the flight control and sub-controllers; when an abnormality occurs, the system preferentially enters a predictable and low-risk protection state, such as decelerating and stalling after the flight control end loses contact, and the wheels stop after the subsystem end ESP-NOW times out. On the other hand, through the pre-flight checklist, Betaflight setting SOP, receiver and UART inspection process, servo power supply and common ground requirements, as well as solder joint damping and mechanical fixation improvements, this team transformed the failure experience in the process into subsequent reproducible engineering specifications.

Therefore, although this system still has risks such as flight stability, landing impact, and solder joint reliability, it already has a clear control architecture, sensing and communication processes, state machine concepts, fail-safe design, layered debugging logic, and operational standardization processes. These design and verification processes can reasonably explain how this group further improves the engineering practice goal from simply "making the system work" to "making the system stable, predictable, debugable and reproducible".

Appendix

1. Team meeting minutes

first week

Meeting time: 2026/03/05 (Thursday) Moderator: CHENG-HSUAN LEE

Secretary: Lin Yueti Attendance status: No one absent

Meeting content: This week we will mainly discuss course topics, site constraints and possible design directions. The team members proposed solutions such as shooting balls, climbing, building bridges, hot air balloons and drones, and compared them in terms of cost, production difficulty, stability and mission success rate. After discussion, it was believed that the error tolerance rate of shooting balls is low, the climbing and bridging mechanisms require high material strength and structural accuracy, and the position of a hot air balloon is difficult to control. Therefore, it was finally decided to use drones as the main solution, and the subsequent focus would be on the design of the flying platform and ball-retrieving mechanism.

second week

Meeting time: 2026/03/12 (Thursday) Moderator: Liao Zhixiang Secretary: Li Wangmao Attendance status: No one is absent

Content of the meeting: This week, the basic structure and core components of the drone were further confirmed, including flight control board, motor, propeller, battery, remote control and receiver. The team initially estimated the thrust-to-weight ratio and endurance time of the drone, and confirmed that the currently planned power system should be able to support the mission requirements. At the same time, we also began to discuss the ball retrieval mechanism, including active clamping jaws, funnel-type mechanisms and other possible designs. We also noticed that the propeller downwash airflow may cause the ball on the platform to move, so we decided that the impact of the downwash flow on the ball retrieval stability must be actually tested in the future.

Week 3

Meeting time: 2026/03/19 (Thursday) Moderator: Yang Zhenguo Secretary: CHENG-HSUAN LEE Attendance status: No one is absent

Meeting content: This week we will discuss and test fuselage and arm materials. The team compared the differences in weight, strength and deformation between MDF and 3D printing materials, and concluded that the machine arm needs to withstand a larger moment, so MDF is more suitable. The body can use 3D printing to reduce weight and facilitate customization. During the meeting, it was also confirmed that the configuration of the motor, battery and propeller still had sufficient safety margins, and they began to plan the frame size, hole configuration and placement of electronic parts in preparation for subsequent physical assembly.

Week 4

Meeting time: 2026/03/26 (Thursday) Moderator: Lin Yuedi Secretary: Liao Zhixiang Attendance status: No one absent

Meeting content: The production and preliminary assembly of most of the body parts were completed this week, and the weight of the body was measured. It was confirmed that the current weight is still lower than the original estimate, which means that there is still some weight room for the subsequent ball-retrieving mechanism. In terms of the ball-retrieving mechanism, the team compared the highly stable clamping jaw with the wide-range ball collecting mechanism, and finally preferred to use a simpler and easier-to-control clamping jaw design. During the meeting, it was also discovered that the flight control board cannot directly control the servo motor, so it was proposed to use ESP32 as an additional control module to control the ball retrieval mechanism or subsystem.

Week 5

Meeting time: 2026/04/02 (Thursday) Moderator: Li Wangmao Secretary: Yang Zhenguo Attendance status: No one is absent

Meeting content: This week, the overall architecture of the UAV and ball-retrieval subsystem was confirmed. Since the drone would be easily affected by downwash and ground effects if it hovered above the platform to retrieve the ball, the team decided to change the direction of the parent-subsystem to "after the drone lands, the subsystem retrieves the ball." The CAD design of the fuselage continues to be optimized, including upper and lower board connections, battery fixation and wiring configuration. As for the ball retrieval mechanism, the first-generation gripper prototype was completed, but after testing and discussion, it was found that the stability of the ball retrieval still needs to be improved. In addition, abnormalities were found during flight control installation and motor testing, which need to be investigated as a priority next week.

Week 6

Meeting time: 2026/04/09 (Thursday) Moderator: CHENG-HSUAN LEE Secretary: Liao Zhixiang Attendance status: No one absent

Meeting content: This week mainly deals with motor failure and sub-car system production. After inspection, it was confirmed that the cause of the motor abnormality was related to the fixing screw being too long. The screw was locked too deep, causing damage to the motor coil. Therefore, the team decided to replace the damaged motor and establish a screw length inspection process. In terms of flight control, the SBUS communication settings of the remote controller, receiver and flight control board have been completed, and basic flight tests have been completed. The ball-picking subsystem was changed to a front fork with a servo baffle design, so that the main force is borne by the structure. The servo is only responsible for limiting the position of the ball, and completes the preliminary ball-clipping test through ESP32 remote control.

Week 7

Meeting time: 2026/04/16 (Thursday) Moderator: Liao Zhixiang Secretary: Lin Yuedi Attendance status: No one is absent

Content of the meeting: This week, we will review the problems after the first flight of the drone, especially the unstable flight caused by the sub-car. The team believes that the main reasons may include a downward shift in the center of gravity, an increase in overall inertia, an unstable fixation method for the sub-car, and the fact that the flight control parameters have not fully matched the new airframe configuration. During the meeting, it was decided to first reduce the weight of the sub-car and improve the fixation method to avoid shaking and amplifying attitude errors during flight. At the same time, it is also planned to increase integrated testing to connect the flight, landing, ball retrieval and recovery processes in series, instead of only testing each subsystem individually.

Week 8

Meeting time: 2026/04/23 (Thursday) Moderator: Yang Zhenguo Secretary: Li Wangmao Attendance status: No one absent

Meeting content: This week's focus is on integration and stabilization before mid-term testing. The team corrected the structural problems that occurred when the sub-car landed, picked up the ball, and recovered, strengthened the baffles and connectors, and adjusted the operating process of the sub-car. In terms of communication, it was discovered that ESP-NOW control had delays and packet accumulation problems, which would affect the immediate response of the sub-vehicle. Therefore, we began to optimize the transmission logic to reduce unnecessary packet transmission. During the meeting, it was also confirmed that the mid-term test strategy is mainly to complete tasks stably and avoid excessive pursuit of speed and increase operational risks.

Week 9

Meeting time: 2026/04/30 (Thursday) Moderator: Lin Yuedi Secretary: Lu Zhaoyu Attendance status: No one is absent

Meeting content: After Lu Zhaoyu joined the team this week, he first became familiar with the drone architecture and current subsystem status. The meeting mainly reviewed the mid-term test results and confirmed that although the current system can complete the ball retrieval task, there are still problems with ball retrieval efficiency, communication delay and operational stability. After discussion, the team believes that simply increasing the flight speed will have limited help in total time, but may increase the risk of errors. Therefore, subsequent improvements should focus on reducing the delay in ball retrieval, improving the response of the sub-car, improving the consistency of ball retrieval, and establishing a clearer operating process.

Week 10

Meeting time: 2026/05/07 (Thursday) Moderator: Lu Zhaoyu Secretary: CHENG-HSUAN LEE Attendance status: No one is absent

Meeting content: A single motor lift experiment was conducted this week. The motor was fixed on the test stand and the motor shaft was parallel to the ground to reduce the impact of the ground effect on the measurement. Experimental results show that thrust and rotational speed have a roughly quadratic relationship, and a more complete correspondence between throttle, rotational speed and lift still needs to be established in the future. In terms of sub-car communication, the team found that the noise input by the remote control will cause a large number of packets to be transmitted continuously, which will make the ESP32 receiving end unable to process it and cause delays. Therefore, the meeting decided to add a dead zone, reduce invalid packets, and change the system to confirm that the previous packet was sent before sending the next packet to improve the control response.

Week 11

Meeting time: 2026/05/14 (Thursday) Moderator: Liao Zhixiang Secretary: Yang Zhenguo Attendance status: No one is absent

Meeting content: Re-evaluate final testing strategy this week. Since the one-ball solution takes about 50 to 60 seconds to transport each time, repeating it multiple times will affect the overall efficiency. Therefore, the team designed a three-ball retrieval mechanism, hoping to collect multiple balls at one time to reduce the number of round trips. During the meeting, it was discussed that the three-ball mechanism requires a larger front entrance and ball storage space, which will also increase the size, weight and control difficulty of the car. Testing the altitude control effects of Surface mode and AltHold mode at the same time, it is initially believed that AltHold is more suitable as the main flight mode, while Surface mode can be used as a record when the barometer is abnormal.

Week 12

Meeting time: 2026/05/21 (Thursday) Moderator: Li Wangmao Secretary: Lin Yuedi Attendance status: No one is absent

Meeting content: Practical testing of the three-ball mechanism will be conducted this week. Although the three-ball solution can theoretically reduce the number of task cycles, actual testing found that as the mechanism becomes larger, the sub-car needs more precise alignment and is easily affected by the platform boundary and the position of the ball. Larger mechanisms also increase risks to circuit integration, structural reliability and landing stability. After discussion, the team believed that increasing capacity may not necessarily improve actual task performance, so they decided to change back to a more stable one-ball solution for the final test. At the same time, the assembly of the third drone was completed as a formal test and backup aircraft.

Thirteenth week

Meeting time: 2026/05/28 (Thursday) Moderator: Yang Zhenguo Secretary: Lu Zhaoyu Attendance status: No one is absent

Meeting content: This week, a complete process of practice will be carried out according to the one-ball plan, including taking off, approaching the platform, landing, picking up the ball with the sub-car, recovering the sub-car, taking off again and returning to the release area. The team focused on process stability rather than adding new complexity. The meeting also established formal pre-test inspection items, including flight control settings, remote control channels, battery fixation, propeller direction, motor status, sub-car battery and ESP32 communication status. After discussion, it was decided to fix the operator and auxiliary inspection personnel to avoid errors caused by unclear division of labor during the formal test.

fourteenth week

Meeting time: 2026/06/04 (Thursday) Moderator: CHENG-HSUAN LEE Secretary: Liao Zhixiang Attendance status: No one absent

Meeting content: This week we will conduct integration testing and filing drills before the end of the term. The team discussed possible problems one by one, including barometer abnormalities, communication delays, sub-car stuck, ball rolling out, landing position deviation, body collision and backup machine switching. During the meeting, the operational SOP was revised to clearly stipulate pre-takeoff inspection, landing angle, sub-car forward distance, baffle closing timing and retry method in case of mission failure. Finally, it was confirmed that the main machine was responsible for the formal test, and other machines were used as backup to reduce the risk of overall failure caused by a single hardware failure.

Week 15

Meeting time: 2026/06/11 (Thursday) Moderator: Lin Yuedi Secretary: Li Wangmao Attendance status: No one absent

Meeting content: Final confirmation before completing the formal test this week. The team re-checked the drone hardware, battery, motor, propeller, flight control mode, remote control channel, AltHold mode and sub-car mechanism to confirm that all systems were functioning normally. During the meeting, it was finally confirmed that the final strategy was a one-ball conservative plan to reduce the difficulty of control and improve repeatability. The team also organized the entire semester's design process, from early stage program development, drone parts selection, fuselage production, sub-car ball pickup, communication optimization to the establishment of SOP at the end, confirming that the report content is consistent with the testing process, and taking the stable completion of the task as the ultimate goal.

2. **Team meeting minutes**

1. Punctuality and standards

Team members are expected to participate in meetings and discussions on time. If you are more than five minutes late, you will be considered late. Latecomers must be compensated with snacks or drinks to maintain team discipline and cooperation efficiency.

2. Work completion specifications

Each member should complete the assigned work within the time limit according to the division of labor. If the designated tasks are not completed, the entire group must be treated to a meal as compensation; if the circumstances are serious or if the progress of the overall report is affected, their names may not be included in the report.

3. Team goals

Our team does not pursue absolute perfection, but rather aims to steadily complete most of the course requirements. With limited time and resources, the team will prioritize feasibility, stability and task completion.

4. Meeting method

In principle, team meetings are conducted simultaneously online. The meeting time is discussed by all members and decided through voting to ensure that the majority of members can participate.

5. Work distribution

Various tasks will be allocated according to member abilities, time and project needs. The division of labor is determined by internal discussion and coordination within the team, and can be readjusted according to actual progress if necessary.

6. Conflict resolution

If team members have differences of opinion on design direction, work allocation, or other matters, they should first be resolved through discussion and negotiation. If consensus cannot be reached, the final plan will be decided by voting.

7. Work review

The work completed by each member will be reviewed and discussed by all members during the meeting. If problems are found, suggestions for modifications should be made and the responsible members should make the corrections.

8. Time investment

Each member should devote about three hours per week to team-related work, including meeting discussions, data search, design and production, test analysis, and report writing.

9. Teamwork Principles

Team members should take the initiative to report progress and raise difficulties as early as possible to avoid affecting the overall progress. All members should share project responsibilities and have the common goal of completing course tasks.